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Product information.

Principles of Hybrid Technology



BMW Service

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General information

Symbols used

The following symbol / sign is used in this document to facilitate better comprehension and to draw attention to particularly important information:



contains important safety guidance and information that is necessary for proper system functioning and which it is imperative to follow.

Information status and national variants

The BMW Group produces vehicles to meet the very highest standards of safety and quality. Changes in terms of environmental protection, customer benefits and design make it necessary to develop systems and components on a continuous basis. Consequently, this may result in differences between the content of this document and the vehicles available in the training course.

As a general principle, this document describes left-hand drive vehicles in the European version. Some controls or components are arranged differently in right-hand drive vehicles than those shown on the graphics in this document. Further discrepancies may arise from market-specific or country-specific equipment specifications.

Additional sources of information

Further information on the individual topics can be found in the following:

- in the Owner's Handbook
- in the Integrated Service Technical Application.

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The information in the document is part of the BMW Group technical training course and is intended for its trainers and participants. Refer to the latest relevant BMW Group information systems for any changes/supplements to the technical data.

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1. Introduction

1.1. BMW EfficientDynamics



BMW EfficientDynamics is the collective term used in communication to refer to a raft of technologies and innovations that interwork to produce a combination of reduced pollutant emissions plus high-performance power and sheer driving pleasure. With BMW EfficientDynamics, BMW has assumed a global leadership role in the race to implement measures to reduce fuel consumption and CO₂ emissions. In the period since spring 2007, one-million-plus BMW new-car owners have already benefited from these endeavours. Worldwide at this time the range consists of 70 models with CO₂ emissions figures of 140 g/km or lower: no other premium carmaker anywhere in the world can produce comparable figures. But BMW EfficientDynamics is more than just high precision injection, engine start/stop function, or brake energy recovery. In the long term, BMW EfficientDynamics also includes zero-emissions motoring with hydrogen fuel, and in the medium term BMW ActiveHybrid – the combination of internal combustion engine and electric motors in the powertrain. The first production BMW ActiveHybrid models, the hybrid versions of the BMW X6 and the BMW 7-series, matured to production status by late 2009. Consequently, BMW – together with Mercedes-Benz – ranks as Europe's first automobile manufacturers to commence hybridizing their model range in one year. Today BMW offers three ActiveHybrid models: F30H, F10H and F01H/F02H, all of which incorporate BMW EfficientDynamics as well.



BMW EfficientDynamics strategy

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Index	Explanation
1	Measures for immediate implementation: reduction of fuel consumption and consequently CO ₂ emissions. Concerted efforts have brought fuel consumption figures down 27 percent over the period 1995 to 2008. In the EU, the BMW Group's fleet average of 156 grams of CO ₂ per driven kilometer gives it a considerably better ranking than the other premium manufacturers. This has been achieved as the cumulative effect of the implementation of engine start/stop function, brake energy recovery, electric coolant pump, electromechanical power steering, optimized engines, active aerodynamics in the radiator zone, and other measures too numerous to mention.
2	Medium-term solutions such as hybrid and all-electric vehicles
3	Hydrogen-power technology is the long-term solution

What does BMW ActiveHybrid stand for and what distinguishes it from the hybrid offerings of our competitors? In principle, BMW ActiveHybrid refers to all hybrid-technology-related endeavours undertaken by BMW to achieve even higher efficiency – in this respect "Active" underscores the highly dynamic performance-motoring claim that will remain a keynote characteristic of all hybrid BMW models to come. BMW ActiveHybrid also refers collectively to the three different modes of hybridization, namely micro hybrid, mild hybrid and full hybrid.

Here in brief are the basic differences between the three:

Micro Hybrids:

- Intelligent Alternator Control
- Automatic Engine Start/Stop function

Mild hybrid:

- Small electric motor, designed to assist the combustion engine (boost function)
- Not enough power for all-electric motoring
- Moderate potential for reduction in consumption. Advantages particularly in urban-traffic conditions
- Slight increase in weight and space requirement because the hybrid components are smaller and not as heavy as their full-hybrid counterparts
- Production costs higher than those for conventional Internal Combustion Engine design, but lower than those for full-hybrid configurations.

Full hybrid:

- Powerful electric motor or motors capable of delivering enough power to propel the vehicle without the assistance of the internal-combustion engine
- High potential for consumption and emissions reduction, particularly in urban-traffic conditions
- Vehicle weighs more and offers less luggage capacity because the hybrid components take up space and the batteries are heavy
- High component costs, high production costs.

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BMW's hybrid strategy calls for hybrid technology only in circumstances in which the ratio of additional cost to savings potential justifies the outlay. The launchpad is provided by the high-end models with big engines like the BMW ActiveHybrid X6 and the BMW ActiveHybrid 7, because this is where the efficiency potential is largest. Hybrid technology for smaller models will be a medium-term development. BMW has decided that the best way to implement this approach will be to adopt what it calls the 'Best of Hybrid' building-block system, which will enable the optimum hybrid components (full hybrid or mild hybrid, depending on circumstances) to be integrated into the individual vehicle concepts.

1.2. Definition of and Justification for Hybrid Vehicles

1.2.1. Definition of hybrid vehicles



This 1906 Griffon Sport was a hybrid, a bicycle fitted with an engine

The term hybrid is derived from the Greek and means of two different origins or of mixed origin. As used in the automotive industry, the term refers to vehicles with two means of propulsion (utilizing energy from two different sources) and having two different energy supplies. What makes these vehicles so special is that the elements combined in this way are established technological solutions in their own right, and the combination of the two can be utilized to produce new and beneficial features. In adopting the hybrid approach, virtually all carmakers have opted for a combination of internal-combustion engine and electric motor or motors drawing energy from a conventional fuel tank on the one hand and a battery bank on the other.

Justification for hybrid vehicles

The advantages of hybrid-drive technology lie primarily in low fuel-consumption figures, but also in the fact that the electric propulsion unit can provide power in all circumstances in which the combustion engine is inherently unable to operate within its most efficient range. By the same token, the power characteristics of the electric motor or motors and the combustion engine supplement each other all the more effectively in that the lower torque of the combustion engine (at the low end of the rpm range) is made up for by the higher torque of the electric unit being available for pullaway.

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1. Introduction

Other advantages are that the vehicle does not need a starter motor or an alternator (depending on the hybridization mode), because the functionality can be discharged by the electric motor(s) [starter motor/alternator]. What is more, brake energy recovery has a positive effect on brake wear (minimisation of the rate of brake wear).

Energy recovery in motor vehicles

The principle of energy recovery in motor vehicles is based on the law of the conservation of energy.

The law of the conservation of energy states that the total energy in any closed system does not change over time. This energy can be converted into different forms, for example thermal energy can be transformed into kinetic energy, in other words heat can be converted into motion. However, within any closed system it is not possible to produce energy and it is also not possible to destroy energy. The total amount of energy in a closed system always remains constant. In other words, energy cannot be produced out of nothing and conversely, it cannot simply disappear. Energy in its various forms, including kinetic energy, thermal energy, radiation energy, nuclear bonding energy and so on can merely change or be converted from one form to another and can then be used in this new form, but in virtually all instances considerable technological sophistication is involved both in the conversion process and in the subsequent utilisation of the converted energy.

Instead of permitting its kinetic energy or potential energy to be more or less wasted as thermal energy in braking, a hybrid vehicle employs its electric motor to transform the energy back into electrical energy that can be stored in a battery. This stored electrical energy is then available when needed to start the combustion engine, for example, or to propel the vehicle.

How can we calculate a vehicle's kinetic energy or its potential energy?

Kinetic energy is calculated by the formula: $E_{kin} = \frac{1}{2} \cdot m \cdot v^2$. The vehicle's mass is represented by m and v is the velocity of the vehicle.

Potential energy is calculated by the formula: $E_{pot} = m \cdot g \cdot \Delta h$. With m being the mass of the vehicle, g being the gravitational acceleration (9.81 m/s^2) and Δh being the difference in height.

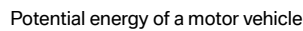
Worked examples:

Kinetic energy: given a vehicle mass of 2500 kg and a velocity of 100 km/h, the formula gives a kinetic energy of $\frac{1}{2} \cdot 2500 \text{ kg} \cdot (100 \text{ km/h})^2$. In order to make the units compatible, velocity has to be converted and expressed in m/s. 100 km/h corresponds to 27.77 m/s.

$$E_{kin} = 1250 \text{ kg} \cdot (27.77 \text{ m/s})^2 = 964\,506 \text{ Nm (Ws)}$$

If it takes 10 seconds to brake the car to a complete standstill, then available power $P = 96\,451 \text{ W}$. Electric power is calculated using the formula $P = U \cdot I$. If the voltage is 12 V, the current obtained in this way would be somewhat more than 8000 A. There is no way in which currents of this magnitude can be utilized effectively in passenger cars. The technique adopted to render the power usable is to increase the voltage and thus reduce current to the range of intensity at which it can be handled in automotive-engineering applications. This is the reason why the voltages used in hybrid vehicles are higher than those used in conventional cars. If we can assume a voltage of 400 V for the purposes of this worked example, the current is 241 A.

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Potential energy: given a vehicle mass of 2500 kg and a difference in height of 100 m, potential energy is

If it takes 10 seconds to descend this height, then available power $P = 245\,250\text{ W}$. Given a battery voltage of 400 V, the current works out at 613 A.

These calculations are highly simplified, but they impart some impression of the magnitudes of energy, power and electric current involved.

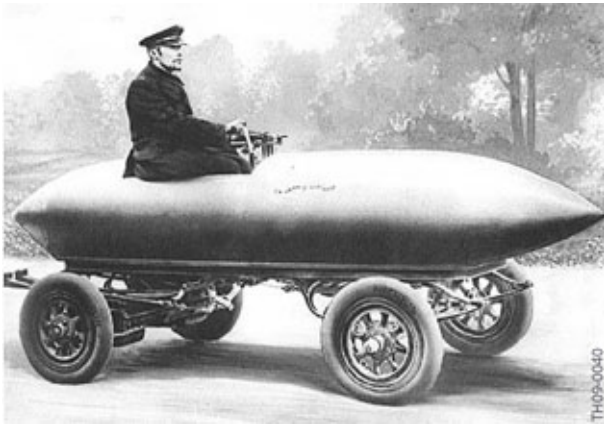
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1. Introduction

1.3. History

1.3.1. Early electrically powered vehicles

There is nothing new about the notion of using electricity as a means of propulsion for wheeled vehicles. In the pioneering days of automobile development (pre-1900), vehicles with electric propulsion units actually outnumbered those powered by combustion engines. It was an electrically powered vehicle, not a petrol-engined car, that was the first to break the magical speed barrier of 100 km/h.

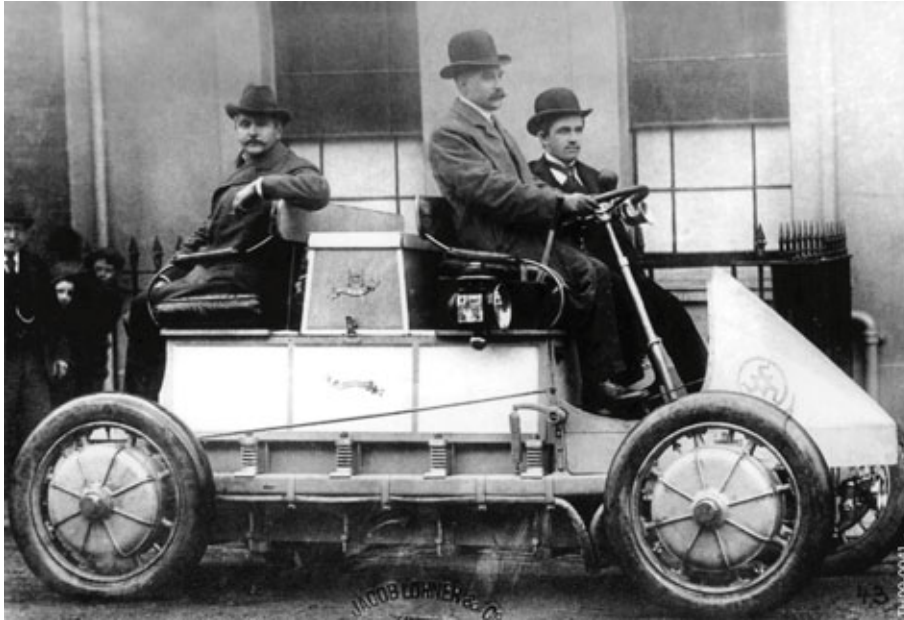


The 'La Jamais Contente' electric vehicle

The vehicle that has gone down in history as 'La Jamais Contente' (French: The Never Satisfied) was the first road-going vehicle capable of reaching speeds in excess of 100 km/h. It was an electrically powered vehicle with a torpedo-shaped body that tipped the scales at 1450 kg. It achieved the record speed of 105.8 km/h on April 29, 1899 on a run near Paris, France. The vehicle boasted two 25 kW motors, rated for 200 volts and 124 amperes. Despite these early achievements, on account of limited range (about 85 kilometers) and the deficiencies of battery technology of the day, electric units were soon supplanted by combustion engines in the course of technological progress. The death-knell was sounded for electrically powered vehicles when it became common practice to use small electric motors merely as starters for spark-ignition or diesel propulsion engines.

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Lohner-Porsche electromobile at the 1900 World Exhibition in Paris, France

Ferdinand Porsche is generally considered the father of the hybrid drive. As long ago as 1896, he applied for a patent for electric wheel-hub motors. Taking motors of this type as his starting point, he worked with Viennese coachbuilder Jacob Lohner and produced the "Mixte" hybrid drive. The propulsion unit consisted of a dynamo powered by a petrol engine. The dynamo generated the electricity for the battery. The hub motors meant that the car dispensed with gearbox and complicated mechanical drivetrain.

1.3.2. History of BMW electric cars



BMW 1602 with electric drive

Back in the early 1970's, BMW engineers were fully aware of the potential of electric propulsion for cars used in urban areas. Several approaches were considered at the time, and two prototypes based on the 1602 drew all eyes when they were used as official vehicles at the 1972 Summer Olympics to accompany the Olympic walkers and marathon athletes. At the time, however, the technological boundary conditions were not propitious for a serious effort geared toward development of a series-produced car. The lead batteries alone added 350 kilograms to the weight of the car and even though the motor was capable of delivering only 12 kW/16 hp, their capacity was sufficient for distances of only 45 to 65 kilometers. Top speed was 100 km/h. The first step had been taken, however, and in 1981 the company again launched an electric-car project. By this time high-voltage sodium-sulphur

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technology was available for the storage batteries. The configuration sufficed to give the eight 4x4 325ix test mules acceptable levels of performance and a reasonable range. Experience amassed with the new battery technology laid the foundation for another milestone in the development story.



BMW E1

The BMW E1 was the first all-electric BMW and it debuted at the 1991 Frankfurt car show. The lightweight city car was designed specially for the electric motor and drivetrain; it was only 3.4 metres long and was propelled by an electric motor rated at 32 kW/45 hp. The speedy little compact boasted a top speed of about 125 km/h. The high-temperature battery bank (sodium-sulphur batteries) had a capacity of about 20 kWh, so the car had a range of some 140 kilometers.



BMW E36 with electric drive

As work on developing the concept continued, the early 1990's saw the production of a small number of electric cars based on the 3-series coupé of the day. In the period up to 1995, the BMW brand contributed eight test cars to a large-scale trial that took place on Germany's Baltic island of Rügen. This co-operative venture involving a number of battery manufacturers and carmakers demonstrated the practicality of the zero-emission propulsion concept, but it also proved that certain problems

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persisted with regard to the use of the high-temperature battery systems. High weight, restricted range and low power density and problems with the durability of the batteries were other factors that hampered efforts to move toward series production of electrically powered vehicles at the time.

In the years that followed, the BMW Group engineers attentively followed the advances in battery technology, which was progressing by leaps and bounds. Today, lithium ion batteries (used in the MINI E, for example) render the whole idea of electric propulsion far more attractive. Millions of these batteries are used in laptop computers and in mobile phones, and the technology combines sustained power delivery, ample storage capacity and comparatively compact dimensions in an extremely convincing package.



MINI E

The air-cooled energy accumulator in the MINI E tips the scales at 260 kilograms and with a total capacity of 35 kWh it provides a range of some 250 kilometers, a significantly above-average figure for electric vehicles today. The compact electric motor with 150 kW/204 horsepower runs on 380 V direct-current (DC) voltage.

1.4. The Competition

Other automotive manufacturers have already introduced hybrid cars or have announced that they will have hybrid cars rolling off their production lines in the coming years. Japanese carmaker Toyota is a leader in this respect and was the first to build hybrid cars in large numbers. The Prius is the hybrid that Toyota has been producing in series since 1997; the car now in the showrooms is the third-generation model marketed under this name.

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Toyota Prius

The petrol engine and the electric motor are coupled to the driven axle by planetary gearing. The combustion engine of the Prius can be shut down completely when not needed. Under uniform load the petrol engine provides all the propulsive power, while at the same time turning a generator to charge the batteries. When load increases, both petrol engine and electric motor provide power in tandem to propel the vehicle.

Hybrid technology has also been introduced at the top end of the market. The LS 600h from Lexus has been available since June 2007.



S400 Hybrid

In 2005, the BMW Group entered into partnership with DaimlerChrysler, as it was then known, and with GM to pool resources in developing a hybrid drive. This joint venture made it possible to bring fully fledged hybrid cars onto the market within a very short space of time. A few months in advance of BMW, Daimler AG debuted its Mercedes-Benz S 400 HYBRID. This car ranks as an implementation of the 'mild' hybrid approach. It has on board a lithium ion battery rated for 0.9 kilowatt hours and so it can claim to be the world's first hybrid car with this high-end battery technology. The battery powers the 15 kW electric motor, which is mounted as a starter/generator on the crankshaft between the V6 combustion engine and the automatic transmission. The BMW ActiveHybrid 7 will have the same hybrid technology.

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2. Principles and Driving Situations

2.1. Classification by Power Rating

In terms of the power ratings of the electric motors they use, hybrid cars can be classified as belonging to one of three groups:

- Micro-hybrid cars
- Mild-hybrid cars
- Full-hybrid cars

The principal characteristics of each group are listed in the table below:

	Power of electric motor	Voltage range	Possible functions	Fuel saving
Micro-hybrid	2 to 3 kW	12 V	- Start/stop function	Less than 10%
Mild hybrid	10 to 15 kW	42 V to 150 V	- Start/stop function - Boost function - Energy recovery	Less than 20%
Full hybrid	> 15 kW	> 100 V	- Start/stop function - Boost function - Energy recovery - All-electric propulsion	More than 20%

2.1.1. Micro-hybrid vehicles

General description

Micro-hybrids constitute the first level in the hybrid-car hierarchy. Micro-hybrid cars have a generator power of 2 to 3 kW and use conventional 12-V battery technology. Power and voltage are both comparatively low, imposing a natural limit on energy recovery during braking and in the overrun phase. In a micro-hybrid car, the recovered electrical energy is fed into the vehicle's on-board 12 V electrical system. In some cases these systems also have a start/stop function with conventional starter or integrated crankshaft starter generator. A drawback of the start/stop functionality is that frequent run-ups mean accelerated wear of the crankshaft, which has frictionless bearings and is designed for constant rotation. Complexity, additional weight and costs for micro-hybrid cars are not beyond reason. All in all, however, the concept cannot justifiably be expected to return more than 10% savings in energy. Strictly speaking micro-hybrid cars are not true hybrids at all, since they have only one means of propulsion.

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2. Principles and Driving Situations

Micro-hybrid solution from BMW



BMW 1-series with automatic engine start/stop function

Intelligent alternator control

BMW has introduced various technologies to reduction the fuel consumption of all BMW Group vehicles. One of these measures is partial recovery of the vehicle's kinetic energy (sometimes also known as energy recuperation). Intelligent alternator control alone can reduce CO₂ by as much as 3 %, depending on the driving profile, with a corresponding reduction in fuel consumption. As far as the customer is concerned, this is achieved without perceptible effects on the power-development aspect of the car's performance. Intelligent alternator control is basically an extension of the charging strategy for the battery. Instead of simply being charged to its full capacity, the battery is charged to a defined extent that depends on various environmental conditions (ambient temperature, battery age, etc.). In contrast to conventional charging, this strategy restricts battery charging to the car's overrun phases. At these times the alternator is run up to full power, electricity is generated and stored in the vehicle's battery. The alternator is not in action while the car is under acceleration. Consequently, no power is diverted to generate electricity, which means that none of the energy in the fuel is utilized for this purpose. This subject is discussed in more detail in the information bulletin entitled "Intelligent alternator control".

Automatic engine start/stop function

The automatic engine start/stop function (Motor-Start-Stopp-Automatik, MSA) is one of the measures that contributes to company endeavours to meet the ACEA voluntary commitment subscribed to by BMW to reduce CO₂ to a defined level by 2008. The reduction in consumption is achieved by having the engine switch itself off automatically when the vehicle is at a standstill. Engine restart is also automatic and takes place as soon as the defined switch-on conditions are satisfied. The automatic engine start/stop function (MSA) is integrated into the E8x, E9x and R5x models in combination with manual gearbox and 4-cylinder engine. MSA is standard in these cars. The MSA function is implemented in the electronic engine management system (DME/DDE). Information available on the bus system for other functions is co-utilized for the MSA function. In order to function correctly, the system also requires certain dedicated sensors. This subject is discussed in more detail in the information bulletin entitled "Automatic engine start/stop function (MSA)".

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2. Principles and Driving Situations

2.1.2. Mild hybrid

General description

Mild-hybrid systems typically use voltages higher than 42 V. Today, some of these systems operate at voltages of 160 V or even higher. The power of the electrical machine represented by the electric motor is in the 10 to 15 kW range. Mild-hybrid systems are generally built around electrical machines that transform some of the kinetic energy in retardation/braking into electric energy that can be stored in batteries. Mild-hybrid systems usually implement a start/stop function with the electrical machine being used to restart the combustion engine when the latter is needed again after shutting down. The electrical machine of a mild-hybrid car is also used to some extent as a booster along with the combustion engine for pullaway and acceleration. Some mild-hybrid systems cut off the fuel supply to the combustion engine if the high-voltage accumulator is adequately charged and the car is travelling at a steady speed up to about 50 km/h. Under these circumstances the car is propelled only by the electrical machine so as to economise on fuel.

BMW's implementations of the mild-hybrid concept



BMW concept 7-series ActiveHybrid

The BMW ActiveHybrid 7 will be in production as a mild hybrid by the end of 2009. Its hybrid system operates at a voltage of 120 V. Within the framework of the mild-hybrid concept, the electric motor boosts the dynamic potential of the 8-cylinder petrol engine and reduces consumption and CO₂ emissions by up to 15 percent, most effectively in urban-traffic conditions and primarily through brake-energy recovery. The design dispenses with the large, heavy hybrid components and batteries of a full-hybrid car, and this in turn has advantages in terms of gross weight and available luggage capacity. The functions implemented in the BMW ActiveHybrid 7 will be as follows:

- Engine start/stop function
- Brake energy recovery
- Boost
- Climate control with engine OFF
- But **not** all-electric propulsion.

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2. Principles and Driving Situations

2.1.3. Full hybrid

General description

It is characteristic of the full-hybrid concept that the vehicle can pull away from rest and drive without the combustion engine turning over. Full-hybrid systems use high-voltage accumulators with voltages in some cases well in excess of 200 V. In a configuration of this nature the electric motor is easily capable of overcoming inertia to set the car in motion, and when sharp acceleration is called for the torque of the combustion engine and that of the electric motor are used in parallel. This is the mode known as boost.

BMW's implementations of the full-hybrid concept



BMW concept X6 ActiveHybrid

The BMW ActiveHybrid X6 has the V8 Twin Turbo with high precision injection of the BMW X6 xDrive50i, combined with two electric motors. Electric motors are inherently capable of developing full torque from rest, and this specific advantage of the electric propulsion unit is ideally combined in this configuration with the V8 Twin Turbo, an engine designed and built specifically for efficiency. The combination of these two different modes of propulsion gives the BMW concept X6 ActiveHybrid exceptional acceleration and performance. Intelligent control of the harmony between electric drive and combustion engine, moreover, ensures that the two offer the benefits of reduced fuel consumption not only in stop-and-go traffic, but also over a much wider roadspeed bandwidth. Along with the functions implemented in the mild-hybrid BMW ActiveHybrid 7, the BMW ActiveHybrid X6 is capable of all-electric driving, in other words it can be driven without the combustion engine turning over.

2.2. Classification by Drive-unit Configuration

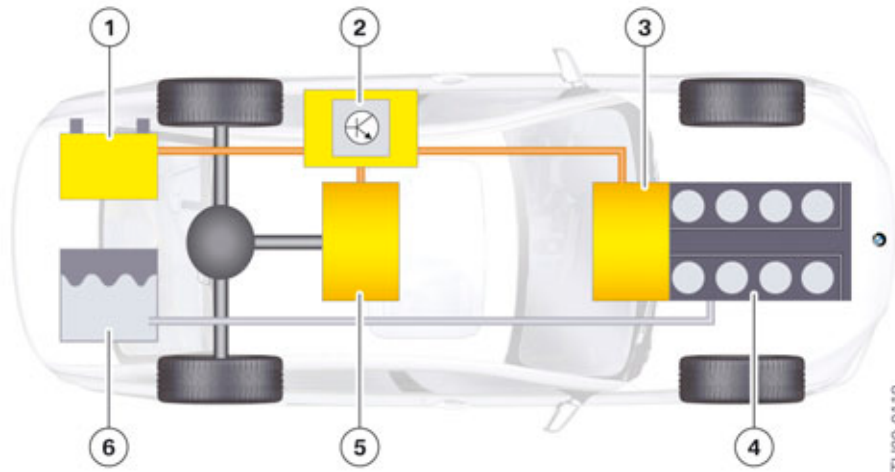
In terms of drive-unit configuration, hybrid cars can be classified as belonging to one of four groups:

- Serial hybrid
- Parallel hybrid
- Power-split hybrid
- Plug-in hybrid

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2. Principles and Driving Situations

2.2.1. Serial hybrid



Components of a serial-hybrid drive

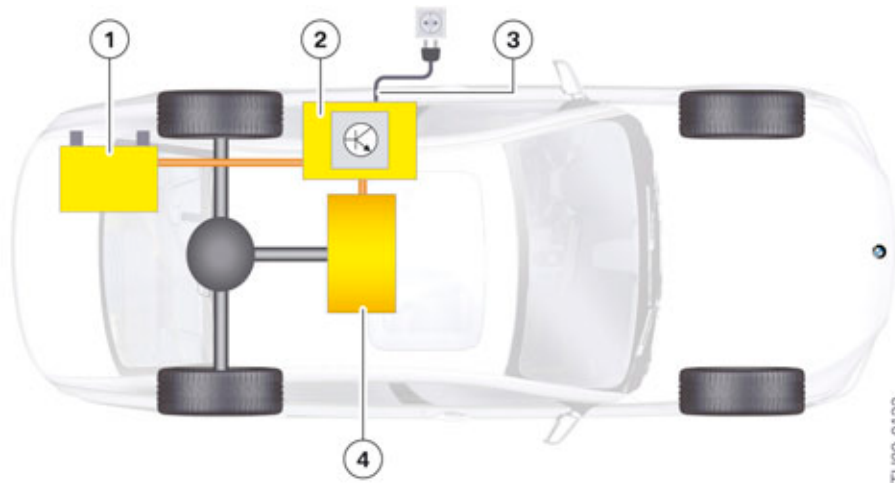
Index	Explanation
1	High-voltage battery
2	Power electronics
3	Alternator
4	Combustion engine
5	Electric motor
6	Fuel tank

A serial-hybrid car has an electric motor and a combustion engine. The characteristic of this concept is that only the electric motor acts directly on the driven wheels. The components in the powertrain are arranged one after the other and this is why the configuration is termed serial. The combustion engine drives an alternator that generates the energy for the electrical propulsion unit and for charging the battery. The flow of electrical energy is controlled by the power electronics. The size of the alternator and the electric accumulator (the battery, in other words) depends on the operating-mode and charging strategy, range and performance. The need for an extra alternator adds complexity to the design, but this is compensated to some extent by the absence of a gearbox. A serial-hybrid configuration offers a great deal of flexibility in terms of the positioning of the individual components. The single most telling drawback of the serial-hybrid concept is the double transformation of energy and the diminished efficiency this inherently entails. Combustion engine and alternator both have to be designed for maximum propulsive power. By comparison with parallel hybrids, the combustion engine is more powerful, but it therefore produces more emissions and has a higher fuel consumption.

Fully electric drives are similar to serial drives to some extent. Instead of being inside the vehicle, however, the generator is external. The battery is recharged from the public grid or a private source to which the vehicle is connected while not in use.

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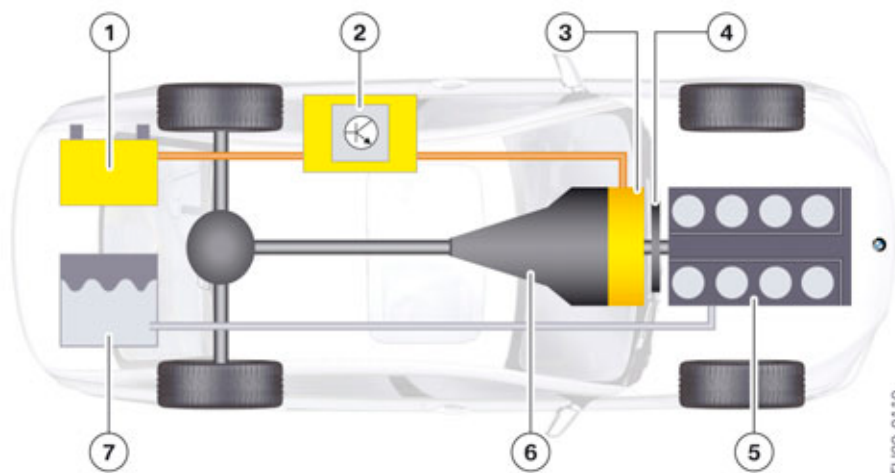
2. Principles and Driving Situations



Components of an all-electric drive

Index	Explanation
1	High-voltage battery
2	Power electronics
3	Mains-supply plug
4	Electric motors

2.2.2. Parallel hybrid



Components of a parallel-hybrid drive

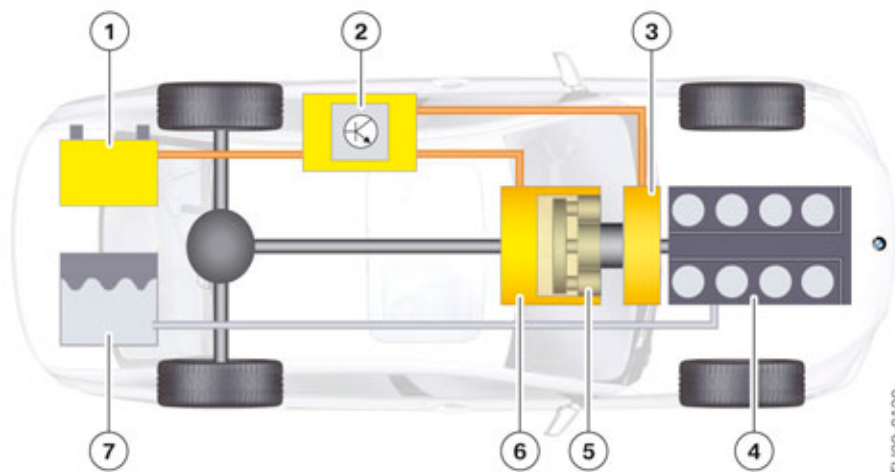
Principles of Hybrid Technology

2. Principles and Driving Situations

Index	Explanation
1	High-voltage battery
2	Power electronics
3	Electrical machine
4	Clutch
5	Combustion engine
6	Gearbox
7	Fuel tank

Unlike serial hybrids, parallel hybrid systems have both the combustion engine and the electric motor mechanically connected to the driven wheels. The two propulsion systems can be used either separately or together to propel the car. The introduction of force from the two sources to the powertrain is parallel, hence the term 'parallel-hybrid system'. Both electric motor and combustion engine can be made smaller and lighter, because the power of both is combined. Savings are the net result, for example in terms of weight, fuel consumption and CO₂ emissions. An option open to designers looking to maximise performance is to install a full-size, full-power combustion engine and then utilise the boost available from the electric motor. This approach too can produce a reduction in fuel consumption. The electric motor can also operate as a generator, which is why the term "electrical machine" is more apt than "electric motor". In overrun phases and when the vehicle brakes the electrical machine generates electricity which – controlled by the power electronics – is stored in the high-voltage battery so that the car can use less fuel. In terms of production outlay, a cost comparison actually puts parallel hybrids ahead of mild hybrids.

2.2.3. Power-split hybrid



Components of a power-split hybrid drive

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2. Principles and Driving Situations

Index	Explanation
1	High-voltage battery
2	Power electronics
3	Electrical machine 1
4	Combustion engine
5	Planetary gearing
6	Electrical machine 2
7	Fuel tank

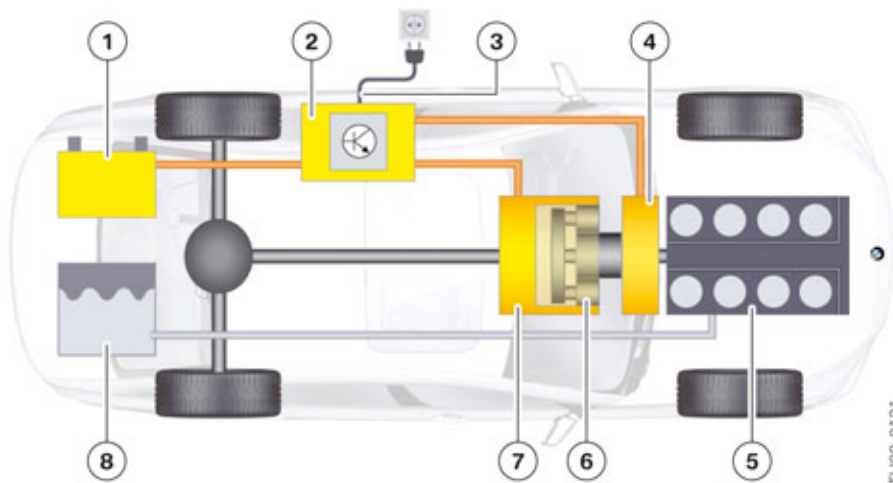
Power transmission can be either serial or parallel, so hybrid systems of this nature are referred to as serial-parallel or power-split systems.

Various operating modes are possible and individual modes can be engaged to suit driving conditions:

- Combustion engine drives alternator (electrical machine 1) to charge the high-voltage battery bank.
- Combustion engine drives alternator (electrical machine 1). The electricity generated in this way is used to power electrical machine 2 (serial hybrid).
- Combustion engine is mechanically connected to the input shaft, as are the electrical machines. Both propulsion units drive the vehicle (parallel hybrid).

In this combined hybrid drive mode a coupling provides the means of switching from one propulsion unit to the other. A vehicle with power-split hybrid drive can be electrically powered up to a certain speed. The electronics also control the way in which the two propulsion modes are combined to keep the combustion engine operating in its most efficient range. The drawbacks of power-split hybrid drives are the sophisticated electronics needed for drive control and the high costs. Power-split systems are usually implemented as full-hybrid configurations.

2.2.4. Plug-in hybrid



Components of a plug-in hybrid

Principles of Hybrid Technology

2. Principles and Driving Situations

Index	Explanation
1	High-voltage battery
2	Power electronics
3	Mains-supply plug
4	Electrical machine 1
5	Combustion engine
6	Planetary gearing
7	Electrical machine 2
8	Fuel tank

Plug-in hybrids are another extended branch of hybrid technology. A plug-in configuration further reduces fuel consumption by having the electrical energy accumulator charged not only by the combustion engine but also off the grid supply when the vehicle is not in use. If the battery charge level is sufficient the vehicle can undertake short journeys in zero-emission electrical mode, with the combustion engine in reserve and needed only for lengthy trips or when the battery is drained. A plug-in hybrid always has a large-capacity battery bank so that more energy from the grid can be stored to make the vehicle less reliant on the combustion engine and maximise the proportion of all-electric driving.

2.3. Driving Conditions

The driving conditions that can apply in the case of a full-hybrid configuration are shown below. The corresponding driving conditions for other hybrid vehicles can be derived by analogy from this summary.

The individual driving conditions can be subdivided into four phases:

- 1 **Accelerating:** the electronic drive control takes power from the high-voltage battery to overcome inertia and set the vehicle in motion. In this phase the car accelerates from rest under electrical power only.
- 2 **Steady driving:** at a certain speed the electrical machine reaches the limit of its power or speed and the electronics switch to the combustion engine as the main propulsion unit.
- 3 **Overtaking:** for overtaking or bursts of speed, the electrical machine and the combustion engine operate together, combining their output to maximise available power.
- 4 **Braking:** the car is initially slowed by electric braking (energy recovery). If braking deceleration is not sufficient the mechanical brake is also applied. During electrical braking the electrical machine operates as a generator and converts a proportion of the braking energy into electrical energy that is stored in the high-voltage battery.

2.3.1. Start/stop function

When the vehicle is at a standstill with the combustion engine at its operating temperature, for example because the traffic has been held up at traffic lights, the combustion engine is switched off. This means that the engine is not emitting CO₂ and fuel consumption is reduced. While the car is at a standstill the high-voltage battery supplies power for climate control, vehicle lights etc.

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2. Principles and Driving Situations

If the charge state of the high-voltage battery is not sufficient to cover these loads the combustion engine is started to charge the high-voltage battery off the electrical machine and provide enough electrical energy to power the various consumers. When the brakes are applied on the approach to traffic lights, the combustion engine is switched off even before the car comes to a complete standstill (the engine is switched off when the car slows to a certain defined speed).

2.3.2. Pullaway

The ability of the electric propulsion unit to deliver high torque at low speed is utilized for pullaway from rest. The car accelerates away from rest solely under electric power, with the electric motors drawing the energy they need from the high-voltage battery. The combustion engine remains switched off (engine at operating temperature).

2.3.3. Accelerating (boost function)

When sharp acceleration is needed to lift the car away from rest at a junction, to overcome a steep gradient or to pass slower traffic, if the high-voltage battery is sufficiently charged its power can be tapped and the additional force made available as kinetic energy through the electrical machine. This is known as the boost function. Combining the power output of the combustion engine and the electric motors produces driving dynamics and acceleration similar to those of a vehicle with a more powerful engine.

2.3.4. Steady driving

The combustion engine and the electrical machine can both contribute to propulsion; the extent to which power is drawn from either source depends on the vehicle's speed and the battery bank's state of charge at a given time.

Combustion engines do not operate at maximum efficiency when required to propel the vehicle at low to medium speeds. The electrical machine, in contrast, is capable of delivering full torque at very low rpm. If the high-voltage battery is adequately charged, electrical energy is tapped from the battery to drive the car under electric power. The combustion engine is not switched on frequently unless the charge level of the high-voltage battery is low: under these circumstances some power is tapped from the engine to recharge the battery.

A combustion engine can operate at maximum efficiency when the car is being driven at a steady, relatively high speed. At this end of the performance band the electrical machine would have to draw too much energy from the high-voltage battery. Consequently, the combustion engine is the primary mover for driving under these conditions. If the high-voltage battery is in a low state of charge part of the power developed by the combustion engine is used to recharge the battery through the electrical machine.

2.3.5. Braking

The ability to utilise the energy that would otherwise be shed on descents or wasted when the car is decelerating is one of the main advantages of a hybrid drive. This mode is known as energy recuperation or more frequently energy recovery. Instead of being dissipated by conversion into thermal energy at the wheel brakes, excess energy is converted into electrical energy by the electrical machine operating as a generator, and stored in the high-voltage battery.

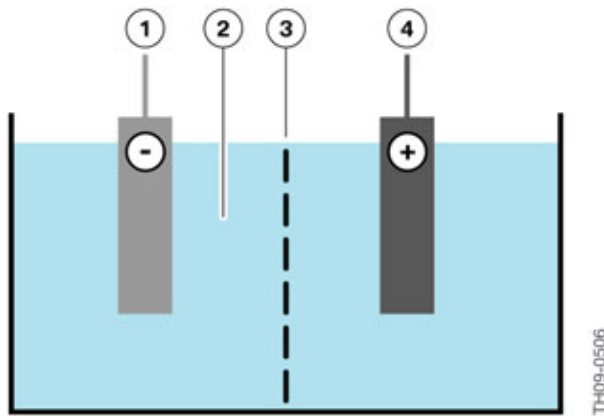
Principles of Hybrid Technology

3. System Components

3.1. Energy Storage Devices

The energy storage device stores accumulated energy so that it will be available for use when needed at some subsequent point in time. Energy is often converted from one form to another so that it can be stored and it then has to be converted again when needed; this process of conversion is unavoidable in order to minimise losses during stoppages. For example, chemical energy (fuel) is stored in the fuel tank and subsequently converted into thermal and mechanical energy by the combustion engine. Losses invariably occur in any process of energy storage or energy transformation. There are many kinds of energy-storage device (mechanical, thermal, chemical, magnetic and electrostatic). Chemical and electrostatic storage devices are explained in more detail below, because these are the types of storage device most frequently used as the secondary source of energy in modern hybrid cars.

3.1.1. Physical principles of chemical storage



Basic structure of a galvanic cell

Index	Explanation
1	Negative electrode
2	Electrolyte
3	Separator
4	Positive electrode

Broadly speaking, a galvanic cell consists of the electrolyte, the battery housing and of course the two electrodes. A separator through which ions can pass but which holds back electrons forms an additional insulating barrier between the electrodes. Chemical reactions that take place in the galvanic cell produce an electron excess at one electrode and an electron deficit at the other. Consequently, there is an electric potential - a voltage, in other words - between the two electrodes.

As the battery discharges, therefore, the chemical reaction converts the energy stored in chemical form to change into electrical energy. The reaction that supplies energy is known as discharge and consists of two subreactions (electrode reactions) which, although separate in space, are interlinked. The electrode at which the corresponding subreaction takes place at a lower ORP (oxygen-reduction potential) than the other electrode is the negative electrode; the other is the positive electrode. As the cell discharges an oxidation process takes place at the negative electrode and electrons are released; at the same time at the positive electrode the corresponding electrons are absorbed in equal numbers by a reduction process.

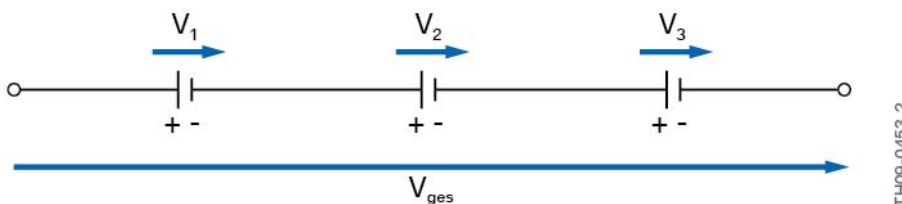
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3. System Components

The electronic current flows through an external consumer circuit from the negative to the positive electrode. Inside the cell the current is carried between the electrodes by ions in the ion-conductive electrolyte (ion current), and ion and electronic reactions in/at the electrodes are interlinked.

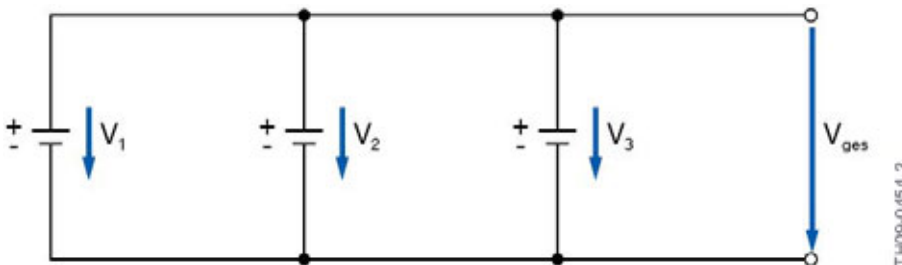
Galvanic cells are DC voltage sources. The name indicating the type of galvanic cell is generally derived from the combination of electrode materials, so for example we have nickel metal-hydride cells. The chemical make-up of the electrolyte and the electrode materials changes, depending on the charge state of the cell. The type of material from which the electrodes are made determines the rated voltage of the cell.

What we generally term a "battery" usually consists of an array - or battery - of galvanic cells interconnected in such a way as to be usable as an energy source. However, in everyday parlance it is also usual to refer to an individual galvanic cell as a "battery". Galvanic cells transform the chemical energy they contain directly into electrical energy. A distinction is drawn between rechargeable and non-rechargeable batteries. The difference is that in a rechargeable battery (sometimes and more accurately termed an "accumulator") the reactions that cause discharge are reversible and the battery can repeatedly cycle between discharging and recharging. Consequently, a battery of this nature can repeatedly transform stored chemical energy into usable electrical energy and vice versa.



Series connection of galvanic cells

If a voltage higher than that available from a single cell is needed, an array of cells can be connected in series. The total voltage of the battery produced in this way corresponds to the sum of the individual cell voltages. In the example above, total voltage $V_{tot} = V_1 + V_2 + V_3$.



Parallel connection of galvanic cells

Capacity

Connecting galvanic cells in parallel produces a battery with a higher capacity. The voltage of the battery is equal to that of a single cell.

Capacity is the electrical charge stored in the battery. A battery's capacity is stated in ampere-hours [Ah]. The actual usable capacity (the current that can be drawn from the battery, in other words) depends on the conditions of discharge. An increase in discharge current means a corresponding decrease in usable capacity.

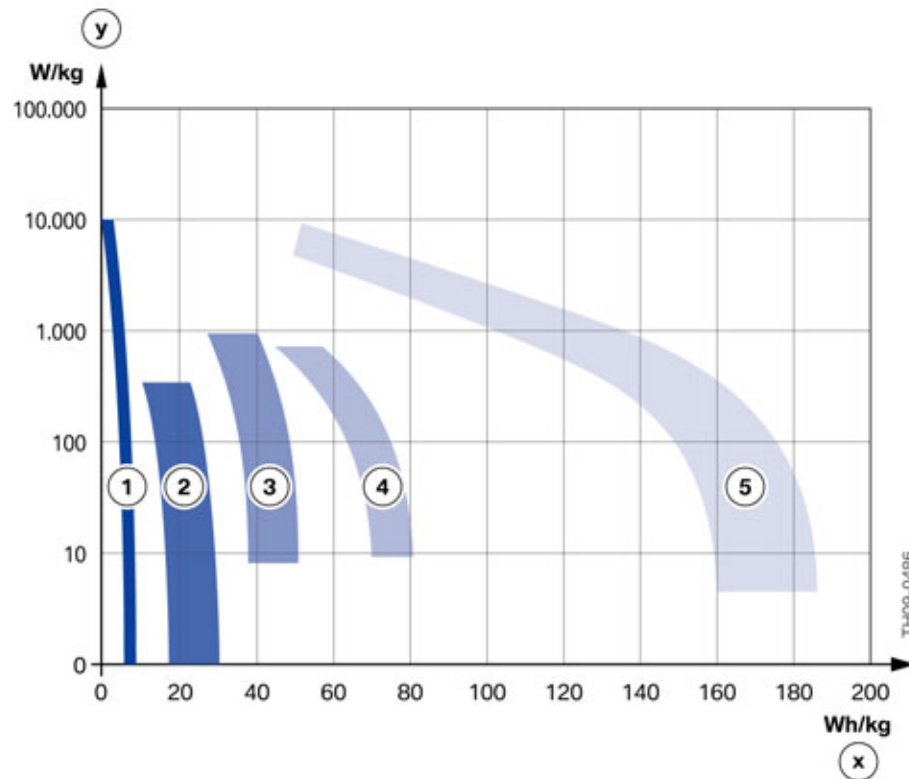
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3. System Components

Power

The power of a battery is stated in watts [W] and is the product of discharge current and discharge voltage. It is not usual to state the energy stored in a battery, but energy per unit of mass or unit of volume is nonetheless an important parameter of battery systems.

Energy density and power density of a battery



Energy and power densities of batteries

Index	Explanation
1	Double-layer capacitor
2	Lead-acid battery
3	Nickel-cadmium battery
4	Nickel metal hydride battery
5	Lithium ion battery
x	Energy density in Wh/kg
y	Power density in W/kg

Energy density is a measure of the distribution of energy over the mass of a substance and is stated in watt-hours per kilogram [Wh/kg]. The energy density of the battery used in a hybrid car is crucial in terms of the range the vehicle is able to cover before the battery has to be recharged.

The power density of a battery defines the battery's electrical power as a function of mass and is stated in watts per kilogram [W/kg].

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3. System Components

The graph above shows the power density and the energy density of various devices capable of storing electrical energy. As the graph shows, double-layer capacitors evince a very high power density but a very low energy density by comparison with other storage devices, so they are able to discharge a very high power over a short period of time. A comparison of nickel cadmium and nickel metal hydride batteries shows that both have about the same power density, but the nickel metal hydride battery has almost twice the energy density. This in turn means that of two batteries capable of storing the same amount of energy, a nickel metal hydride battery would weigh only half as much as a nickel-cadmium battery. Translated into terms of range, a car with a nickel metal hydride battery would be able to travel twice as far as a car fitted with a nickel-cadmium battery of the same size.



Comparison of energy densities of petrol and a 12-V battery

Fossil fuels have a significantly higher energy density. Petrol or diesel, for example, has an energy density of 11.8 kWh/kg. A 12-V lead-acid battery has an energy density of 30 Wh/kg. In other words, the energy density of petrol is about 400 times greater than that of a lead-acid battery. If a 12-V battery had a volume of 12 liters, the amount of petrol needed to provide the same amount of energy would have a volume of 0.03 liters.

Batteries of very many different kinds are available on the market. Only the most important types that are of interest from the point of view of hybrid propulsion technology are described below.

3.1.2. Battery types

Lead-acid battery

The lead-acid battery is one of the oldest kinds of battery system (lead acid batteries have been around since 1850), but even today batteries of this kind supply electrical energy for millions of motor vehicles. Lead-acid batteries are commonly used as starter batteries for the combustion engines of motor vehicles. Over a limited period of time, batteries of this kind are also capable of supplying power to on-board electrical consumers even when the combustion engine is not running.

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3. System Components



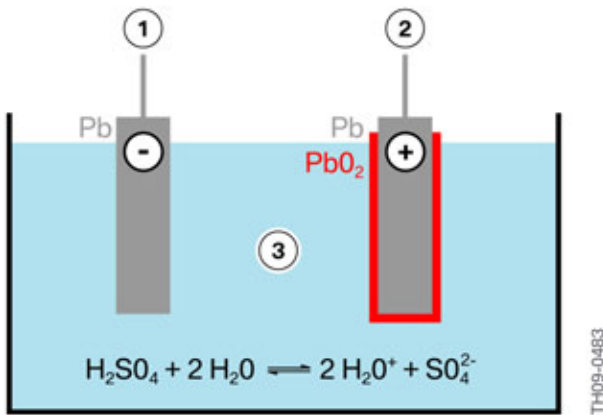
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Structure of a lead-acid battery

Index	Explanation
1	Seal plug
2	Hydrometer (Magic Eye)
3	Carry handle
4	Positive terminal of battery
5	Battery housing
6	Bottom ridge for holding battery in position
7	Plate block, consisting of positive and negative plate sets
8	Negative terminal of battery

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3. System Components



Chemical reaction in a lead-acid battery

Index	Explanation
1	Cathode (negative terminal)
2	Anode (positive terminal)
3	Sulfuric acid

When a lead-acid battery is fully charged, its positive terminal consists of lead oxide (PbO₂) and the negative terminal consists of lead (Pb). The electrolyte is diluted sulfuric acid (H₂SO₄). Both poles of a discharged lead-acid battery consist of lead sulfate (PbSO₄).

The overall reaction that takes place as the battery discharges can be expressed as follows:



The cells consist primarily of positive and negative electrodes and the separators, plus the requisite structural components. Each cell supplies a voltage of two volts. Six cells connected in series deliver the battery voltage of 12 V. The energy density of a lead-acid battery is typically in the order of 30 Wh/kg.

Nickel-cadmium battery

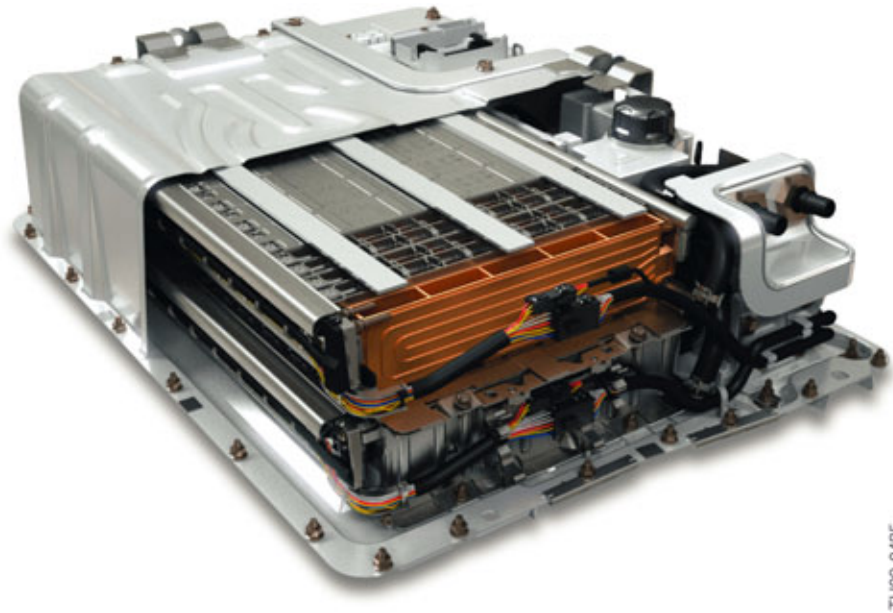
Nickel-cadmium batteries (NiCd) were first developed more than 100 years ago and the technology is still in use today. An important difference between lead-acid batteries and nickel-cadmium batteries is that in the latter the electrolyte remains unchanged through the charge/discharge cycle. The electrodes of a fully charged nickel cadmium cell consist of plates charged with cadmium at the negative terminal and nickel-hydroxide at the positive terminal. The electrolyte is potassium hydroxide. This combination supplies a voltage of 1.2 V. Energy density is comparable to that of a lead-acid battery.

Cadmium is a heavy metal and is classified as an environmentally hazardous substance and these batteries suffer from what is commonly known as the memory effect - two reasons why NiCd batteries have recently been supplanted by new battery systems. The memory effect is the characteristic loss of capacity that occurs when a nickel-cadmium battery is frequently recharged after being only partly discharged. The battery gives the impression of "remembering" the energy levels it reached in previous discharge cycles. Instead of the original nominal amount of energy, the battery outputs only this lower amount of energy and the voltage drops.

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3. System Components

Nickel metal hydride battery



Nickel metal hydride battery in the ActiveHybrid X6

The nickel metal hydride battery (NiMH battery) is often considered the successor to the NiCd battery. An NiMH battery cell has nominal voltage of 1.2 volts. The energy density of an NiMH battery is about 80 Wh/kg, so it is twice that of an NiCd battery. NiMH batteries are virtually free of the memory effect described above. They are capable of releasing their stored electrical energy within a very short time with no more than a negligible tailing off in voltage level.

NiMH batteries are easily damaged by overcharging, total discharge, overheating and reversed polarity. They are also sensitive to changes in temperature. As soon as ambient temperature dips close to the freezing point these batteries evince a significant drop in capacity.

The anode consists of a metal alloy that can reversibly store hydrogen in its crystal lattice to form a metal hydride. The electrolyte is a 20% solution of potassium hydroxide, which also surrounds the nickel-hydroxide cathode.

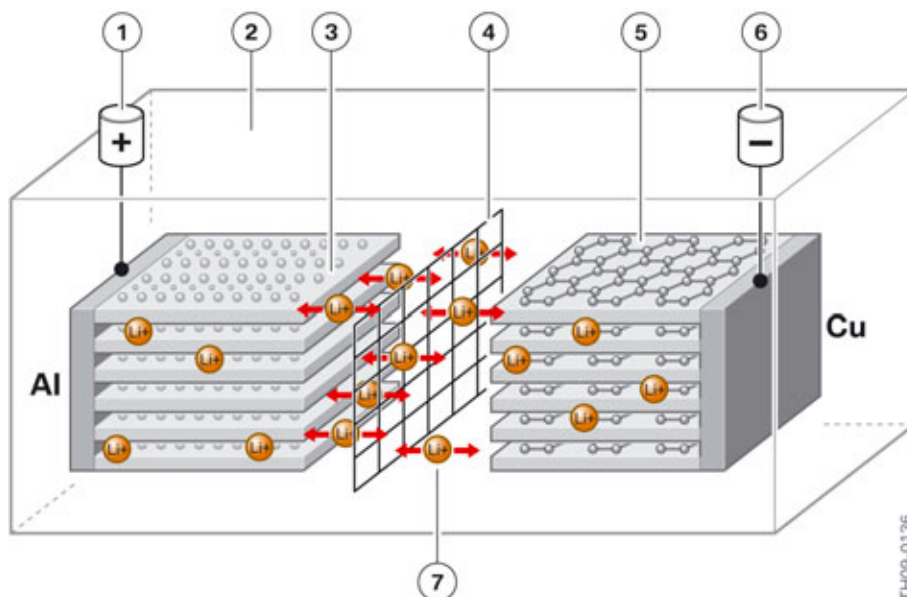
As the battery discharges the hydrogen oxidizes; this chemical process produces a potential of 1.32 volts at the two electrodes. In order to prevent the metal from oxidizing instead of the hydrogen as the battery approaches the end of its discharge cycle, the negative electrode is much larger than the positive electrode.

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3. System Components

Lithium ion battery (Li-ion battery)

In the 1960's, researchers were starting to look at lithium batteries with lithium-metal anodes and non-aqueous electrolytes. The non-rechargeable lithium batteries were initially used in space exploration and for military purposes. Low self-discharge is a feature of batteries of this kind and for this reason they are still used even today in cardiac pacemakers, clocks and cameras. Rechargeable lithium batteries made their commercial breakthrough with the market launch of a cell that dispensed entirely with metallic lithium; this was the lithium-ion battery. Today, lithium-ion batteries are the preferred energy source for portable devices with high energy demand (mobile phones, digital cameras, notebook computers, etc.). Their high energy density also makes them attractive for use in electrically powered and hybrid vehicles. Additional benefits are the ability to supply a constant voltage across the entire discharge period cycle and the fact that they do not suffer from a memory effect.



Structure of a lithium-ion cell

Index	Explanation
1	Positive electrode
2	Housing with electrolyte
3	Lithium-metal oxide
4	Separator
5	Graphite layer
6	Negative electrode
7	Lithium ion

Most Li-ion cells have a positive electrode made of lithium-metal oxides built up in layers (e.g. LiCoO_2 or LiNiO_2). The negative electrode consists of layers of graphite. The two electrodes are in a non-aqueous electrolyte. There is a separator between the two electrodes.

The lithium-ion cells generates a source voltage by the movement of lithium ions. When the cell is charging, positive lithium ions migrate through the electrolyte from the anode and insert themselves into the graphite layers of the cathode.

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The lithium ions combine with the graphite (carbon) without destroying the graphite's molecular structure. As the cell discharges the lithium ions extract from the graphite and move back to the metal oxide and the electrons can flow through the external circuit to the positive electrode. The lithium ions cannot combine with the graphite of the cathode in the process known as intercalation unless the negative electrode has a protective coating through which the small lithium ions can pass, but that constitutes an impermeable barrier to molecules in the electrolyte.

Lithium-ion batteries have a low self-discharge rate and their efficiency is approximately 96%, due primarily to the high migration capability of lithium ions. Efficiency is affected by temperature and drops off sharply at low temperatures.

A conventional lithium-ion cell supplies a nominal voltage of 3.6 V. The tension of a lithium-ion cell, therefore, is three times that of a nickel metal hydride battery. Total discharge to below 2.4 V causes irreversible damage and loss of capacity to the cell, so it is essential to ensure that this threshold is not reached.

Specific power density is in the range from 300 to 1500 W/kg. Energy density is about twice that of a nickel-cadmium battery, for example, and is about 95 to 190 Wh/kg.

It is best not to allow lithium ion batteries to discharge to below 40 %, because at lower levels significant loss of capacity can occur due to irreversible reactions in the electrodes. Another aspect is that the higher the cell voltage, the more rapid the ageing process in batteries of this kind. Consequently, it is best to avoid keeping lithium-ion batteries constantly 100 % charged. The optimum state of charge is in the range between 50 % and 80 %.

There are a number of points that have to be borne in mind with regard to handling lithium ion batteries. Mechanical damage can short the cell. The high current level would melt the casing and cause the cell to catch fire. Li-ion batteries are hermetically sealed, but even so they should not be immersed in water. Lithium-ion cells evince a severe reaction to water, particularly when they are fully charged. If a battery of this kind catches fire it is important not to attempt to extinguish it with water: smother the fire with an inert substance such as sand, for example.

For production-related reasons scatter affects a number of important parameters of lithium-ion cells (capacity is a case in point) and each battery is an array consisting of many individual cells, so each cell has to be individually monitored. This is precisely what the battery management system is designed to do. This system ensures that the individual cells are not overcharged and not discharged too severely, and it also balances charge over individual cells, if necessary.

3.1.3. Electrostatic storage devices

3.1.4. Double-layer capacitors

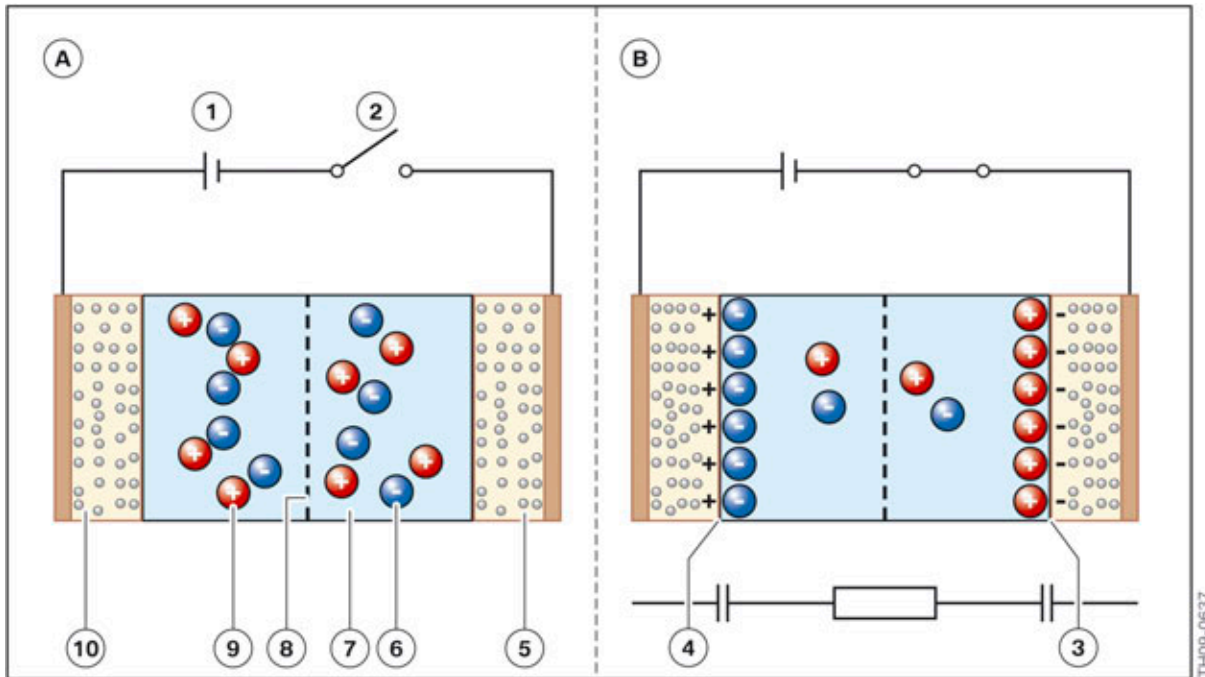
The principle of the double-layer capacitor was discovered in 1856 by German physicist Hermann von Helmholtz. He described the make-up of double layers of charge carriers forming on electrodes contained in an electrolyte when a voltage is applied from an external source. Today's double-layer capacitors are marketed under many different names (the names vary from manufacturer to manufacturer), including for example Goldcaps, Supercaps, Boost Caps and Ultracaps.

A double-layer capacitor is an electrostatic energy storage device for electrical energy with a very high power density of up to 10 kW/kg, but an energy density of about 5 Wh/kg, which is low by comparison with chemical storage devices. The advantages of a double-layer capacitor are its very high efficiency (approaching 100 %), its low self-discharge rate and its durability. These devices, moreover, are completely free of the memory effect.

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3. System Components

The inherently low energy density renders double-layer capacitors unsuitable for use as sole energy accumulators for vehicle propulsion, but in combination with chemical storage devices the weight savings are considerable and an additional benefit is that the chemical accumulators have an extended service life.



Structure of a double-layer capacitor

Index	Explanation
A	Double-layer capacitor is discharged
B	Double-layer capacitor is charged
1	Voltage source
2	Switch
3	Insulating charge-carrier layer
4	Insulating charge-carrier layer
5	Negative electrode
6	Negative charge carriers
7	Electrolyte
8	Separator
9	Positive charge carriers
10	Positive electrode

A double-layer capacitor consists of two electrodes in an electrolyte. When a voltage is applied to the electrodes, ions of opposing polarity migrate out of the electrode and collect at the electrodes. They form zones of immobile charge carriers in layers that are only a few molecules thick. A double-layer capacitor does not have a dielectric in the true sense.

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3. System Components

Instead, two layers of electric charge carriers form at the interface between electrodes and electrolyte and these layers act in much the same way as a dielectric. The two layers of charge carriers are the double layer that give the capacitor its name. The electrodes with the layers of charge carriers as dielectric act like two capacitors connected in series by the electrically conductive electrolyte. Double-layer capacitors store electrical energy purely by electrostatic means.

Capacity depends on the thickness of the insulating layer and the surface area of the electrodes.

$$C = \epsilon_0 \cdot \epsilon_r \cdot A/d$$

Index	Explanation
C	Capacity of the plate capacitor
ϵ_0	Electrical field constant of a vacuum
ϵ_r	Permeability of a material to electric fields
A	Surface area of the electrodes
d	Distance between electrodes or thickness of the insulating charge-carrier layer

The high power density of double-layer capacitors is due to the very thin insulating layer and the large surface area of the electrodes. Activated carbon is used to produce electrodes with this very large surface area. Activated carbon is fine-grade carbon with a very large inner surface (300 to 3000 m²/g). 4 grams of activated carbon aggregate to an inner surface roughly equal to the surface area of a football pitch. The insulating layer is only a few nanometers thick.

The capacity of double-layer capacitors is in the range from 1 to 50 farads and dielectric strength is about 2.5 V. As is the case with chemical accumulators, capacity and operating voltage can be increased by connecting multiple capacitors in series or in parallel, respectively.

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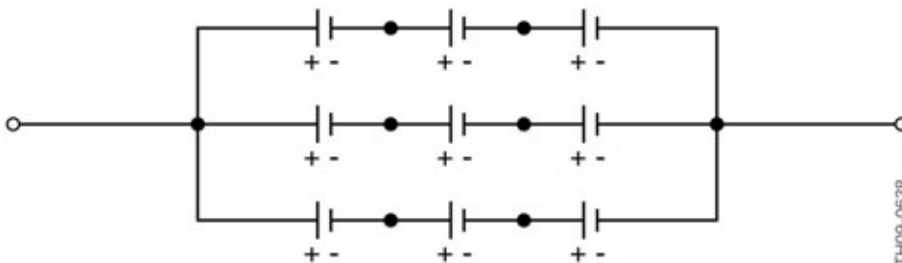
3. System Components

3.1.5. Summary notes on energy storage devices

As we have seen, the galvanic cell is the heart of any useful energy-storage system. Consequently, selecting a cell of suitable type is of crucial importance in terms of the system's ability to store and provide energy. The table below is an overview of the properties of the mainstream electrical energy-storage devices.

Energy storage device	Cell voltage [V]	Power density [W/kg]	Energy density [Wh/kg]	Memory effect	Operating temperature [°C]
Lead-acid battery	2	Up to 500	30	-	Up to 45
Nickel-cadmium battery	1.2	Up to 1,000	40	Yes	Up to 65
Nickel metal hydride battery	1.2	Up to 1,000	80	Low	Up to 60
Lithium ion battery	3.6	300 to 1,500	95 to 190	No	Up to 50
Double-layer capacitor	2.3 to 2.7	Up to 10,000	5	No	Up to 65

If voltage and/or capacity of the cell is not sufficient for a given application, multiple cells can be connected in series or in parallel, respectively.



Series and parallel connection of battery cells

3.2. Electrical Machines

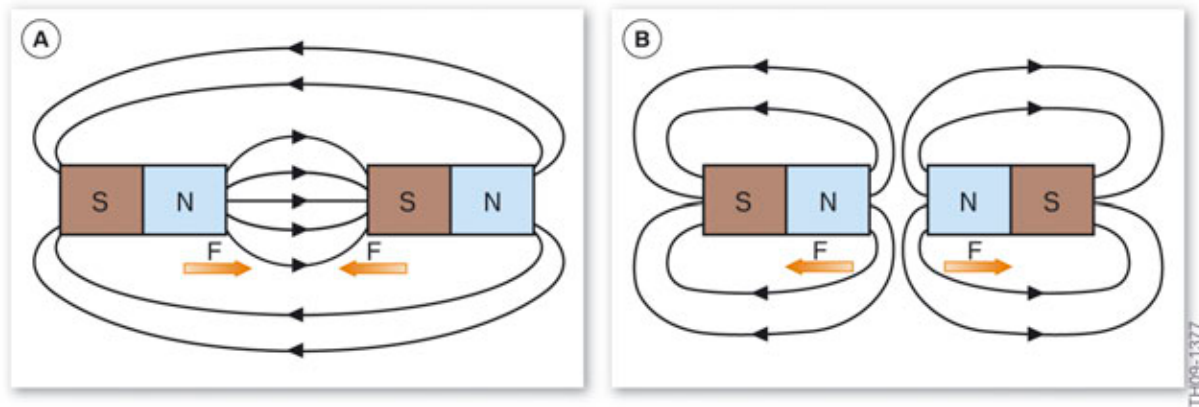
3.2.1. Important terms and laws of physics

Magnet

A permanent magnet (a magnet made from what is termed a magnetically "hard" material) is usually made of an alloy primarily of iron and often containing cobalt or nickel. It is magnetized in the process of production and it retains this magnetism permanently, hence the name. Permanent magnets are used for a wide variety of purposes, including that of generating energizing fields in fractional-horsepower electric motors.

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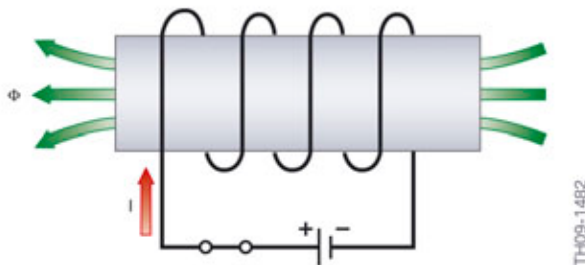
3. System Components



Behavior of magnetic poles

Index	Explanation
A	Opposite magnetic poles are mutually attractive
B	Like magnetic poles repel

Electromagnet



Magnetic field of an electromagnet

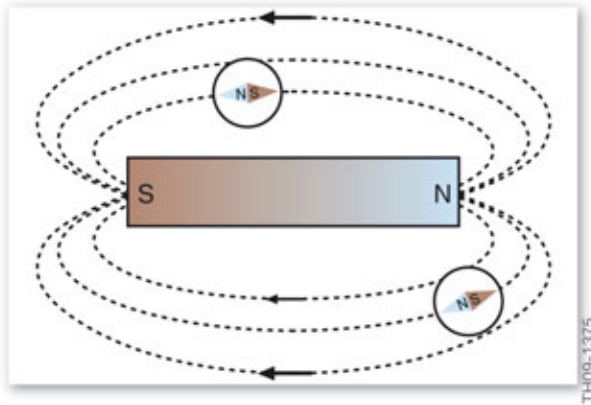
Index	Explanation
Φ	Magnetic flux
I	Current through coil

A coil carrying an electric current generates a magnetic field very similar to that of a bar magnet. Consequently, it can be used as an electromagnet. Unlike the magnetism of a permanent magnet, this magnetism can be switched on and off, because it occurs only while current is flowing through the coil. In direct current motors with electric excitation such as series-wound or shunt-wound motors, electromagnets are used to generate the magnetic field in the stator. Electromagnets are also used for the coils in relays.

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3. System Components

Magnetic field



Magnetic field lines

A magnetic field can be generated with the aid of a permanent magnet or an electromagnet. The magnetic fields can be visualized as magnetic field lines running from the magnet's north pole to its south pole. The stronger the magnetic field, the more densely packed are the magnetic field lines in the diagram.

Faraday's law of induction

Electromagnetic induction was discovered in 1831 by Michael Faraday. Electromagnetic induction is one of the basic phenomena of electrophysics. Faraday's law of induction describes a relationship between magnetic fields and electrical voltages and is crucial to our understanding of how electrical machines work.

The law of induction states that a U_{ind} will be induced by the reversing magnetic flux Φ flowing through a coil with the number of turns N . This relationship can be expressed as: $U_{\text{ind}} = N \cdot \Phi/\text{dt}$.

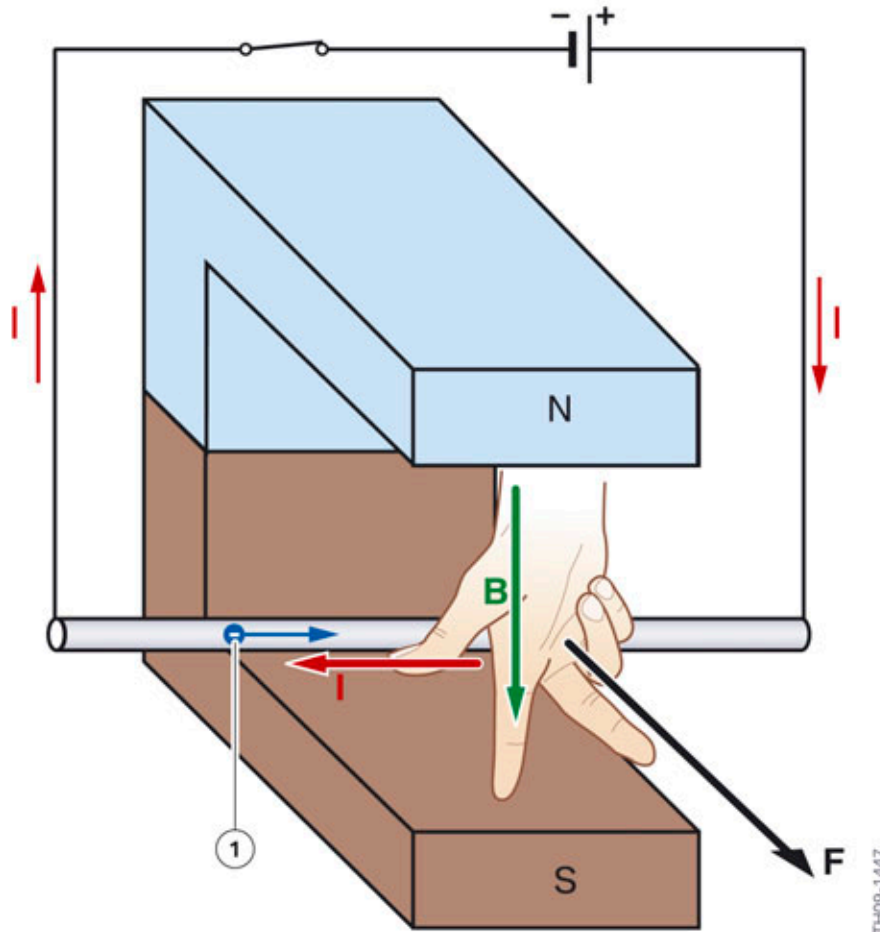
Even in a magnetic field without a time rate of change a voltage is induced in the coil if the coil itself is moved inside the magnetic field. Induction is utilized primarily in electrical machines such as generators, electric motors and transformers. The direction of flux of the induced voltage can be determined by application of Lenz's law: the direction of flux of the induced current is always in such a direction as to oppose the motion or change causing it.

If the current flowing through the coil changes in intensity, its magnetic field also changes and induces in the coil a voltage that is opposite the change in current direction causing it. This is generally termed self-induction. The faster the magnetic field changes and the stronger the changes, the higher is the resultant induction voltage.

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Lorentz force



Lorentz force acting on an electron

Index	Explanation
1	Electron
B	Magnetic flux direction
F	Force acting on an electron
I	Electric current
F	Force acting on an electron
N	North pole
S	South pole

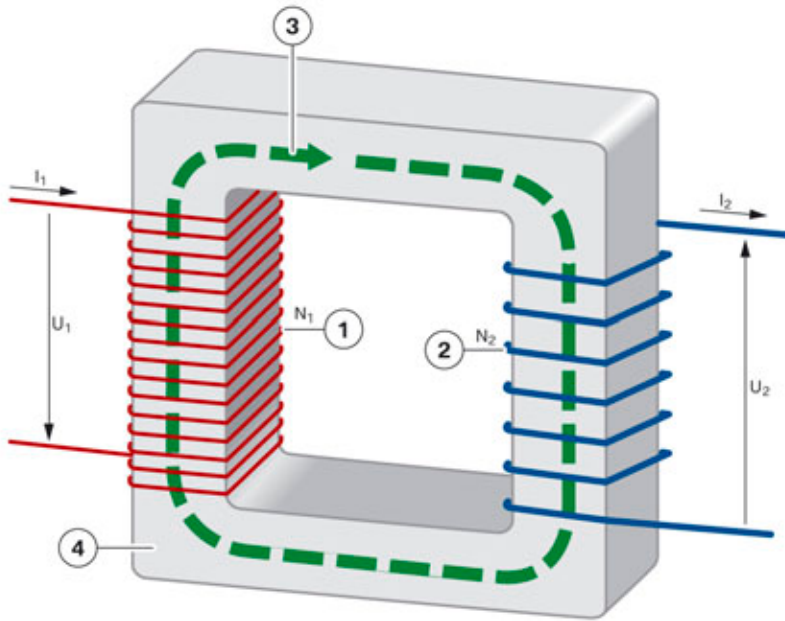
The force known to physics as the Lorentz force is the force that a magnetic field exercises on a moved electrical charge. A force F acts on the electrons in the conductor, causing the entire conductor to move in one direction. The orientation of the vectoring force can be determined by the "right-hand rule" or by the principle of cause/agent/effect. Applied to the graphic above, this means:

Cause = current $\rightarrow$ agent = magnetic field $\rightarrow$ effect = force on electrons

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3. System Components

Transformers



Structure of a transformer

Index	Explanation
1	Primary coil with number of turns N_1
2	Secondary coil with number of turns N_2
3	Magnetic flux
4	Iron core

A transformer consists of two coils around a common iron core. There is no conductive connection between the two coils, in other words they are galvanically insulated from each other.

If an alternating current is applied to one of the coils, the law of induction requires that an alternating current will be induced in the second coil. A voltage can then be measured between the ends of the second coil. The strength of the induced voltage depends on the primary voltage and the number of turns in the two coils.

$$U_1/U_2 = N_1/N_2$$

If the primary and secondary coils have the same number of turns, secondary voltage U_2 will be equal to primary voltage U_1 . If the secondary coil has twice as many turns as the primary coil, the secondary voltage will be twice the primary voltage. A load connected to the secondary coil draws energy from the circuit that has to be made up for on the primary side. In an ideal (imaginary) transformer, the energy input on the primary side is equal to the energy tapped on the secondary side, in other words an ideal transformer is loss-free. The currents I_1 and I_2 are inversely proportional to the voltages, because in this ideal transformer power is equal on the primary and secondary sides.

$$U_1 \cdot I_1 = U_2 \cdot I_2$$

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Ideal transformers, however, do not exist in the real world, because losses always occur. Every transformer, therefore, outputs less electrical energy than it receives. The losses in electrical energy are partly the result of heat generated in the coils on account of electrical resistance and partly the result of what are known as eddy currents. The eddy currents can be minimized by using a transformer core made up of a large number of thin iron sheets instead of a solid-iron core. These sheets of iron are insulated by a lacquer coating and this arrangement interrupts the flux of eddy currents.

Electrical machines, general

Electrical machines are devices capable of converting electrical energy into mechanical movement and mechanical movement into electrical energy. Depending on how the energy is transformed, we talk about electric motors (electrical to mechanical), or generators (mechanical to electrical). All electrical machines make use of the fact that like magnetic poles repel and opposite magnetic poles attract. At least one of the magnetic fields is produced by an electric current. Electrical machines can also be classified by type of current, for example direct current (DC), alternating current (AC) or rotary current (more commonly called three-phase current) machines, or by the operating principle, which gives us the classifications of synchronous or asynchronous machines.

Out of all these various classifications, the following electrical machines are discussed in more detail below:

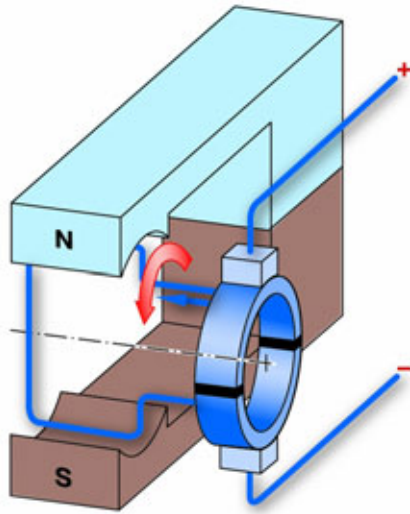
- Direct-current machines
- Synchronous machines
- Asynchronous machines

3.2.2. Direct-current electrical machines

Direct-current electrical machines were the earliest electromechanical converters, primarily because of historic development and the availability of electrical energy in the form of direct current from galvanic cells. The first direct-current machine was built in about 1830. The introduction of rotary current (three-phase current) in 1890 or thereabouts meant that the DC machine was ultimately supplanted by the synchronous machine. Even today, however, direct-current machine designs are still with us and are used in a very wide range of applications. Direct-current motors with rating up to about 100 W are used in huge numbers in vehicle electrical systems as motors for wipers, power window lifters and blowers and as servomotors.

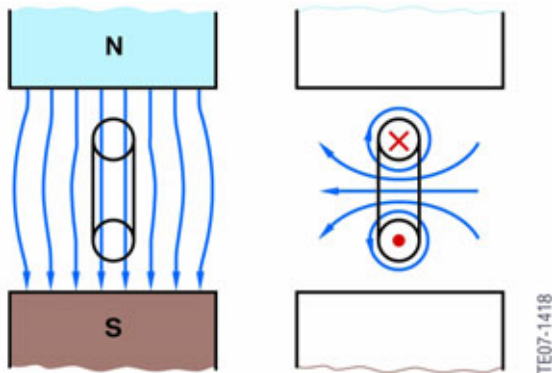
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3. System Components



Operating principle of the direct-current electrical machine

A direct-current electrical machine converts electrical energy (in the form of direct current) into rotational energy. The machine consists of a stationary part, known as the stator, and a part mounted in bearings so that it is free to rotate and known as the rotor (armature). Most direct-current machines are of the internal-rotor type, in other words the rotor is the inner part and the stator is the outer part of the electrical machine. The stator consists of an electromagnet or, in some very small DC machines, a permanent magnet.



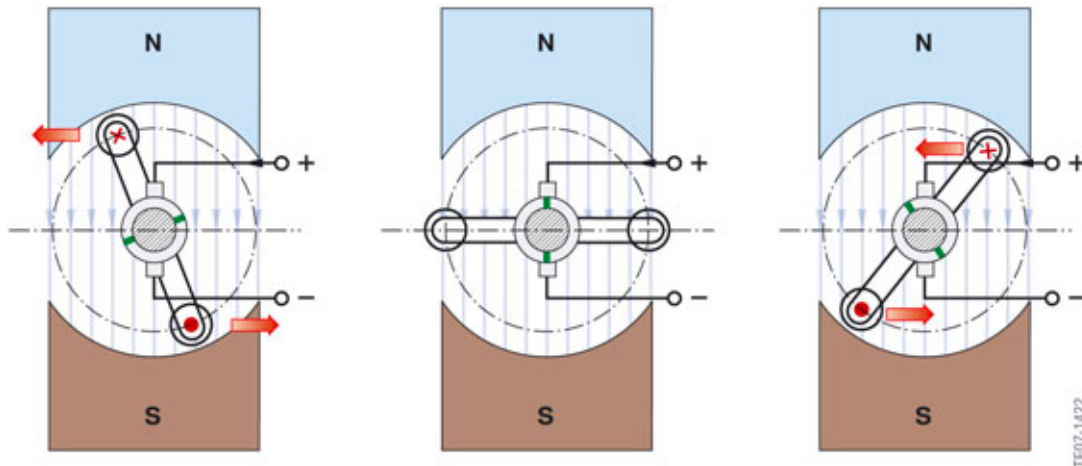
Left: magnetic field between two poles
Right: magnetic field of a conductor carrying a current

Electric motors operate on the principle that a force is applied to a conductor carrying a current in a magnetic field. The magnetic field of the conductor carrying the current and the magnetic field of the permanent magnet interact. If the permanent magnet is immovable and the conductor is mounted in rotary bearings, a force is applied to the conductor. The effect of this force is to cause the conductor carrying the current to rotate. The force acting on the conductor depends on:

- the strength of the electric current flowing through the conductor
- the strength of the magnetic field
- the effective length of the conductor (number of turns).

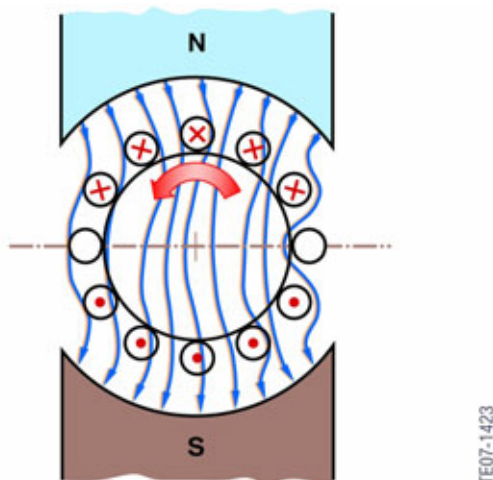
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3. System Components



Rotation of the current-carrying conductor

In order to increase the effect of the force, the current-carrying conductor is replaced with a coil with an iron core. For the sake of simplicity the graphic above shows only a conductor. When a voltage is applied to the coil, the current flowing through the coil generates a magnetic field (coil field). The magnetic field between the poles of the permanent magnet and the coil field produce a resultant magnetic field. A counter-clockwise or clockwise torque is produced, depending on the direction of current flux through the coil. The coil rotates until the coil field is vectored in the same direction as the magnetic field between the poles of the permanent magnet. The coil then comes to a standstill in what is known as the neutral zone of the pole field. In order to keep the coil in constant rotation, the direction of current flux through the coil has to be reversed. The direction in which the current flows is reversed by a commutator to which the start of the coil and the end of the coil are connected. Every 180° of rotation the commutator reverses the direction of current flow and the coil remains constantly in rotation.



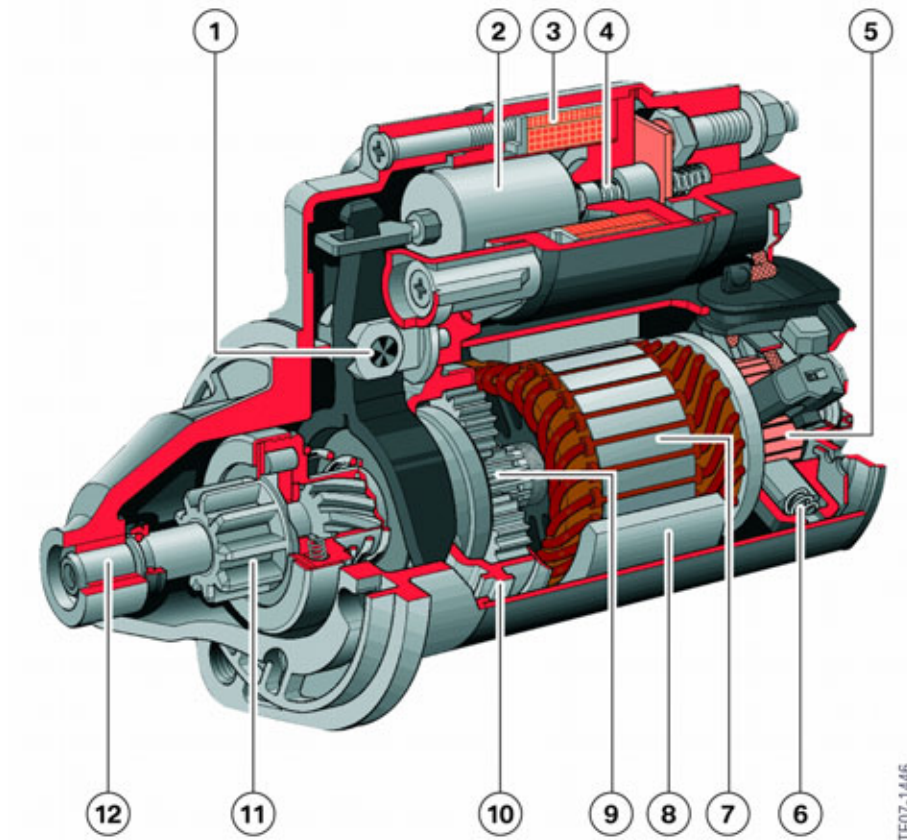
Multiple coils with a common iron core

The technical application of this principle calls for current flowing to the armature through a segmented commutator and wiper contacts (carbon brushes). The commutator is made up of segments of metal separated by thin strips of non-conductive material (plastic, air gaps) to form a cylindrical or circular surface with multiple interruptions. Carbon brushes (in most cases there are two brushes are pressed against the commutator by springs to transmit the current. As the rotor rotates

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the current direction through the armature windings is reversed and the conductors carried into the stator's magnetic field are those through which current is flowing in the direction that produces torque. The speed of the motor depends on voltage and direction of rotation. Typical applications for direct-current motors in automobiles include wiper motors and starter motors, for example.



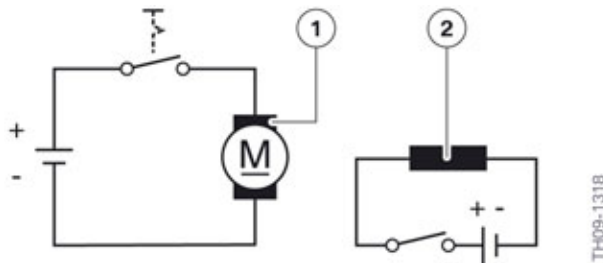
Structure of a starter motor

Index	Explanation
1	Forked lever
2	Relay armature
3	Relay coil
4	Relay springs
5	Commutator
6	Carbon brush
7	Rotor (armature)
8	Permanent magnet
9	Planetary gearing
10	Centered hollow ring gear with damper
11	Pinion
12	Drive bearing

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3. System Components

Until now, we have been looking at DC machines that use a permanent magnet to generate the primary magnetic field. However, the primary magnetic field in a DC machine can also be generated by an electromagnet. If the voltage supply for the excitation winding is independent of the voltage supply to the armature, the electrical machine is said to be of the separate-excitation type. Speed control for machines of this type is simple, because armature voltage and excitation voltage can be varied independently of each other.

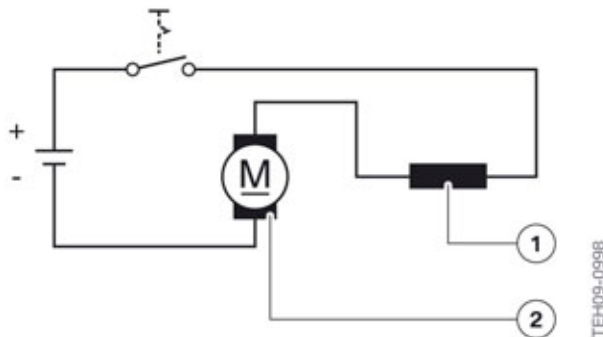


Direct-current machine with separate excitation

Index	Explanation
1	Armature winding
2	Excitation winding

If excitation winding and armature circuit are interconnected, the electrical machine is of the self-excitation type. The way in which excitation winding and armature circuit are connected provides a further means of distinguishing the type, namely as a series-wound or shunt-wound machine.

Series-wound machine



Block diagram of a series-wound machine

Index	Explanation
1	Excitation winding
2	Armature winding

A series-wound machine (also termed a series-excited machine) has the excitation winding and the armature winding connected in series. The excitation winding must be of the low-resistance type. An alternating-current supply causes both excitation field and armature current to reverse after every half-wave, which means that motors of this type can also be operated with alternating current. In this case, however, the iron core of the stator must consist of a stack of metal plates in order to avoid the losses that would otherwise be caused by eddy currents. The speed of series-wound motors depends heavily on load (series-wound load dependence). Consequently, series-wound motors are suitable

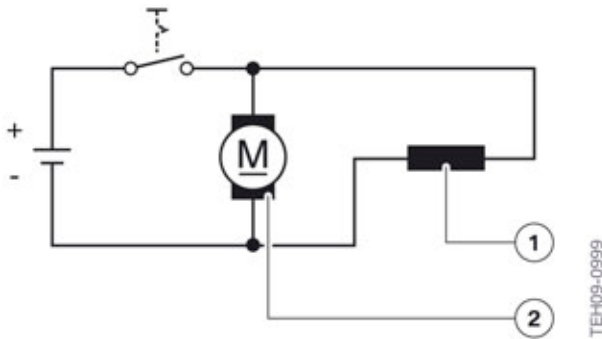
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only for base-load operation as otherwise a reduction in specified torque would be accompanied by a correspondingly sharp rise in speed. If the base load were to be disconnected the motor could accelerate to the point at which excessive centrifugal forces would cause irreparable damage.

Series-wound motors have tremendous pullaway torque, but a disadvantage inherent to the operating principle is this close relationship between speed and load torque. If load torque is allowed to drop, motor speed increases.

Shunt-wound machine



Block diagram of a shunt-wound machine

Index	Explanation
1	Excitation winding
2	Armature winding

The main advantage of shunt-wound motors is their "speed stability" - in other words changes in speed in response to fluctuating load are no more than minor. Speed can be varied over a wide range, however: this is accomplished by varying armature voltage without any change in field excitation. The design of the field windings is such that even when the motor is stopped (armature voltage = zero) excitation can remain applied for an indefinite period of time.

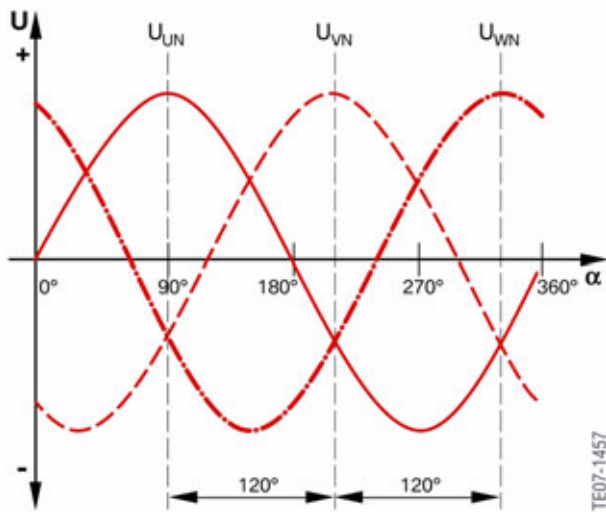
3.2.3. Rotary-current (three-phase) machines

General description

Three-phase machines are electromechanical converters that can operate either as motors or as generators. In motor mode the rotating magnetic field is generated by the three-phase rotary current. In generator mode the machine rotary current. Rotary current is an alternating current with three phases (live conductors). The term "rotary current" derives from the mode of generation, but the term "three-phase" motor is more common, simply because these machines have three phases.

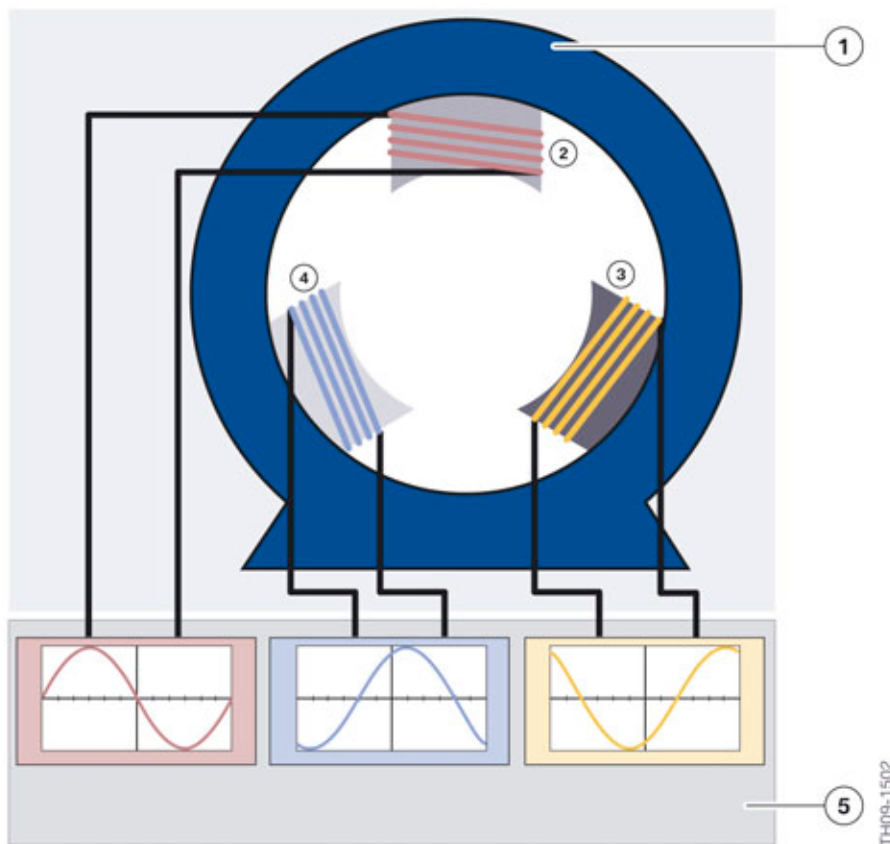
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Phases of the three alternating voltages

At any point along the time axis of this diagram, the sum of the three alternating voltages is always zero.



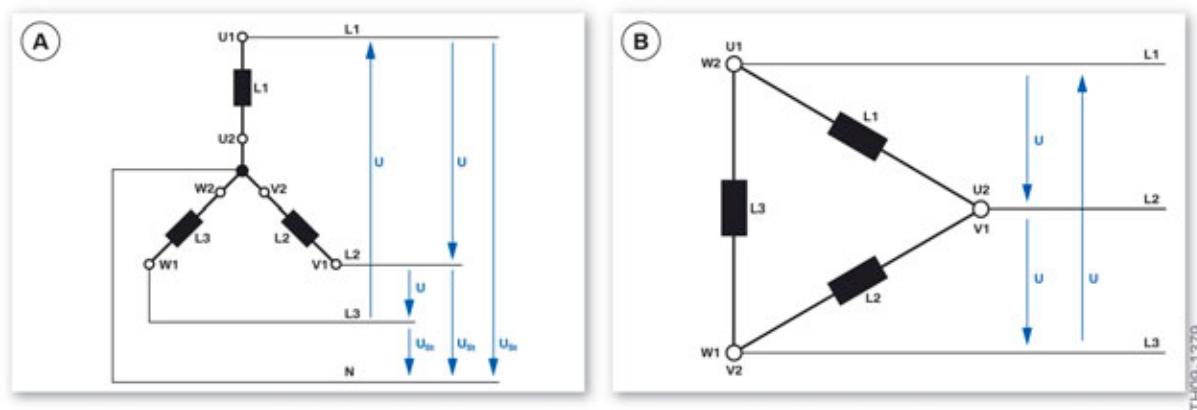
Structure of a stator

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Index	Explanation
1	Stator
2	Winding U
3	Winding V
4	Winding W
5	Phases of rotary current

Three coils with their axes offset 120° are necessary to generate a rotating magnetic field. The three coils are generally mounted on the stator of the three-phase motor. The three coils receive three alternating voltages with a 120° offset. The coils are connected in a star circuit or a delta (three-point) circuit. The type of circuit used is of no significance at this point of our observations. The important point to bear in mind, however, is that the currents flowing through the three winding legs have a 120° phase offset. The naturally rotating field produced in this way is the reason why three-phase motors are very simple in their structure.



Star and delta circuits of winding legs

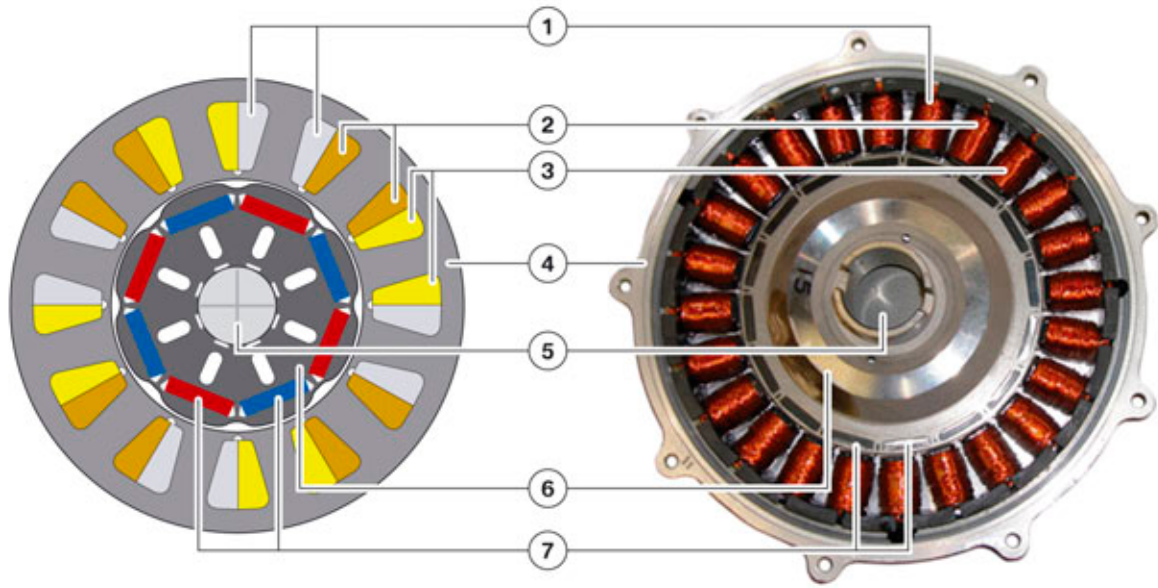
Index	Explanation
A	Star circuit
B	Delta circuit

In a star circuit, leg ends U2, V2 and W2 are all connected to point N. Outer conductors L1, L2 and L3 of the star circuit run from leg starting points U1, V1 and W1, respectively. In a delta configuration the start of one coil leg is connected to the end of another coil's leg. Broadly speaking, the coils are all connected one behind the other. Outer conductors L1, L2 and L3 run from the connecting points to the consumer. Because the coils are concatenated in this way, only three conductors are needed to wire the three phases L1, L2 and L3. In principle, both types of three-phase machine have the same stator structure with a three-leg rotary-current winding. They differ only in the structure of the rotor. The structure of the rotor is the feature that enables us to distinguish between synchronous and asynchronous machines.

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Synchronous machine



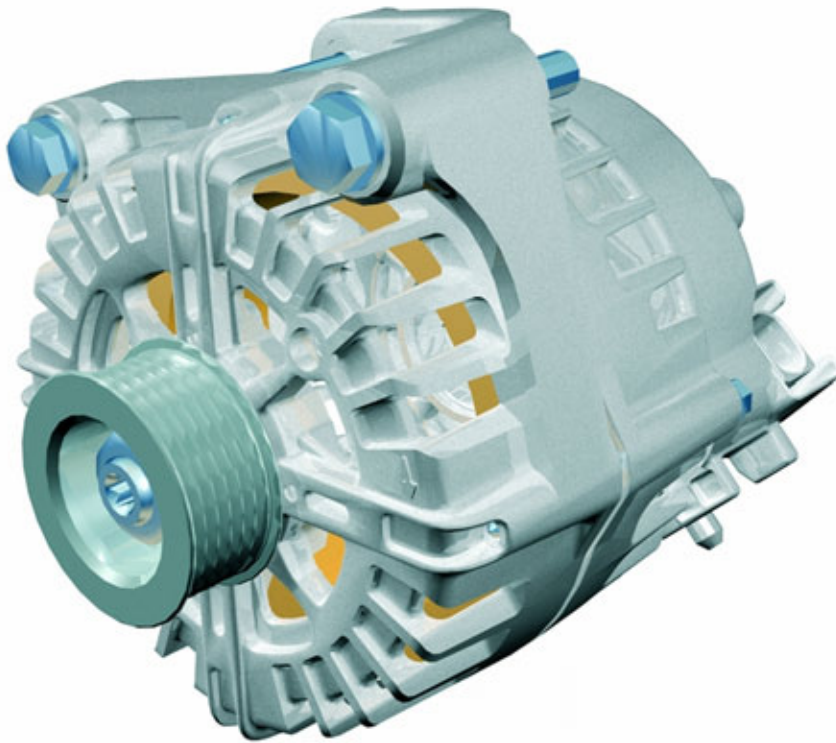
Structure of a synchronous machine

Index	Explanation
1	Winding W
2	Winding V
3	Winding U
4	Stator with rotary field windings
5	Shaft
6	Rotor with permanent magnets
7	Permanent magnets

A three-phase synchronous machine is an electromechanical converter that operates with a three-phase supply as an electric motor, or as a generator producing a three-phase supply for a consumer. Synchronous machines are used primarily as generators to produce electrical energy in power plants. A synchronous machine is generally the first choice nowadays as well as the generator in a motor vehicle to supply electrical consumers and charge the battery. In the medium-power range synchronous machines have become rare these days although this is set to change because hybrid-vehicle technology is turning more and more to synchronous machines.

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3. System Components

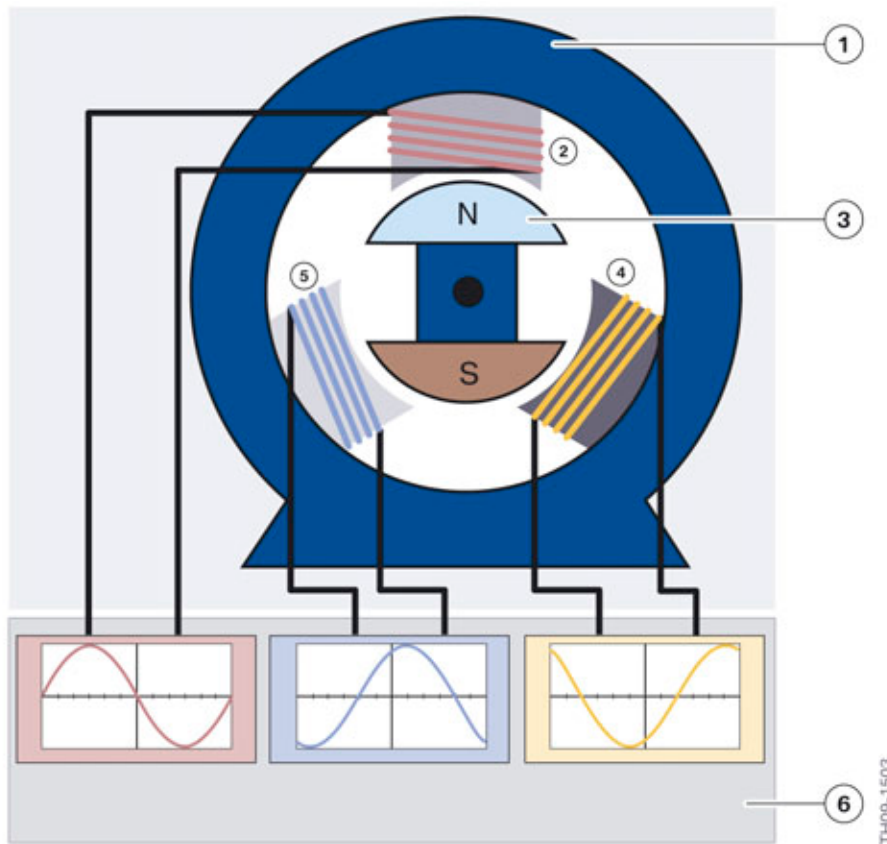


A typical automotive alternator

The magnetic field in the rotor of a synchronous machine is generated by permanent magnets (in small-scale machines) or by electromagnets (in larger machines). In the latter case wiper contacts are necessary, but the current flowing through these contacts is comparatively low. Unlike direct-current machines, a synchronous machine does not need a commutator.

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Structure of a synchronous machine with permanent magnets

Index	Explanation
1	Stator
2	Winding U
3	Rotor
4	Winding V
5	Winding W
6	Phases of rotary current

Most synchronous machines are of internal-pole design. Some designs have the stator windings inside the machine and the rotor with permanent magnets outside. A machine of this kind is referred to as an external-rotor machine.

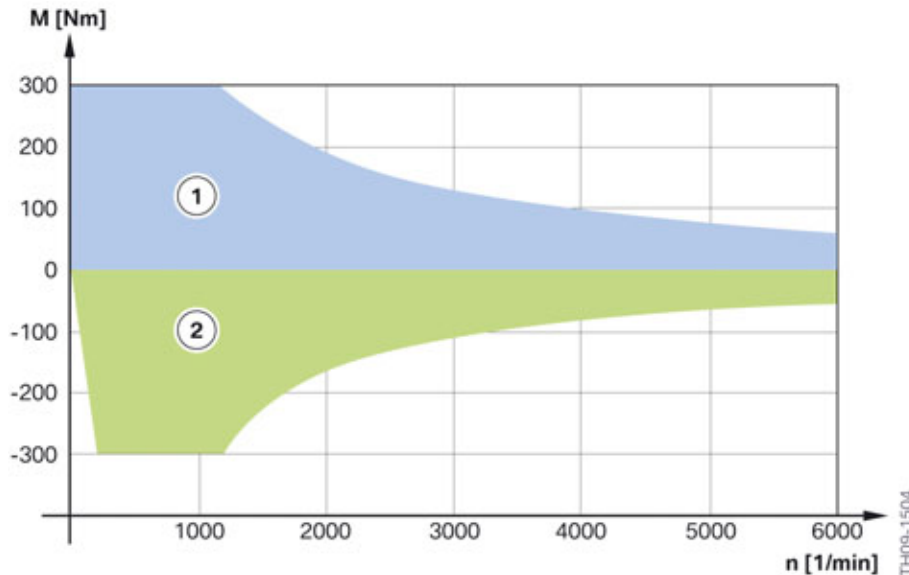
Principle of synchronous machines

When a rotary current is applied to the windings of the stator, a corresponding rotating magnetic field is generated. The magnetic poles of the rotor follow the direction of the rotating field. This causes the rotor to rotate. The rotor moves at the same speed as the rotating magnetic field. This speed is known as the synchronous speed. Hence the term "synchronous machine". The speed of the synchronous motor is precisely defined by the frequency of the rotary current and the number of pole pairs.

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A frequency converter has to be used in order to achieve stepless speed control of a synchronous motor. Synchronous machines do not start up under their own power: they must be set in motion either mechanically or with the aid of frequency converters, accelerated to their rated speed, and synchronised.



Working range of a permanently excited synchronous machine

Index	Explanation
1	Operation in motor mode
2	Operation in generator mode

Most of the three-phase machines used in hybrid automotive technology are of the synchronous type. The permanent magnets generate the magnetic field of the rotor, so it is not necessary to tap energy into the system from outside. This is why these machines have a very high power density, combined with a very high degree of efficiency (> 90%).

Other advantages of permanent-magnet synchronous machines include:

- Low mass moment of inertia
- Low maintenance
- Speed independent of load.

Disadvantages of synchronous machines include:

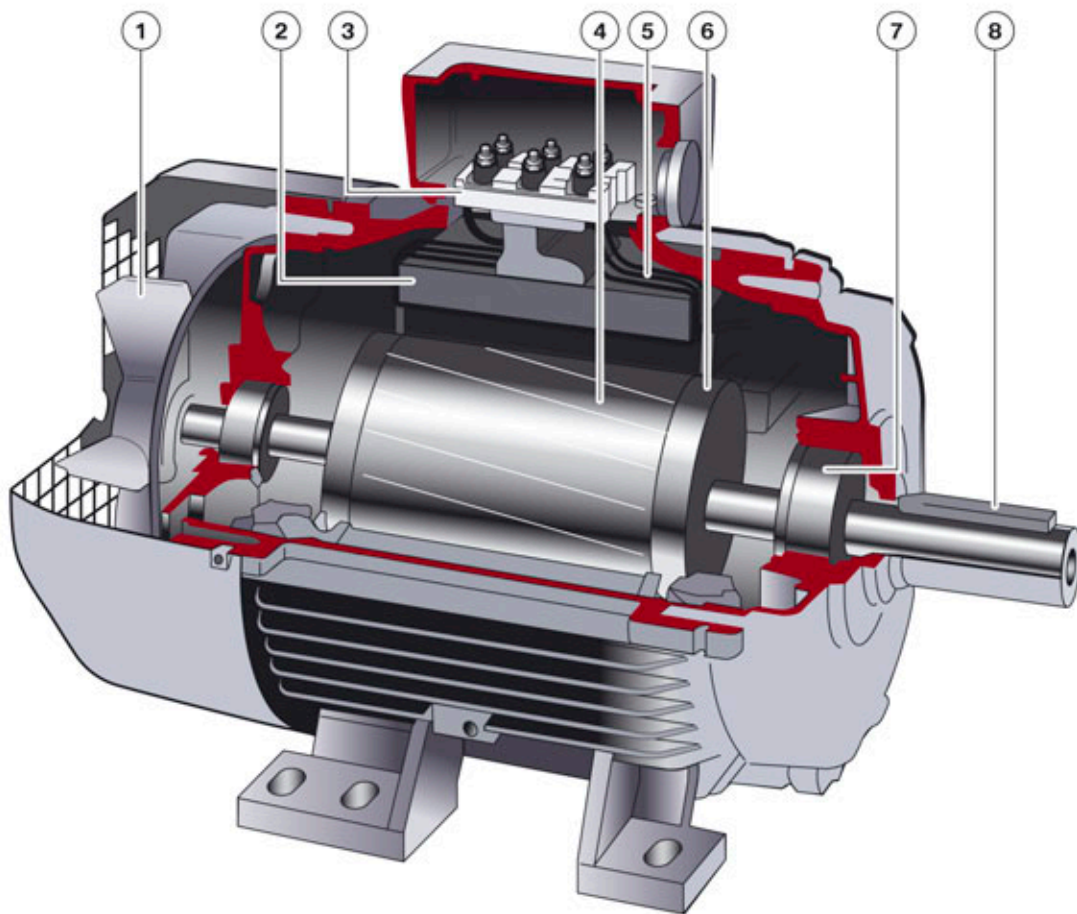
- High initial procurement costs for magnetic material
- Sophisticated control electronics necessary
- No self-start capability.

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3. System Components

Asynchronous machines

A three-phase asynchronous machine can be used as either a motor or a generator. The characteristic feature of an asynchronous machine is that the rotor does not have a direct supply of electricity. The motor's magnetic field is built up by induction from the rotary field of the stator. Another term frequently used for machines of this type is "induction machine", because the current flowing through the rotor is always induced by the rotary field of the stator. The rotor is often shaped like a round cage with bar conductors that make up the cage short-circuited at the ends.



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Structure of an asynchronous machine

Index	Explanation
1	Fan
2	Stator sheetpack
3	Terminals (line in)
4	Rotor sheetpack with rotor bars

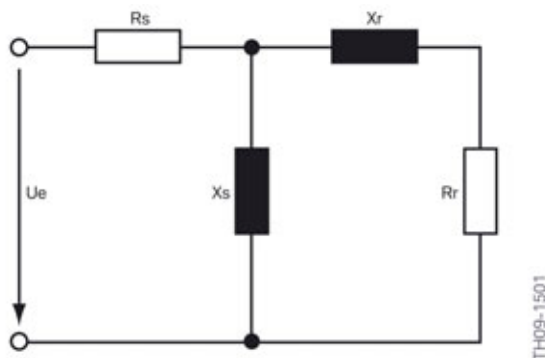
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Index	Explanation
5	Stator winding
6	Short-circuit ring
7	Roller bearing
8	Shaft

The rotary field in the stator winding reverses the magnetic flux in the conductor loops of the rotor. This induces a voltage that causes current to flow in the short-circuited conductor bars. This current generates a magnetic field of its own. Lenz's law states that the direction of flux of the induced current is always in such a direction as to oppose the motion or change causing it. The outcome is a torque that causes the rotor to spin in the direction of the stator's rotary field. Consequently, induction is caused by the difference in the relative speeds of the rotary fields of the stator and the rotor. However, the rotor cannot be permitted to achieve the same speed as the stator's rotary field because under these circumstances the magnetic flux change in the conductor loops would be zero and no voltage would be induced. The difference between the speed of the stator's rotary field and the rotor is known as the slip speed, but it is also referred to as the asynchronous speed. It is dependent on load. Stator field and rotor rotate at different speeds so they are not in synchronisation - this is why a machine of this type is referred to as an asynchronous motor. One advantage of asynchronous motors over synchronous designs is that they are inherently much simpler and therefore more robust. This derives primarily from the fact that the principle dispenses with both commutator and brushes. Simplicity of design also makes for lower cost of manufacture and low-maintenance operation. Asynchronous motors are the most commonly used kind of electric motor.

In terms of electrical layout, an asynchronous machine corresponds to a transformer. The stator winding is the primary side and the short-circuited conductor bars are the secondary side. Current settles to a level that is dependent on speed.



Equivalent circuit diagram of an asynchronous motor

Index	Explanation
U_e	Line voltage
R_s	Ohmic resistance of the stator winding
X_s	Inductive resistance of the stator winding
X_r	Inductive resistance of the rotor
R_r	Ohmic resistance of the rotor

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R_s and X_s are the most significant features of the equivalent circuit diagram of an idling asynchronous motor, and this is the reason why the input of a machine of this nature is almost entirely reactive power. As long as the rotor is not in motion, the machine is a transformer with short circuit on the secondary side. This means high currents and powerful magnetic fields. In this pullaway range the machine is not operating efficiently and the motor heats up rapidly and severely. As soon as the armature starts to rotate and adjust to the rotary field in which it is spinning, the currents drop and efficiency improves. Speed control for asynchronous machines is generally implemented with power electronics and frequency converters and speed varies as frequency is increased or decreased.

Advantages of asynchronous machines:

- Durability
- Low maintenance, because structure is straightforward and brushless
- Comparatively low manufacturing costs
- Self-start capability
- High short-term overload capability
- Inherently robust design.

Disadvantages of asynchronous machines:

- Less efficient than synchronous machines with permanent magnets and high torque utilisation
- Low pullaway torque unless coupled to a frequency converter with run-up control.

3.3. Power Electronics

Power electronics is a term often applied to a subsection of electrotechnical engineering, but it also refers to the electronic components that are used to switch, control and convert electrical energy. The components and circuits of power electronics are made up of silicon diodes and silicon thyristors. These components can handle very high voltages and currents (up to 4500 V and 1500 A).

Silicon transistors and silicon triacs are used at lower powers. Thyristors are switching elements with a variable switching time adjustable by means of the control voltage at the control electrode, or gate. A component consisting of two thyristors joined in an inverse-parallel configuration and having common control is known as a triac. The conversion of electrical energy by means of transformers or rotating machines is not considered part of power electronics. In the main, power electronics enable electrical energy to be converted in terms of the shape of the waveform, the voltage and current levels, and the frequency.

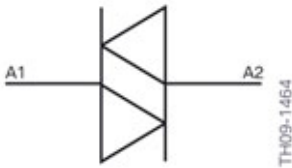
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3.3.1. Important components in power electronics

Diac

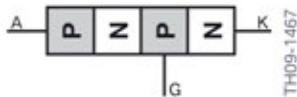
A diac is an electronic component capable of conducting electric current when the voltage at its two trigger connections exceeds a certain value. This voltage is known as the breakdown voltage. Diacs are sometimes also called "bidirectional diodes" or more commonly "diodes for alternating current". Hence the name: Diac is an abbreviation derived from "diode for alternating current".



Symbol used in circuit diagrams to represent a diac

The two trigger connections are designated anode 1 and anode 2, because polarity cannot be specified. A diac is bidirectional, so it can switch AC voltages. As soon as the voltage at the trigger connections (A1 and A2) exceeds a certain voltage threshold, it gates and the links between p n layers go conductive. The current flowing through the diac has to drop to a certain value before resistance increases and the diac stops allowing current to pass through. Diacs are used primarily in the trigger circuits of triacs to produce trigger pulses with very steep flanks.

Thyristors



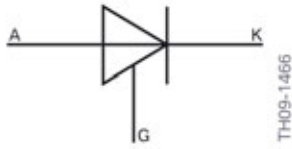
Structure of a thyristor

Index	Explanation
A	Anode
G	Gate
K	Cathode
p	p layer
n	n layer

A thyristor is a semiconductor consisting of four or more semiconductor layers with alternating n and p doping. "Thyristor" is a portmanteau word, i.e. a term coined by combining two words, in this case "thyatron" and "resistor".

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3. System Components



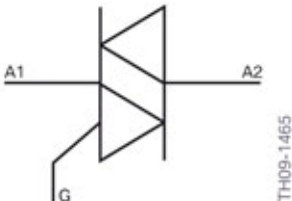
Symbol used in circuit diagrams to represent a thyristor

Index	Explanation
A	Anode
G	Gate
K	Cathode

A thyristor has a cathode, an anode and a control electrode, the gate. In its normal state, a thyristor is blocking in both directions. In the forward conducting direction it blocks until a current pulse is applied to the gate. A positive pulse at the gate makes the thyristor go conductive. The thyristor returns to its blocking state when the voltage is the load circuit is switched off or reversed. In the reverse blocking direction it blocks current like a normal diode. Thyristors are often used in three-phase circuits. For example, a thyristor actuator can be used as a soft-start device to enable squirrel-cage asynchronous motors to run up with controlled starting currents and torques. Thyristor rectifiers are used to control variable-speed three-phase motors.

Triac

A triac is an electronic component with semiconductor layered structure that basically consists of two inverse-parallel thyristors. Consequently, a triac can carry current in both directions.



Symbol used in circuit diagrams to represent a triac

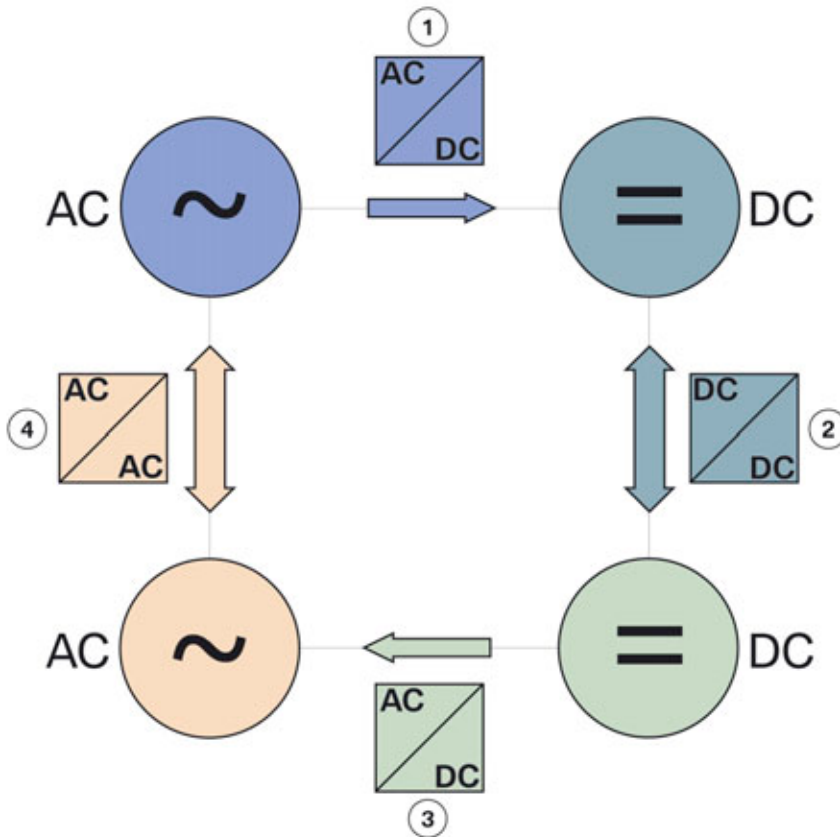
Index	Explanation
A1	Anode 1
G	Gate
A2	Anode 2

A triac has a control electrode G (the gate) and two anodes A1 and A2. Application of a trigger voltage to the gate makes the triac go conductive in both directions. In order for one gate to suffice for both thyristors, triacs have two triggers so that they can be switched to conductive mode by either a positive or a negative pulse. Triacs cannot be manufactured in a form that makes them suitable for very high current levels. Consequently, single thyristors are still used in power electronics.

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3.3.2. Current inverters, general



Current inverters and converters

Index	Explanation
1	Rectifier
2	DC/DC converter
3	Inverter
4	AC/AC converter

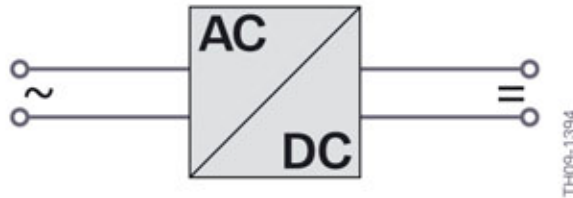
Circuits designed to convert electric energy in terms of the voltage waveform and the voltage and current levels are known as current converters. Distinctions are drawn between rectifiers, inverters and converters (often called transformers).

Rectifiers convert AC voltage to DC voltage. Conversely, a DC voltage can also be changed into an AC voltage. Inverters are used for this purpose. Transforming a DC voltage to a higher or lower DC voltage is a job for a converter. Converters are also more frequently, but not necessarily accurately, called DC/DC transformers. AC converters (also known as AC transformers) are used to transform an AC voltage into a different AC voltage of a different amplitude. Frequency converters are used if the frequency of the AC voltage has to be changed. In hybrid cars the power electronics are used to convert direct current voltage into alternating current voltage and vice versa. Moreover, power electronics are also used for highly flexible adaptation of the operating points of electrical machines.

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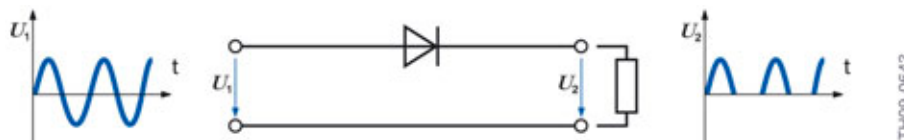
3.3.3. Rectifiers



Symbol used in circuit diagrams to represent a rectifier

Rectifiers are used to transform AC voltage into DC voltage. A rectifier consists of multiple diodes arranged in such a way as to function as current rectifiers. The diodes deflect the corresponding half-waves of the AC voltage to the same direction, producing a pulsating direct current voltage. In order to obtain true constant-voltage DC, the voltage output by the rectifier has to be smoothed by capacitors or chokes. Rectification can be uncontrolled by semiconductor diodes, or controlled by thyristors. A controlled rectifier needs a control voltage, a trigger that defined the time at which a given electronic switch has to be opened and closed in order to achieve the rectifying effect. Controlled rectifiers are implemented using electronic gates such as thyristors and MOSFET's. An uncontrolled rectifier rectifies the input AC voltage without extra control electronics.

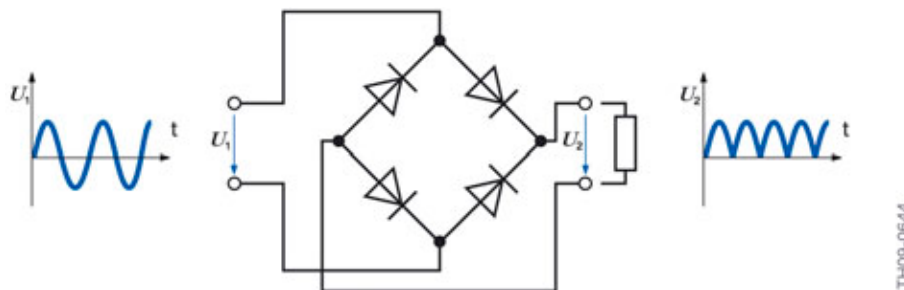
Half-wave rectifiers



Half-wave rectifier

A half-wave rectifier works by allowing one half of each AC wave to pass. The other half wave is blocked. The drawbacks of this circuit are the ripple at its output and the very poor degree of efficiency. The rectified voltage has to be smoothed before it can be used. The ripple is of the same frequency as the input voltage.

Full-wave rectifiers



Full-wave rectifier circuit

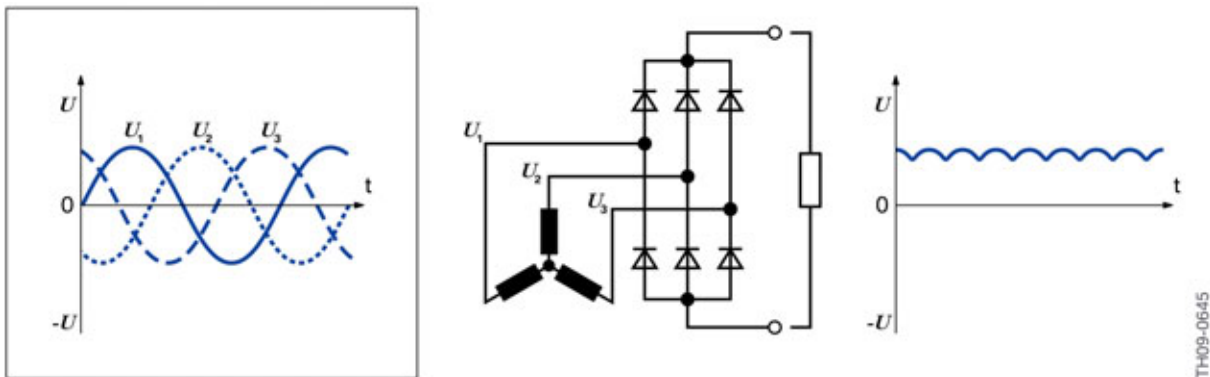
The drawbacks of a half-wave rectifier can be overcome to a large extent by using full-wave rectifiers (also known as bridge rectifiers or - less commonly - two-pulse bridge connections). There are four diodes in the circuit. The AC voltage applied on the left is converted into a pulsating direct current voltage (shown here on the right). This is a full-wave rectifying circuit, so the negative half-wave of the AC voltage always appears as exclusively positive in the DC circuit at consumer R. The ripple

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frequency is twice that of the input voltage, and the circuit for smoothing the output voltage need not be as complicated as in the case of half-wave rectification. Similarly, this circuit also has a better degree of efficiency.

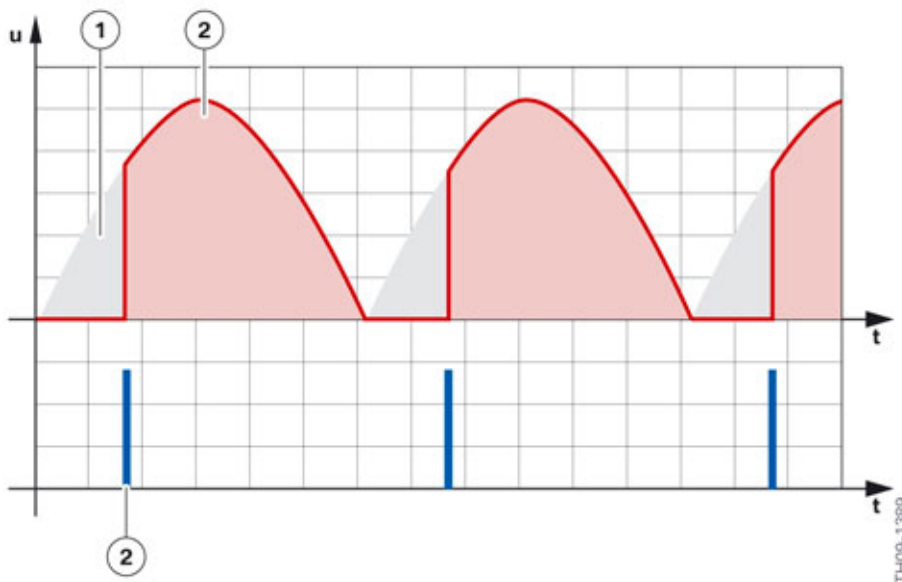
Three-phase current rectifiers



Three-phase full-wave rectifier

Three-phase current can also be rectified by what is known as a six-pulse bridge circuit. This circuit has six diodes, so all the half-waves of the three phases are used. The rectified direct current voltage evinces no more than slight ripple. In motor vehicles, circuits of this nature are used to rectify the alternator voltage, for example.

Controlled rectifiers



Power control of a variable-control rectifier by shifting the thyristor trigger timing

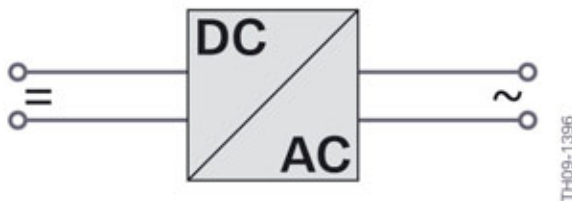
Index	Explanation
1	Proportion of input voltage not used for rectification
2	Proportion of input voltage used for rectification
3	Trigger pulses for thyristor

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Controllable rectifiers add power control to their rectifying capabilities. They can be used for example to implement speed control of variable-speed direct current motors. Controllable rectifiers with thyristors as control valves block current in both directions until a trigger pulse occurs at the gate of a control valve. In the graphic above, the triggers pulse for the thyristors are represented by the blue spikes at the bottom. When the trigger pulse dies the current flux (shown red) is sustained. The thyristor does not return to its blocking state until the current has dropped below a certain threshold value; once it blocks the thyristor has to be fired again in the next half-wave.

3.3.4. Inverters

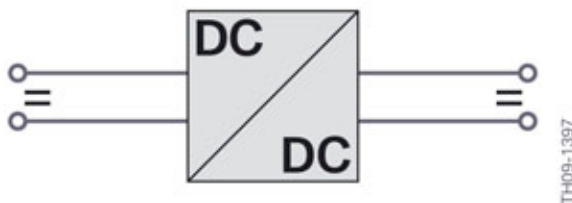


Symbol used in circuit diagrams to represent an inverter

The converters that convert direct current voltage into alternating current voltage are called inverters. Inverters can be designed to provide single-phase alternating current or three-phase AC. They are capable of very high degrees of efficiency in the region of 98 percent. Inverters are used in situations in which electric consumers need an AC supply in order to operate, but only a DC source is available. In hybrid cars, for example, which store their electrical energy in high-voltage batteries but the electrical machine that provides propulsive power needs three-phase electricity.

Another typical application is in a photovoltaic array. The power from the DC voltage source has to be fed into the AC or three-phase grid.

3.3.5. DC/DC converter



Symbol used in circuit diagrams to represent a DC/DC converter

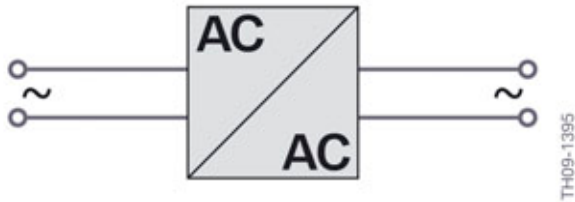
A DC/DC converter (often - though inaccurately - termed a transformer in common parlance) is a converter that converts a constant-current input voltage to a higher or lower DC voltage by periodic switching. DC/DC converters are commonly used in electrical propulsion technology. The basic types are step-down transformers, step-up transformers and inverters. The switches are generally either power MOSFETs or thyristors. Direct current voltage as such cannot be transformed, so a DC/DC converter works much like an electronic switched-mode power supply, initially transforming the direct current voltage into an AC voltage. This AC voltage then passes through a transformer in which it is stepped up or down, as appropriate, and then transformed back into a direct current voltage and smoothed. On account of the principle involved, current can flow in only one direction through a DC/DC converter. In mild-hybrid and full-hybrid vehicles a DC/DC converter is needed to step the voltage of the high-voltage battery down to 12 volts. The DC/DC converter is a dual-mode implementation: it

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has to be bidirectional so that the high-voltage battery can be charged with jump leads or a charger in the event of it running flat. This means that the converter can handle direct current voltages and convert them in both directions.

3.3.6. AC/AC converter

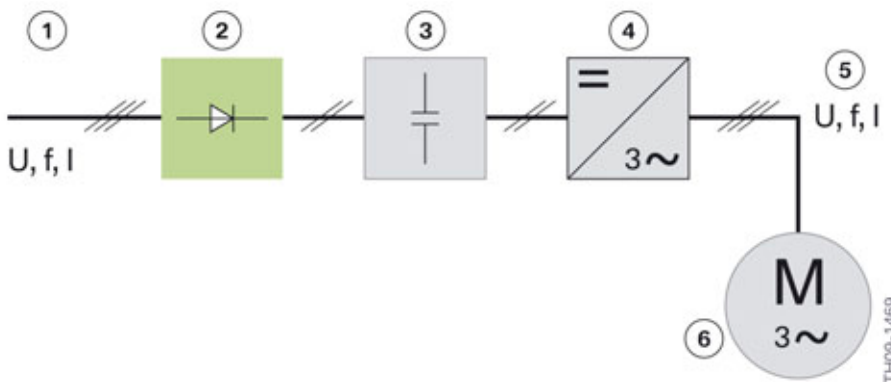


Symbol used in circuit diagrams to represent an AC/AC converter

An AC/AC converter is used to convert an input AC voltage into an output AC voltage of a higher or lower level. It is also possible to convert alternating current voltages with transformers. But transformers are not included in the scope of power electronics. In other words, although an AC/AC converter does the same job as a transformer it does not consist of coils with an iron core; instead, it is designed as a circuit comprising power-electronics components.

3.3.7. Frequency converters

Frequency converters convert an AC or three-phase input with constant voltage amplitude and frequency into a new three-phase output with variable voltage amplitude and variable frequency. Voltage/frequency control implemented in this way enables stepless speed control of three-phase motors.



Basic structure of a frequency converter

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Index	Explanation
1	Line voltage, line frequency, line current strength
2	Rectifier
3	Capacity
4	Inverter
5	Variable voltage, variable frequency, variable current strength
6	Three-phase motor

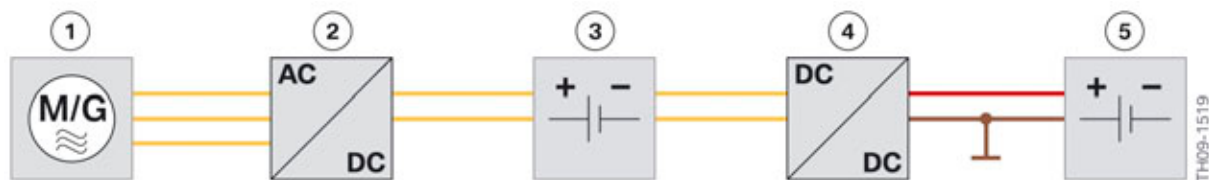
The frequency converter converts the constant voltage and frequency of the line supply into a direct current voltage. It then uses this DC voltage to generate a new, three-phase supply for the three-phase motor; this output-side supply has a variable voltage and variable frequency.

The conversion of voltage and frequency amplitudes produced by the rapid switching actions in the power section causes interference that can be transmitted either in the line-borne supply or by the electromagnetic field. This can result in malfunctions in instrumentation and control equipment, data-processing systems, and so on. Consequently, all electrical equipment must be compliant with the requirements for electromagnetic compatibility (EMC) and this applies in particular to frequency converters.

3.3.8. Examples

Mild-hybrid configuration

Some examples of how power electronics can be used in hybrid vehicles are outlined below.



Electrical powertrain of a mild-hybrid vehicle

Index	Explanation
1	Electrical machine
2	AC/DC converter
3	High-voltage battery
4	DC/DC converter
5	12-V battery

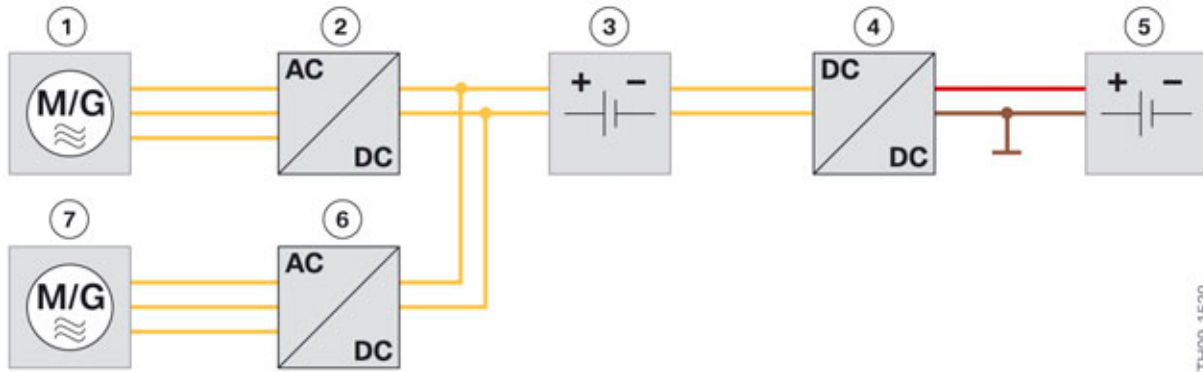
The powertrain of a mild-hybrid vehicle has at least one electrical machine with a power rating of about 10 to 15 kW. In a configuration of this nature a large proportion of the vehicle's kinetic energy can be recovered during braking and retardation and stored in the high-voltage battery. The three-phase current from the on-board alternator is converted into direct current voltage by an AC/DC converter. The AC/DC converter is bidirectional, in other words the direct current voltage from the high-voltage battery is also converted into three-phase current by the AC/DC converter so that the electrical machine can utilise this supply and function as a motor for propulsion purposes. In a hybrid

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vehicle the function of the conventional starter motor and the conventional alternator are generally discharged by the electrical machine, so the 12 V vehicle electrical system has to be supplied by a DC/DC converter. This DC/DC converter also has to be bidirectional so that the high-voltage battery can be charged via the DC/DC converter and the 12 V vehicle electrical system.

Full-hybrid configuration



Electrical powertrain of a full-hybrid vehicle

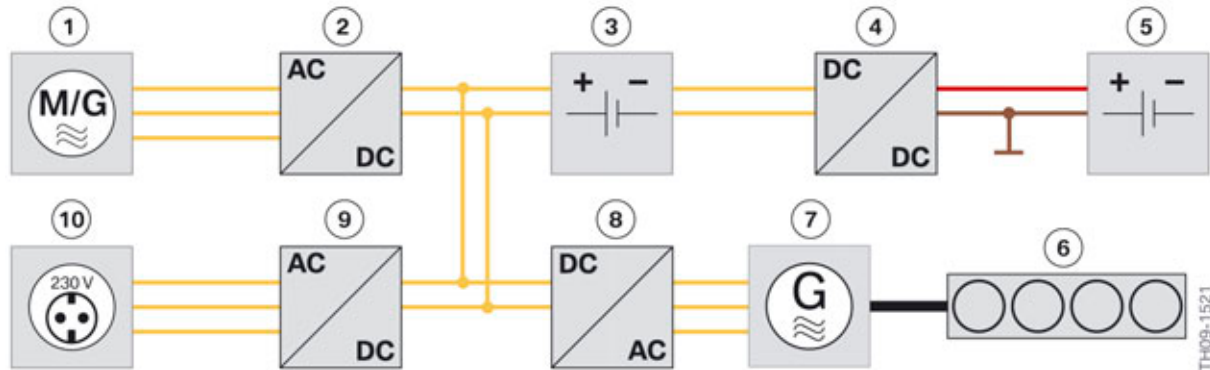
Index	Explanation
1	Electrical machine 1
2	AC/DC converter 1
3	High-voltage battery
4	DC/DC converter
5	12-V battery
6	AC/DC converter 2
7	Electrical machine 2

A full-hybrid vehicle has an electrical configuration similar to that of a mild-hybrid car. The difference is that in most cases a second electrical machine is needed and the two electrical machines together aggregate the higher power (20 to 50 kW) necessary to enable the vehicle to be driven under electrical power only. Consequently, the layout also calls for a second AC/DC converter.

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Plug-in hybrid configuration



Powertrain of a serial plug-in hybrid

Index	Explanation
1	Electrical machine
2	AC/DC converter 1
3	High-voltage battery
4	DC/DC converter
5	12-V battery
6	Combustion engine
7	Alternator
8	AC/DC converter 2
9	AC/DC converter 3
10	External power supply

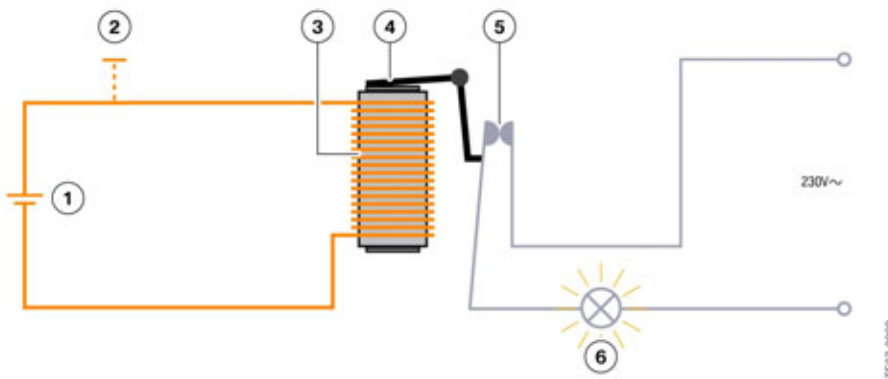
The AC/DC converter in a plug-in hybrid is used to transform the 230 V alternating current line voltage into a direct current voltage for the high-voltage battery so that it can be charged off the public grid. The high-voltage battery in a plug-in hybrid must evince a significantly higher energy density than comparable batteries in other hybrid cars. This is essential in order for plug-in hybrids to achieve their characteristically much greater range in all-electric mode. Frequently, the all-electric range of a plug-in hybrid is extended by the provision of a small-displacement combustion engine. If the charge state of the high-voltage battery drops below a defined limit, the combustion engine is used to power a generator. The three-phase power from the generator is converted by another AC/DC converter into direct current voltage so that the high-voltage battery can be recharged.

3.4. Electromechanical Switch Contactor

An electromechanical switch contactor is an electric switch for high-power switching actions. A switch contactor operates on much the same principle as a relay. The main difference is that a switch contactor can switch effectively at much higher power ratings than a relay; switch contactors operate in ranges from 500 watts up to several hundred kilowatts.

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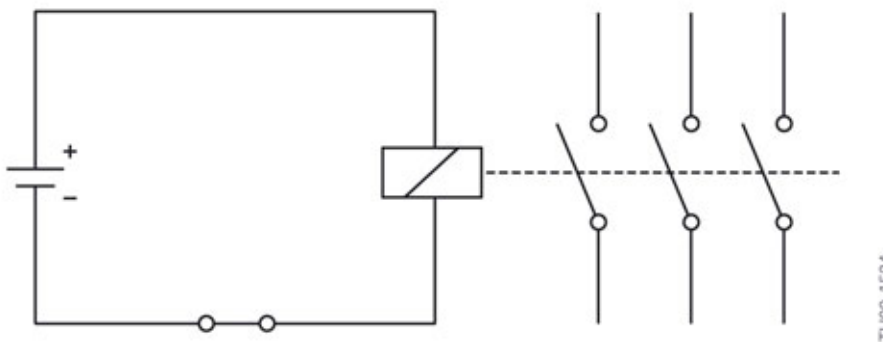
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Operating principle of a relay

Index	Explanation
1	Voltage source
2	Button (pressed)
3	Coil
4	Armature
5	Working contacts
6	Load circuit is closed (consumer is switched on)

Another difference between switch contactors and relays is that a switch contactor is always a normally closed (NC) contact or normally open (NO) contact. To refresh your memory: relays can also take the form of flip-flops. The switch contacts of relays, moreover, are single-throw interrupters whereas the contacts of switch contactors are usually double-throw interrupters.



Symbol used in circuit diagrams to represent a switch contactor

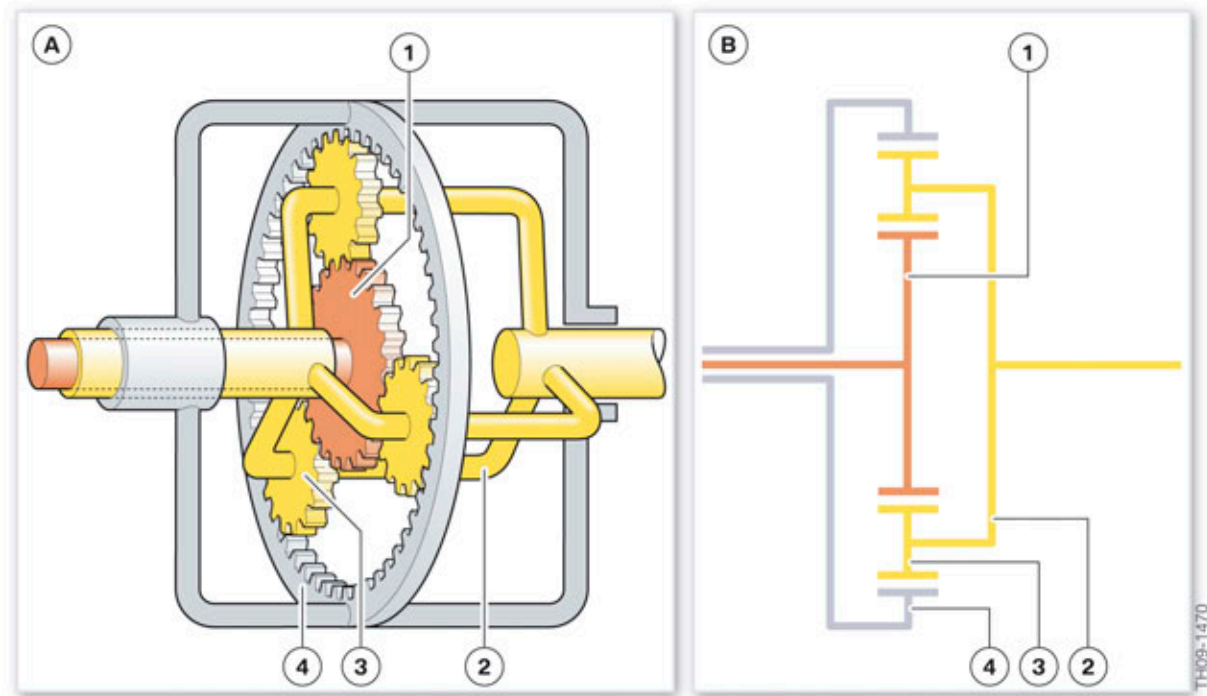
The actuating coils of contactors can be designed for AC or DC voltage. On account of the high-speed tripping of the solid switch contacts, a switch contactor can be a source of mechanical vibration and noise when it throws. When a contactor switches off the actuating coil operating as an inductive load generates interference in the form of a voltage spike. An overvoltage protector has to be built into the circuitry to protect the control electronics. In alternating-current circuits a combination of resistor and capacitor (RC) is used for this purpose. A freewheeling diode can be used in DC circuits. RC combinations are also used to avoid breakaway sparking and electric arcs at the switch contacts.

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3.5. Planetary Gearing

The basic structure and the method of operation of a planetary gear train is described in the basic information publication on power transmission/gearing.



Planetary gearing

Index	Explanation
A	3D view
B	Skeleton diagram
1	Sun gear
2	Planet carrier
3	Planetary gear
4	Ring gear

A basic property of all planetary gear trains is that they offer various transmission ratios. The assembly accomplishes this by having different shafts that can be used as input and output, while a third shaft is held so that it does not rotate.

Examples:

- High ratio:
The sun gear is driven, the ring gear is held motionless, the planetary gears rotate on the internal gearing of the ring gear. The planet carrier rotates in the same direction as the driven sun gear. The ratio is high because the driven sun gear rotates considerably faster than the planet carrier:

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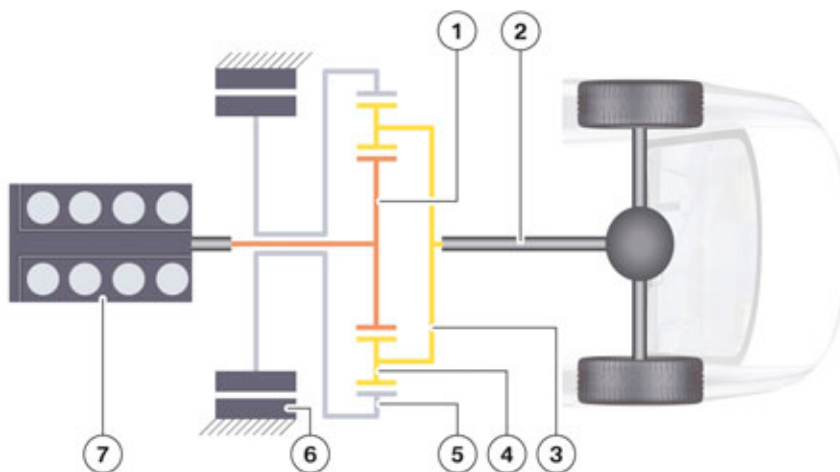
3. System Components

$$i = n_{\text{input}} / n_{\text{output}} = n_{\text{sun gear}} / n_{\text{planet carrier}}$$

- **Low ratio:**
The ring gear is driven, the sun gear is held motionless, the planetary gears rotate on the external gearing of the sun gear. The planet carrier turns in the same direction as the driven ring gear. Transmission is in low ratio, because the speed ratio between input and output is lower than in the first example:
$$i = n_{\text{input}} / n_{\text{output}} = n_{\text{ring gear}} / n_{\text{planet carrier}}$$
- **Negative ratio:**
The sun gear is driven, the planet carrier is held motionless. The planetary gears reverse the direction of rotation of the ring gear vis-à-vis the input. The ratio is high and the direction of motion is reversed:
$$i = n_{\text{input}} / n_{\text{output}} = n_{\text{sun gear}} / n_{\text{ring gear}}$$

Automatic transmissions, for example, make use of the various transmission ratios offered by planetary gearing. These assemblies incorporate nested planetary gears. Fixed links are used, for example between a planet carrier of the first and the sun gear of the second planetary gear set. Shifted links are also used, both between individual planetary gear sets and between planetary gear sets and the casing. Planetary gear trains of this nature are also found in mild-Hybrid cars, for example, incorporating automatic transmission configurations of conventional design.

However, hybrid technology also makes use of another property of planetary gear train, namely their ability to split torque or power to all three shafts. The term often used in this context is power splitting by the planetary gear train, and this is how power-split hybrid drive gets its name. Power splitting is technically usable in hybrid vehicles on account of the fact that the speed and torque of electrical machines can be controlled through their entire bandwidth, and this applies to both the generator and motor modes. Of course electronic control of the combustion engine's speed and torque is also an essential prerequisite without which it would not be possible to implement power splitting in hybrid vehicles. The graphic below will serve to illustrate the principle of power splitting. For the sake of simplicity, all losses (such as losses due to friction, for example) are ignored for the purposes of this explanation.



Planetary gear train with combustion engine interfaced to sun gear, electrical machine interfaced to ring gear and planet carrier interfaced to output shaft

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Index	Explanation
1	Sun gear
2	Output shaft
3	Planet carrier
4	Planetary gear
5	Ring gear
6	Electrical machine
7	Combustion engine

We can now consider the various operating states of the two drives and the effects on the output shaft.

- 1 The combustion engine is running and supplies its drive torque to the planetary gear set via the sun gear.

The electronics act on the electrical machine to keep it at a standstill. The engine speed of the ring gear is therefore zero. This state is comparable to a shifted connection to the motionless housing of the gearbox.

Only the drive torque of the combustion engine is transmitted to the output shaft in accordance with the gear ratio between sun gear and planet carrier. Power output at the output shaft is the same as the power input from the combustion engine, because the path incorporating the electrical machine neither intakes nor outputs power.
- 2 The combustion engine again supplies its drive torque to the planetary gear set via the sun gear. In addition, the electrical machine is now also operating as a motor. Consequently, the sum of the torques of combustion engine and electrical machine is now acting on the output shaft, in accordance with the transmission ratios. The speeds of rotation, too, match the transmission ratios. The electronics controlling the combustion engine and the electrical machine have to be able to allow for this difference in speed in order to have the desired total torque applied to the output shaft so that the shaft turns at the desired speed.

Conversely, this means that changing the speed of the electrical machine changes the combustion-engine/output-shaft transmission ratio. The speed of the electrical machine can be varied steplessly, so this amounts to a means of steplessly varying the transmission ratios. In hybrid vehicles this effect can be used, for example, to keep the combustion engine operating as frequently as possible within its range of maximum efficiency.
- 3 The power-split capability of a planetary gear set is also employed to good effect when the electrical machine is used as a generator instead of a motor. Under these circumstances the electrical machine acts as a "mechanical consumer", in other words it intakes torque. This torque is subtracted from the torque output of the combustion engine (with due provision made for the gear ratio). Consequently, there is correspondingly less torque available at the output shaft. Here again electronic control of the electrical machine's speed and torque keeps the combustion engine deliberately in its efficient operating range (for example high load at midrange rpm).

The mechanical energy that the electrical machine intakes is not lost, however; it is transformed into electrical energy. In hybrid vehicles the power electronics ensure that this electrical energy is stored in the high-voltage battery.

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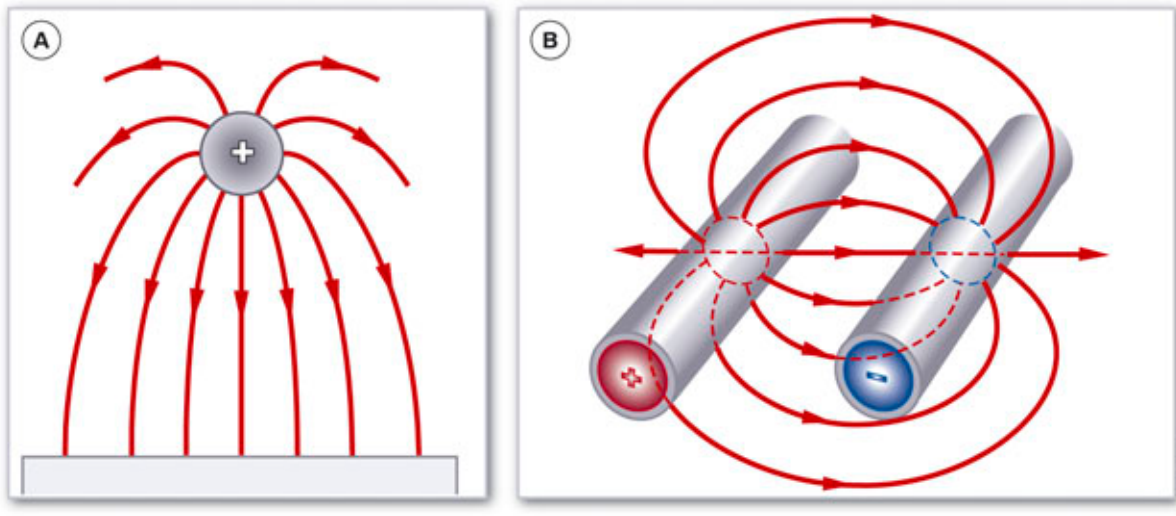
4. Safe Working Practices

4.1. Basic Physical Effects

4.1.1. High electric voltage

Electrical field

Every electrically charged object is surrounded by an electrical field. The effect is familiar and often experienced when a non-conductive object takes on an electrostatic charge. The electrical field is caused by the difference in potential between the electrically charged object and other objects in its vicinity. Instead of "difference in potential", we can also use the terms "electric voltage", or "electric tension".



Electrical fields of charged objects

Index	Explanation
A	Electrical field of a positively charged sphere
B	Electrical field of two electric conductors

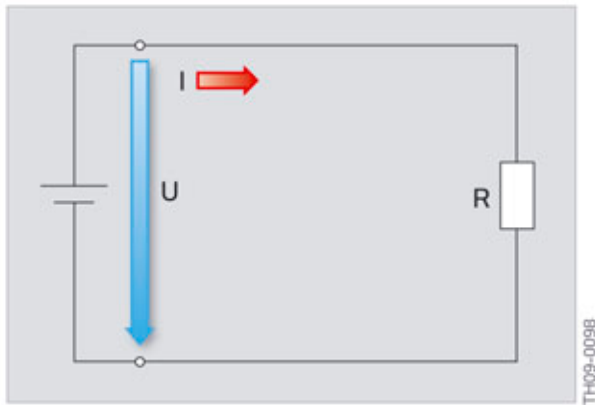
Live electric conductors also generate an electrical field in their vicinity. The electric voltages encountered in the field of hybrid-auto engineering are in the order of magnitude of several hundred volts. However, the field generated in this way does not constitute a direct hazard. The electrical field of an electrostatically charged object can be considerably stronger by comparison. If, for example, you bring your hand close to a live high-tension cable, it is not likely that sparking will occur.

Electric voltage as a cause of current flow

As well as generating electrical fields in the vicinity of objects, electric voltage is also the reason why electric current flows in a circuit. The higher the voltage U the higher the current I for any given electrical resistance. In the high-voltage electrical system of a hybrid vehicle the voltages exceed those in the familiar 12-V systems by several orders of magnitude.

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Simple circuit

Example: $R = 1 \text{ k}\Omega$

Voltage U	Current I
12 V	12 mA
120 V	120 mA
360 V	360 mA

Needless to say, the technical components of the high-voltage electrical system are designed to have the resistances needed for the higher voltages involved, so the current flowing through the system is also of the desired level. The resistance of the human body, however, is independent of the electrical voltage of any given external source to which it is exposed. Consequently, the current that would flow through the human body in the event of contact with live parts of a hybrid car's high-voltage electrical system would be much higher than in the case of a 12-V system. The effect of electric current on the human body is discussed in detail in the section entitled "Hazards involved with electricity".

4.1.2. High electric current

Live conductors are surrounded by a magnetic field. The strength of this magnetic field increases with the current flowing in the conductor.

The interrelationship between magnetic fields and electric circuits (effect of force, induction) were discussed in the section entitled "Electrical machines — Important terms and laws of physics": when a current-carrying conductor is in a magnetic field, the Lorentz force acts on the vectored charge carriers in the conductor. Moving an electric conductor in a magnetic field induces a voltage in the conductor.

When two current-carrying conductors are side by side, the two effects listed here can be observed:

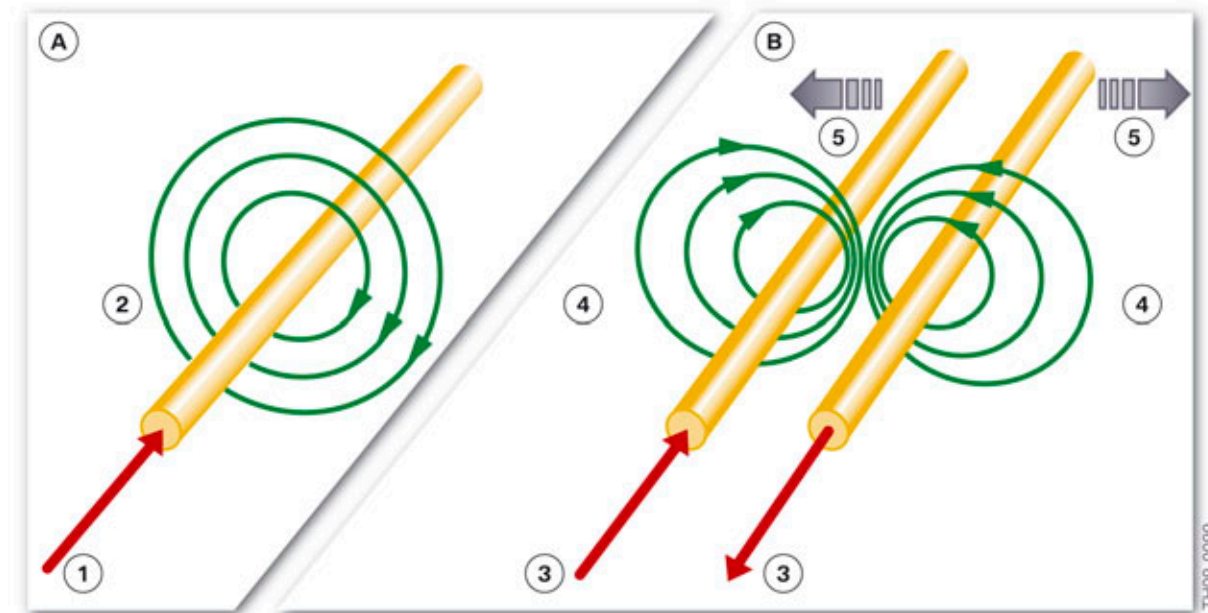
- 1 Force
- 2 Induction

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Force

The magnetic fields of two current-carrying conductors are superimposed. In the high-voltage electrical system of a hybrid car there are parallel-routed cables from/to the high-voltage battery (positive and negative leads). Current flows in opposite directions in these cables. The magnetic fields of the two cables are superimposed in such a way that the field lines between the conductors are closer together. This means that the magnetic field in this region of superimposition is stronger.



Magnetic field around current-carrying conductors

Index	Explanation
A	Single current-carrying conductor
B	Two parallel current-carrying conductors
1	Current flowing in one direction
2	Magnetic field generated by the flowing current
3	Two opposing currents
4	Magnetic field generated by the two opposing currents
5	Force acting on the two conductors on account of the magnetic field

The law of the conservation of energy states that the effect of superimposition must be opposite to its cause (in other words the current flow). Consequently, the two conductors are subject to a force acting in a direction opposite to the superimposition of the magnetic field; this force pushes the two conductors apart.

Example: current in the two conductors $I = 200 \text{ A}$, distance between conductors $r = 2 \text{ cm}$, length of conductors $l = 1 \text{ m}$. Force acting on the two conductors $F = 0.4 \text{ N}$. By way of comparison: the weight of a bar of chocolate ($m = 102 \text{ g}$) is approx. 1.0 N .

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The numerical values used here are similar to those that can occur in a full-hybrid car. The magnitude of the force acting on the conductors on account of the magnetic field is comparatively slight. Nevertheless, as regards the installation of high-voltage cables it is important to bear the following in mind.



Always carefully insert or clip high-voltage cables into the fasteners provided for the purpose.

Always use the strain-relief devices provided for high-voltage cables.

Always tighten the screw connections of high-voltage cables to the specified tightening torque.

Always securing latch the retainers of plug connections on high-voltage cables.

Always make sure that spacing between individual high-voltage cables and the spacing between high-voltage cables and signal lines, for example, is correct and to specification.

When the vehicle is being driven the currents flowing through the system are not always steady: in fact there are very frequent occurrences of (superimposed) alternating currents. This in turn means that instead of being constant, the forces acting on the cables reverse direction in accordance with the direction of current flow. The fasteners and connections therefore have to prevent the high-voltage cables from "being shaken loose" by these alternating forces. If the mechanics servicing the car fail to bear this in mind, vibrations can result in noises audible when the car is being driven. There is also the possibility of an electrical connection working loose and causing unacceptably high contact resistance.

Induction

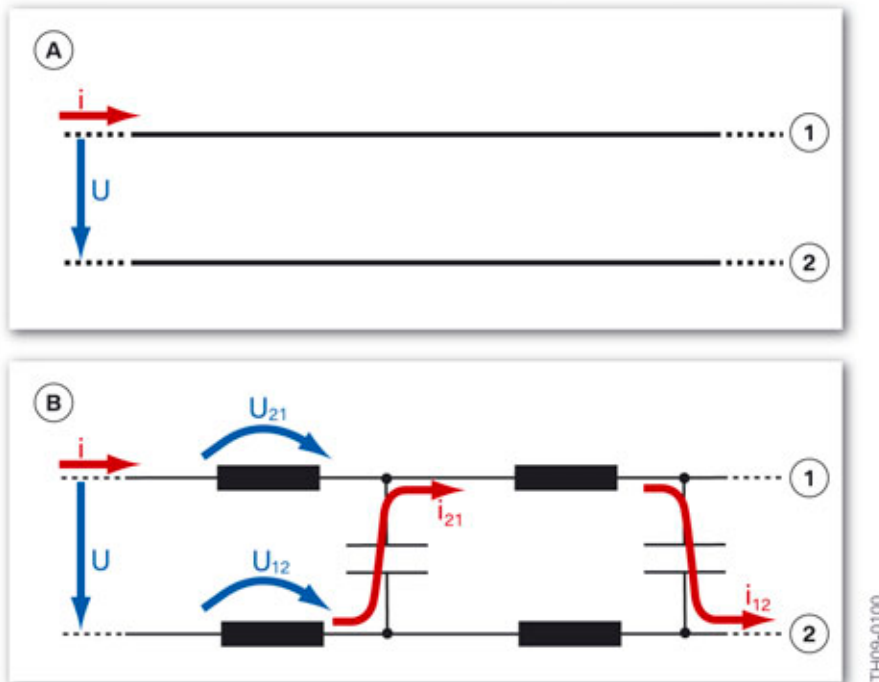
When the current flowing in a conductor changes there is a corresponding change in the magnetic field surrounding the conductor. A voltage is induced in the conductor itself: this phenomenon is of particular significance in terms of wound conductors (coils). If there is another electrically conductive object present in the conductor's magnetic field, a voltage will also be induced in this object. This causes current to flow. The larger the change in current and the faster the rate of change, the higher is the induced voltage. Currents are high in the high-voltage electrical systems of hybrid vehicles and the switching frequencies too are high, so induction is an important consideration. Service employees are not at risk on this account. However, when signal lines are routed in proximity to high-voltage cables, it is important to ensure compliance with the boundary conditions for ensuring electromagnetic compatibility (EMC).

4.1.3. High frequency

The high-voltage electrical systems include inverters and DC/DC converters. These power-electronic circuits operate at high switching frequencies. Consequently, instead of being constant the voltage and current transients of the high-voltage cables evince frequencies that can be as high as several hundred kilohertz. The electrical and magnetic fields produced in this way can lead to circumstances in which interference voltages and stray currents occur in other conductors in the vicinity. These effects have been known since electronic circuits were first developed. These effects can be visualized in the form of inductances in and capacitances between the conductors in a circuit diagram.

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4. Safe Working Practices



Inductance and capacitance of electric cables

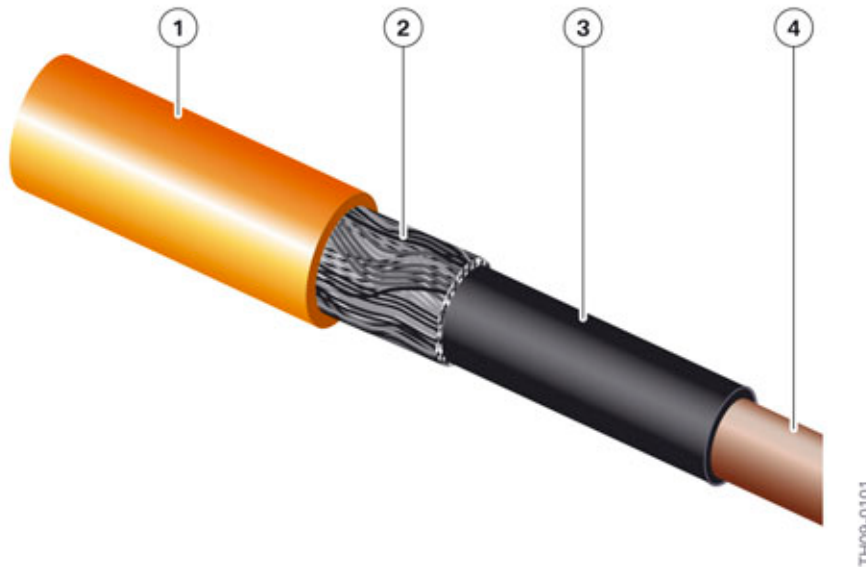
Index	Explanation
1	Cable 1
2	Cable 2
A	Two electric cables running parallel
B	Equivalent circuit diagram for two parallel cables
i_{12}	Stray current produced in cable 2 by capacitance, caused by the voltage on cable 1
i_{21}	Stray current produced in cable 1 by capacitance, caused by the voltage on cable 2
u_{12}	Interference voltage on cable 2 produced by inductance, caused by the current on cable 1
u_{21}	Interference voltage on cable 1 produced by inductance, caused by the current on cable 2

Interference on signal lines can be minimized by suitable countermeasures such as using shielded wires, twisted wires or siting the high-voltage components and cables as far away as possible from signal lines. All these measures fall under the generic term of "electromagnetic compatibility".

A new aspect of hybrid-car technology is that interference voltages and stray currents of this nature are most likely to occur in the vicinity of the high-voltage cables. The high-voltage cables are therefore shielded in order to preserve electromagnetic compatibility.

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Shielded high-voltage cable

Index	Explanation
1	Orange colored outer sheath
2	Braided-wire EMC shielding
3	Insulation of the electric conductor
4	Electric conductor

It is very important to ensure that the measures implemented in the vehicle to preserve electromagnetic compatibility are never impaired by substandard repairs or temporary fixes. If a situation of this nature were allowed to occur the result would inevitably be malfunctions in the sophisticated on-board electronic systems. In hybrid vehicles this also applies to the layout of the high-voltage components such as the high-voltage cables, but also the power electronics. Consequently, never permit any attempt at temporary or provisional repairs to damaged insulation or housing components. Failure to comply with this precaution can put users of the vehicle and Service employees at risk and can also result in malfunction due to interference voltages and stray currents.

4.2. Hazards Involved with Electricity

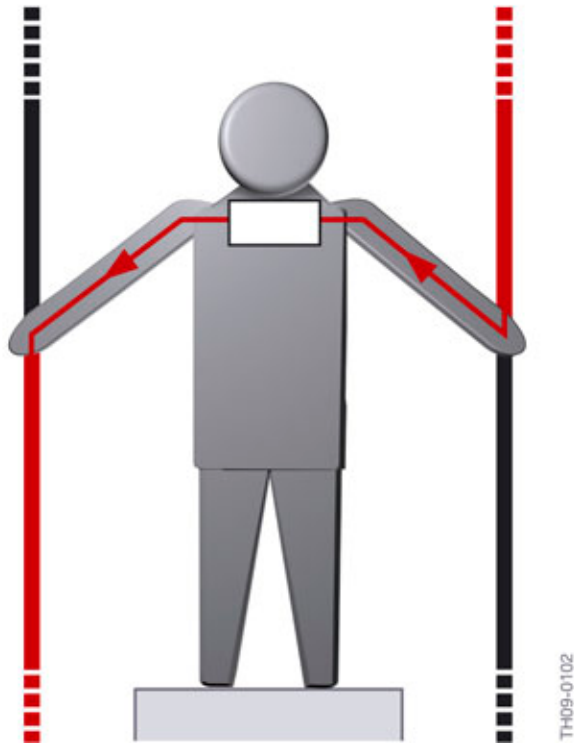
4.2.1. Current flow through the human body

Influencing factors

The cells in the human body are to a limited extent electrically conductive. The high proportion of liquid in the cells is one of the main reasons for this. If a person touches a live component, an electric current can flow through his or her body. The current invariably takes the shortest path through the body. The organs affected by the electricity depends on the path taken by the current through the body. In the graphic below, the heart and the respiratory organs are among those affected.

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Example of how electric current can flow through the human body

Approximate resistances can be given for the various paths that current is likely to take through the human body. The human body's ohmic resistance to current depends on the following factors.

- Clothing
- Wetness of the skin
- Length of and nature of the path through the body.

The denser and drier the items of clothing at the parts of the body at which the electric current enters and exits the body, the higher the resistance. If the skin is wet from water or damp from perspiration, the body's electrical resistance is correspondingly lower. If the current follows a short path through the body resistance is higher than if the current takes a longer route. The table below lists guideline values for the electrical resistance of the human body that result when the influencing factors outlined above are taken duly into account.

Path of electrical current through the body	Ohmic resistance (approximate value)
From hand to hand	Approx. 1000Ω
From one hand to both feet	Approx. 750Ω
From both hands to both feet	Approx. 500Ω
From both hands to the torso	Approx. 250Ω

The strength of the current depends only on the voltage applied and on the bodies ohmic resistance, therefore: $I = U/R$.

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Examples:

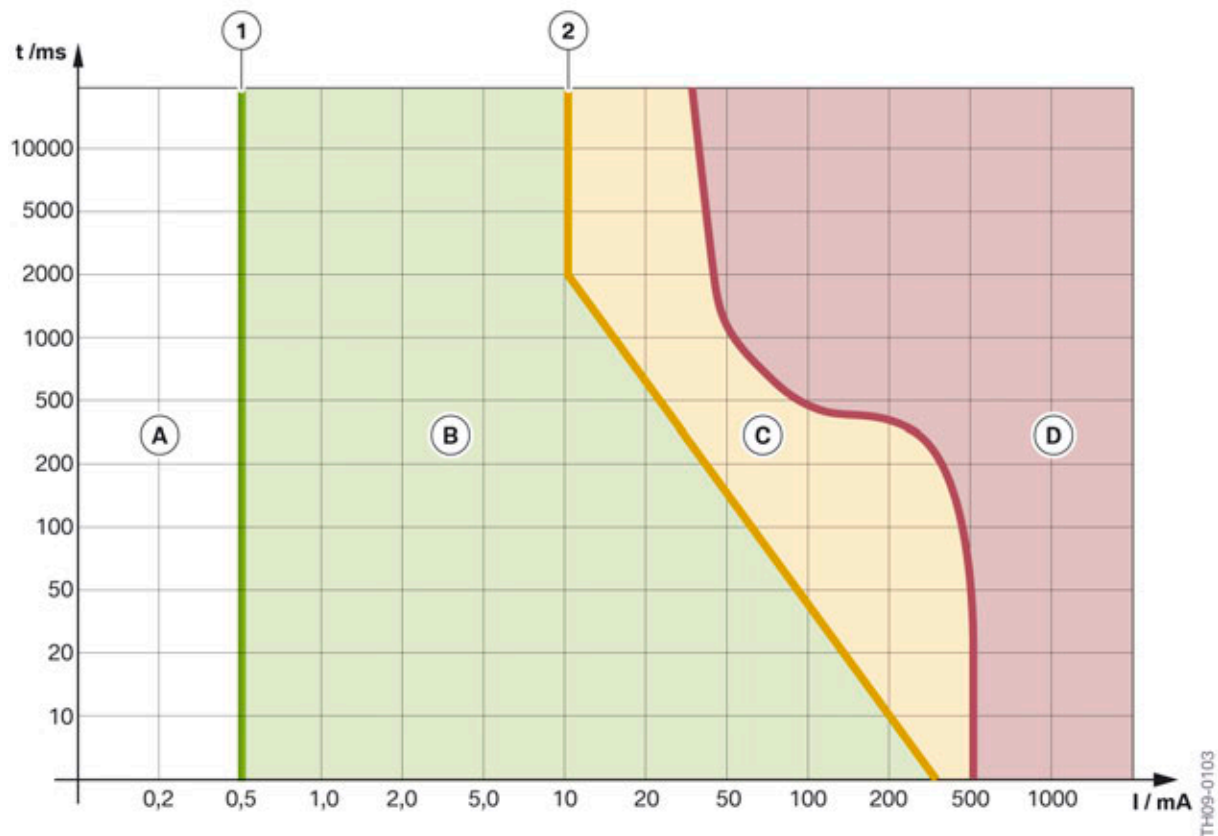
Situation	Applied voltage U	Resistance R of the body	Current I flowing thorough the body
Person touches the terminals of a 12V battery, one hand on each terminal.	12 V	1000 Ω	12 mA
Person touches the terminals of a high-voltage battery, one hand on each terminal.	420 V	1000 Ω	420 mA
Person touches the external conductor of a wall socket with one hand while standing with both feet on the floor.	230 V	750 Ω	307 mA

Effects of electrical current passing through the human body

Electricity does not only have effects that can be used for technical purposes (heat, light, chemical and magnetic effects). Electricity is also capable of producing effects in living organisms, including the bodies of human beings. This is what is termed the physiological effect. The reason for this is that many functions of the human body are controlled by electrical processes. The movements of our muscles are produced by electrical impulses, and this also includes our heartbeat. Similarly, the information provided by the sensory organs is carried by electrical means through the nervous system to the brain. And the human brain itself operates with electrical signals. These signals within the human body evince only minute voltages (mV) and currents (μ A). If an electric current from an external voltage source flows through the human body, it overlays the body's natural electrical signals. This can have a massively disruptive effect on the natural electrical processes that take place all the time in the human body. The effect is usually perceived as an electric shock, and the natural response is to jerk back, away from the source. If the current is strong the reactions of the muscles become uncontrollable. A muscular spasm can occur with the result that the live part is tightly clenched and cannot be released. This let-go limit is important, because once it is passed a very dangerous loop is closed: the longer the current is able to flow through the body the more damaging are its effects.

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Effects of electric current on the human body depending on duration and current strength, source VDE 0100, Part 410

Index	Explanation
1	Limit of perception
2	Let-go limit
A	Effect not perceptible
B	Effect perceptible, can cause muscle spasm
C	Muscle spasm, respiratory difficulties
D	Ventricular fibrillation, apnoea, cardiac arrest

The most dangerous disruption of normal muscular activity affects the respiratory muscles and the cardiac muscle. This can cause apnoea, the cessation of movement of the respiratory muscles. Depending on current strength, the length of time during which current continues to flow, and the frequency (alternating current), ventricular fibrillation. This is a condition of feeble, high-frequency contractions of the heart muscle, which are insufficient to maintain the circulation of the blood.



Apnoea and ventricular fibrillation both cause a breakdown of blood circulation, which means that the supply of oxygen to the body's vital organs is cut off. These are conditions of acute danger to life! Under these circumstances immediate first aid is absolutely essential to rescue the victim from injury to health or fatality!

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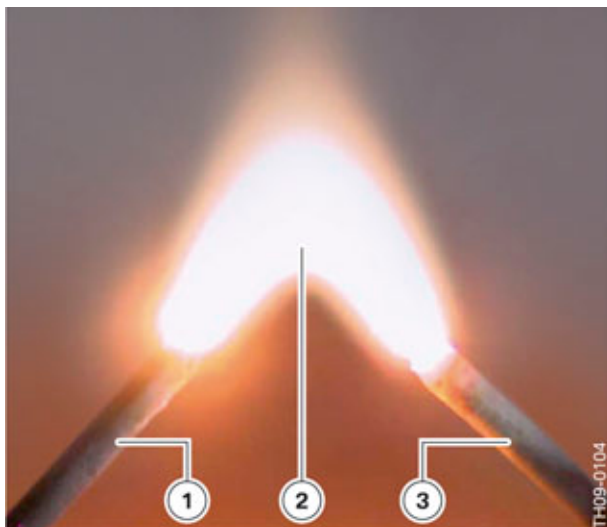
The heat developed by the electric current can also cause injury to and in the human body. External burns can be suffered, primarily due to arcing. Internally, the electric current heats up the tissue through which it flows. The body fluids in particular can become hot enough for evaporation to occur. Injuries of this nature are known as internal burns. The organs can cease to function within a very short time and the blood ceases to circulate. These are conditions of acute danger to life!

These are the immediate effects of severe electric shock, but there are also others that might only emerge some time after the actual incident. Even then it is possible for mortally dangerous situations to develop. For example, it takes some time for body cells destroyed by an incident involving exposure to electricity to be broken down by natural processes. In fact, the process can take several days. The substances produced in this way have to be passed through the kidneys. If dead cells in very large numbers are present the kidneys can become overloaded and kidney failure can be the result. Consequently, it is absolutely essential to ensure that after receiving first aid, the victim of an electric shock is thoroughly examined and re-examined by a professional physician.

4.2.2. Arc flash

Definition

An arc flash occurs when current flows between two conductors across a gap containing a gas (e.g. air). This gap is usually of an insulating or poorly conductive nature. An arc flash can occur when two conductors are initially brought into contact with each other and a current flows. Subsequent separation of the conductors initially produces only a very narrow gap between the two. Since the gap is narrow the electric field is extremely strong and might be higher than the breakdown strength of the gas occupying the gap. Under these circumstances sparkover occurs and gas molecules are ionized. Ions and electrons are violently released from the material of the two conductors, and the result is that after the sparkover the material evinces signs of wear and tear. Another result is that charge carriers capable of migrating are produced: positively charged ions and negatively charged electrons. Because of the presence of the electrical voltage, these charge carriers migrate across the gap to the corresponding conductors. They then react with the conductors. When charge carriers move, their movement means that electric current flows. This mode of producing a current flux in gases is known as gas discharge. In the case of an arc flash this is a continuous process. New charge carriers are produced in succession and the current continues to flow. The gaseous matter between the conductors assumes the state in which it is known as plasma.



Electric arc between two conductors

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Index	Explanation
1	Conductor
2	Electric arc
3	Conductor

A light arc cannot flash between two conductors unless the voltage reaches a certain minimum level and the current flows at a certain minimum strength (before the conductors are separated). These values cannot be positively stated because they depend on the material of which the conductors consist.

In the past, electric arcs were used as a source of light. Today the primary uses are for research purposes and in welding technology (tungsten insert gas welding, TIG).

Effects and hazards

In the plasma between the two conductors the migrating charge carriers collide and cause the gas to heat up. Depending on the material of the conductors and the surrounding gas in the plasma, temperatures can reach approximately 4000 °C or higher. These extremely high temperatures can release other charge carriers from the material of the conductors. This sustains the arc and causes steady attrition of the conductor material as it is "burned up".

This consumption of the material is an effect that has to be taken duly into account in terms of technical design: arcing that occurs at switch contacts as they open erodes the contacts. This is why the manufacturer generally guarantees the operation of relays or switch contactors for only a limited number of switching operations.

Electric arcs are also a source of danger to humans:

- Burns: the very high temperatures involved inevitably cause extremely severe burns if a person comes too close to an arc or penetrates directly into the arc. Keep well clear of electric arcs! Use suitable protective gloves to hold the conductors!
- Ultraviolet radiation: the colliding charge carriers produce not only heat but also light with ultraviolet (UV) components. This UV light can cause injury to the eyes, and more specifically to the retina. This causes the painful condition known among welders as "eyeflash" or simply "flash", otherwise often referred to as "welding flash". Never look directly into an electric arc - always wear a protective facemask!
- Flying particles: ions and electrons are constantly shed from the conductors in the high-temperature zone produced by the electric arc. The process can also cause small particles of the material to break off in an uncontrolled manner. These tiny particles are often tremendously hot! Keep well clear of electric arcs unless you are wearing suitable protective clothing (including protective gloves and eye protection)!

If you find yourself inadvertently confronted by a situation in which an electric arc occurs, it is essential to comply with the following rules:



Try to switch off the voltage source by means of the facilities provided for the purpose (e.g. high-voltage safety switch)!

Keep well clear of the arc and keep your eyes averted!

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If you have to approach the electric arc itself, do so only when wearing the appropriate protective equipment (protective clothing, eye protection, protective gloves) as required for welding work.

Wear protective clothing



Wear eye protection



Wear protective gloves



Avoiding electric arcs

Because of the dangers that can result, it is absolutely essential to prevent electric arcs from occurring whenever possible. Engineers have gone to a great deal of trouble to implement measures to prevent arcing at the components of hybrid vehicles. Service employees, therefore, are unlikely to find themselves confronted by the hazards described in the preceding section. Compliance with the following rules will further reduce the risk of electric arcs occurring:



Never attempt to disconnect a high-voltage cable while the high-voltage system is still in operation and the cables are live. Before disconnecting a high-voltage cable, switch off the high-voltage system and make sure that it is de-energized.

4.3. Safety Rules to Prevent Hazards

4.3.1. Basics

Every individual working in the Service sector bears responsibility for the discharge of certain important duties with regard to health and safety at work:

- Always comply with verbal instructions and written rules concerning health and safety
- Always make full use and correct of the protective facilities provided
- Always use the facilities (tools, vehicles) in accordance with instructions.
- If equipment is found to be defective, make sure that it is correctly restored to full working order. Each individual not qualified to undertake such repairs themselves must immediately notify their superior to have the repairs duly undertaken at the earliest possible opportunity.

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Safety rules apply to all work on components that operate at hazardous voltage levels. In Germany the binding rules are formulated by the professional liability insurers (e.g. BGV A3, §6) and there are also national implementations in individual countries and international standards (e.g. DIN ECE R 100). Hazardous voltages are:

- Alternating current (AC) voltages of 25 V or higher
- Direct current (DC) voltages of 60 V or higher

A voltage is defined as being hazardous on the basis of the consequences resulting when a person touches a live part. The voltage is defined as "hazardous" when the current flowing through the person's body can cause injury to health. In hybrid cars the components that work at hazardous voltages are referred to collectively as "high-voltage components". They are identified either by the warning label shown below or by a high-visibility orange color (as is the case with high-voltage cables).



Warning: hazardous electrical voltage



Warning for high-voltage component, type 1



Warning for high-voltage component, type 2

The principal health and safety rule is simple:



Do not work on live parts!

Consequently, before starting work it is essential to de-energise the system in question and make sure that it cannot be reenergized without the knowledge and consent of the person undertaking the work. Other, specific, more detailed health and safety rules are derived from this principal rule. Each and every individual involved in Service must always apply these rules before starting and while working on high-voltage components. Strict compliance with these rules is absolutely essential in order to ensure health and safety.



These safety rules (DIN VDE 0105) are as follows:

- 1 De-energise the system
- 2 Secure the system so that it cannot be reenergized without your knowledge and consent
- 3 Check that the system is de-energized
- 4 Establish connection to ground and short-circuit
- 5 Use suitable means of shielding off live components in the vicinity to prevent accidental contact.

Rules 1 to 3 are the only rules that apply to the present range of vehicles with high-voltage systems. These rules are considered in more detail below.

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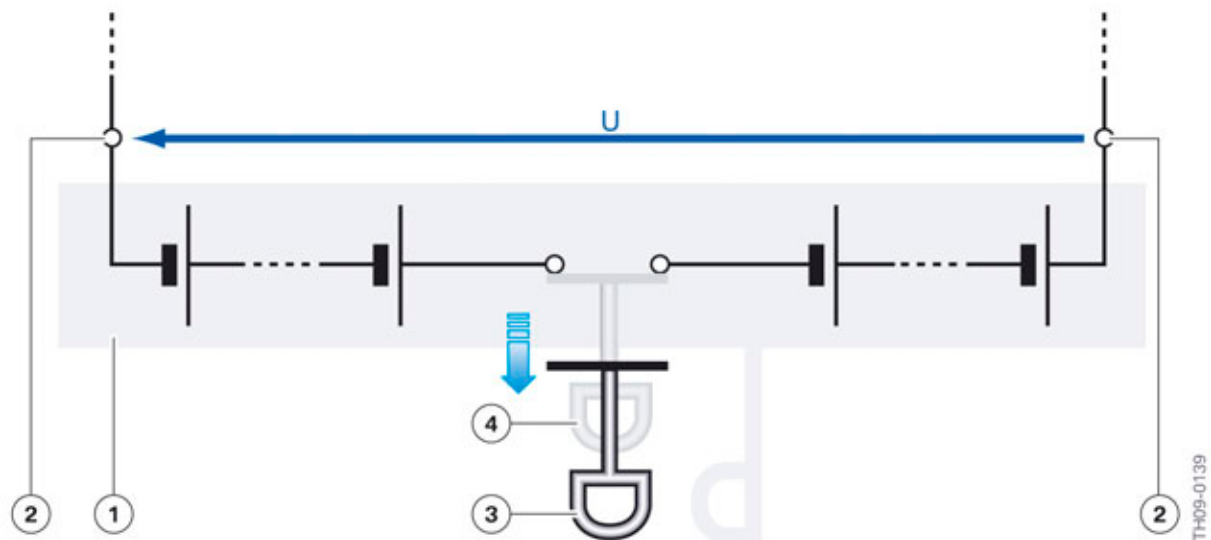
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Rule 4 cannot be applied to any practical purpose because the high-voltage electrical system is an IT network as identified by the two-letter IEC 60364 code for earthing systems. In a network of this kind no conductor is connected to earth or ground. Consequently, there is no additional protection to be gained by earthing or grounding. The same applies to short-circuiting, because this could lead to other, additional hazards. These hazards more than outweigh the potential benefits. Example: if voltage were to be re-applied to the high-voltage battery while it was short-circuited, a very high short-circuit current would flow. This current would be high enough to set the high-voltage battery on fire!

Rule 5 does not apply to the present vehicles because they have only the one high-voltage electrical system. This rule is intended for technical facilities in which there are two or more circuits operating with hazardous voltages and situated in close proximity to each other. In configurations of this kind it is common practice to de-energise only the circuit on which work is actually to be performed. Shielding with suitable materials ensures that the other circuits in the vicinity that remain in operation while work is in progress cannot be sources of danger (e.g. by short to the de-energized circuit).

4.3.2. De-energize the system

A Service employee working on high-voltage components can come into contact with components such as the connections of the high-voltage cables. While the vehicle is in operation these live components carry hazardous voltages. Service employees must be able to work under conditions that do not entail risk to health and safety, so when work is to be undertaken all the high-voltage components must be "dead", in other words free of hazardous voltage. The simplest possible way of achieving this is to remove the energy source, in other words the high-voltage battery, by disconnecting it from the circuit. This can be accomplished as shown in the graphic below.



Safety rule 1: de-energise the system

Index	Explanation
1	Housing of the high-voltage battery
2	External connections of the high-voltage battery
3	High-voltage safety switch pulled (disconnected)
4	High-voltage safety switch inserted (system is live)

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This effectively pulls the plug on the series-connected battery cells. Once the voltage source has been disconnected in this way, the externally accessible poles of the high-voltage battery are no longer live. The connector, the plug at which the connection is broken, is called the "high-voltage safety switch" but in technical jargon it is also frequently referred to as the "Service Disconnect".

In the vehicles in the current range the high-voltage safety switch varies in appearance and configuration from model to model. By way of example, the illustration above shows the type used in the BMW ActiveHybrid X6.



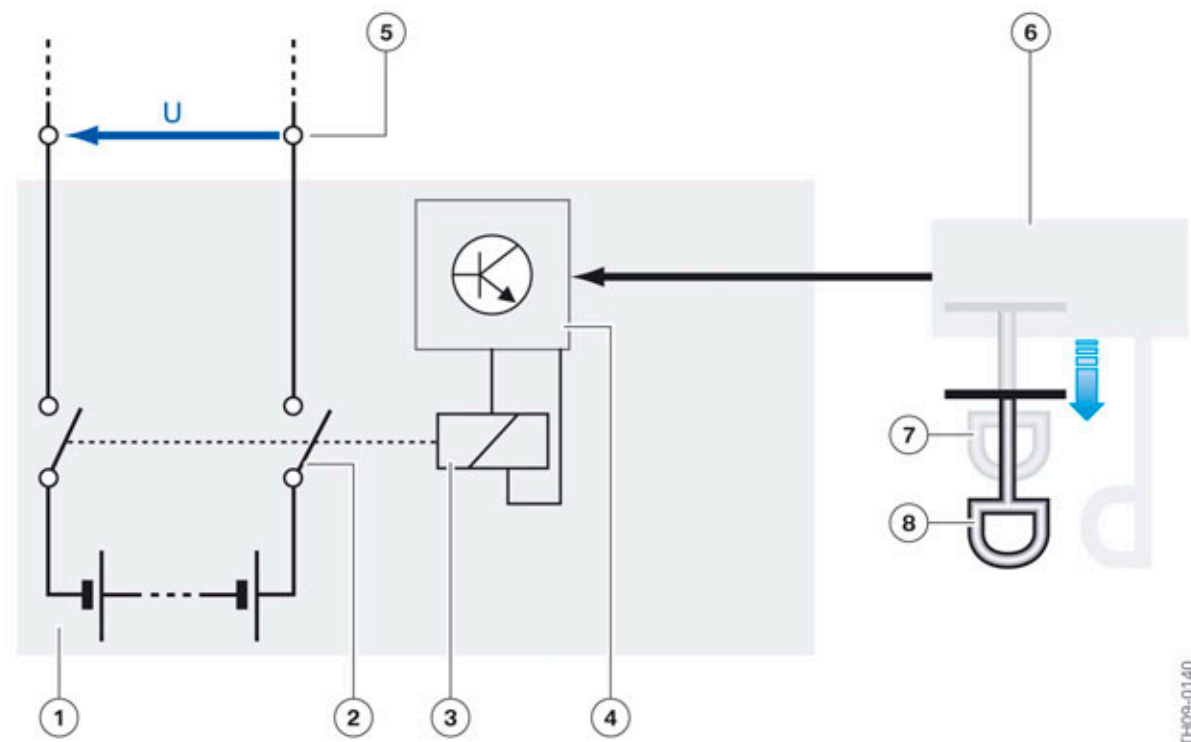
High-voltage safety switch (Service Disconnect) in the BMW ActiveHybrid X6

Index	Explanation
1	High-voltage battery
2	High-voltage safety switch (Service Disconnect)

As an alternative to disconnection of the series-connected battery cells, there is another way in which the high-voltage safety switch can be used. In this respect, the high-voltage safety switch acts as control input for a control unit. The control unit immediately interrupts the supply to the switch contactors, as soon as it establishes that the high-voltage safety switch has been pulled. The contacts of the switch contactors respond by opening automatically. The effect is the same as disconnection of the series-connected battery cells: once the high-voltage safety switch has been pulled there is no longer a hazardous voltage present at the external poles of the high-voltage battery.

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Safety rule 1: de-energise the system (alternative effect of pulling the high-voltage safety switch)

Index	Explanation
1	Housing of the high-voltage battery
2	Two switch contactor contacts
3	Switch contactor solenoid
4	Control unit that evaluates the state (pulled/inserted) of the high-voltage safety switch and actuates the switch contactor accordingly
5	High-voltage connections of the high-voltage battery
6	Separate housing for the high-voltage safety switch
7	High-voltage safety switch inserted (system is live)
8	High-voltage safety switch pulled (disconnected)

When the high-voltage safety switch is pulled, several processes take place automatically and in parallel in the high-voltage system. This ensures that hazardous voltage is no longer present at the poles of the high-voltage battery, at electronic components, or at the electrical machine or machines:

- 1 Disconnection of the series-connected battery cells or/and opening of the switch contactors in the high-voltage battery
- 2 Discharge of the capacitors in the other high-voltage components
- 3 Short-circuiting of the windings of the electrical machines.

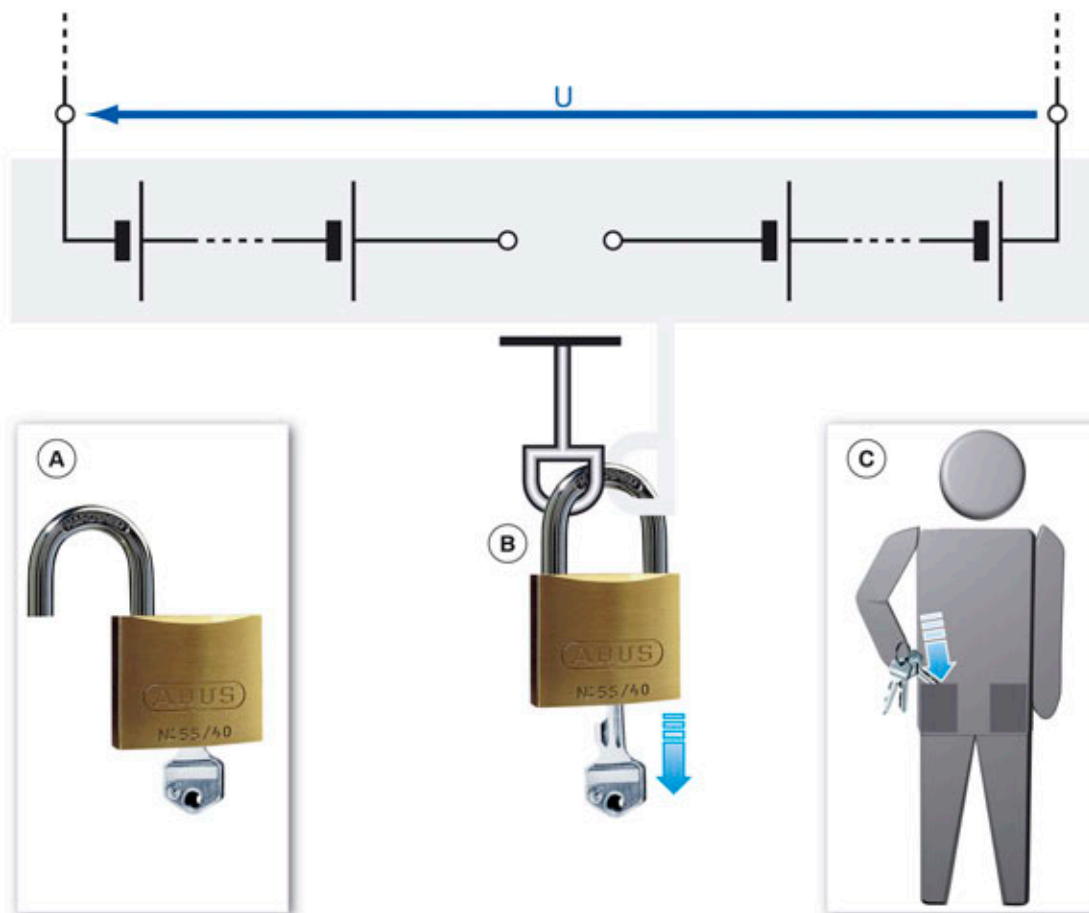
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These measures, once implemented, avert the major hazards that could otherwise threaten Service employees. But what would happen if someone else (intentionally or unintentionally) reinserts the high-voltage safety switch? If this were to happen there would be a possibility of a hazardous voltage being applied to components being worked on or about to be worked on, and this would of course constitute a major hazard to health and safety. Consequently, not only does the high-voltage system have to be de-energized, it also has to be secured so that it cannot be reenergized without the knowledge and consent of the person undertaking the work.

4.3.3. Secure the system so that it cannot be reenergized without your knowledge and consent

Once the high-voltage safety switch has been pulled, the high-voltage system can be secured at the switch to ensure that it not switched on again. This entails installing a conventional padlock. Once the high-voltage safety switch has been pulled and secured in this way, it cannot be reinserted until the padlock has been removed. The person working on the high-voltage system has the key of the padlock in safekeeping until all work on the high-voltage system is completed.



TH09-0141

Safety rule 2: secure the system so that it cannot be reenergized without your knowledge and consent

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Index	Explanation
A	Use an open padlock
B	Secure the high-voltage safety switch in its pulled position by engaging the padlock
C	Keep the key of the padlock in a safe place at all times

This simple precaution suffices to ensure that no-one else can switch on the high-voltage safety system until all work on the system has been completed. This rules out inadvertent endangerment of the persons working on the high-voltage system not only before work commences, but also for as long as it takes to complete all the work in question.

4.3.4. Check that the system is de-energized

Once the person who is going to work on the high-voltage system has de-energized the system and locked the switch so that it cannot be switched on again without the key, there are still certain safety rules that have to be applied. The next step is a check to ensure that the high-voltage system is in fact de-energized and safe. In BMW vehicles, it is not necessary to use an external tester for this step in the procedure. The design of the high-voltage system is such that the system itself can tell that it has been de-energized. Several high-voltage components measure their own voltage, using integrated circuits for voltage measurement. The results of these measurements are transmitted via bus systems to the instrument panel. When the results of all these measurements show a voltage that is below the hazard threshold, the instrument panel shows a symbol indicating that the high-voltage system has been successfully powered down and is no longer live.



Check Control symbol: high-voltage system is de-energized and safe

In this Check Control symbol, the danger symbol representing high voltages (lighting flash) is struck through. The symbol is therefore a visual indicator clearly emphasizing that there is no longer a hazardous voltage present in the system. Depending on the vehicle type, the symbol actually shown in the instrument panel might not necessarily be exactly as illustrated here. Consult the instructions in the repair manual or the training media for the vehicle type in question to ascertain precisely which symbol is used to indicate this state.

This is the third safety rule and once again it is essential to apply it before starting work on high-voltage components. The procedure is a simple a check to ensure that the high-voltage system has been correctly switched off and has safely de-energized. The check ensures that the high-voltage system is not a source of danger for persons working on the vehicle.

4.4. Other Dangers Associated with Hybrid Technology

Chemical energy stores (batteries) are used to provide electric power for propulsion in hybrid vehicles. These batteries contain substances that are classified as hazardous and for which legislation lays down certain rules that have to be observed to ensure that they are handled with due care. The requisite information is laid down in binding form in safety data sheets. This training document merely summarizes the most important underlying points and touches on the possible hazards involved.

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4.4.1. Nickel metal hydride battery

Nickel metal hydride (NiMH) batteries, if used correctly and kept free of damage, are not a direct source of endangerment. The casing is designed to ensure that fluids (electrolyte, for example) cannot escape at any point in the entire usable lifespan of the battery. If the casing is damaged, as can happen for example in the event of an accident or if the battery is not used correctly in accordance with the manufacturer's instructions, NiMH batteries can be a source of endangerment as follows:

- Chemical burns (electrolyte)
- Injury to health (electrolyte, coolant)
- Fire/explosion.

A solution of potassium hydroxide is used as the electrolyte in NiMH batteries. This solution is classified as a caustic irritant. In the event of electrolyte escaping from a NiMH battery, it is important not to come into contact with the spilled liquid. Obtain medical assistance if a person is wetted by or swallows the electrolyte. Notify the fire service in the event of a spillage: the fire service has the special equipment needed to deal correctly with the spilled electrolyte.



Comply with the instructions set down in the safety data sheet on how to proceed in the event of electrolyte escaping from a nickel metal hydride battery.

NiMH batteries can develop full power only within a limited ambient temperature range. Consequently, the battery is connected to an on-board cooling circuit. The coolant is the same as that commonly used in BMW vehicles. It is hazardous to health and under no circumstances should it be swallowed or allowed to come into contact with the skin.



Comply with the instructions set down in the safety data sheet for BMW coolants when working on the cooling system of the nickel metal hydride battery.

When NiMH batteries are charging or discharging, the chemical reactions involved produce gases (oxygen and hydrogen, respectively). These gases can escape through a breathe valve in the casing of the nickel metal hydride battery. When the battery is used correctly in accordance with the manufacturer's instructions the concentration is low and poses no hazard (according to information provided by manufacturer).



In order to preclude the possibility of explosion, always keep fire, sparks and all other sources of ignition well away from NiMH batteries.

NiMH batteries bear labels showing hazard symbols to indicate the possible dangers.

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Index	Explanation
1	Warning: hazardous electrical voltage
2	Warning: explosive substances
3	Warning: battery-related hazard
4	Warning: caustic substances
5	Prohibition: fire, naked flame and smoking prohibited
6	Prohibition: keep out of reach of children
7	Requirement: always comply with operating instructions and repair instructions

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Index	Explanation
8	Requirement: always wear eye protection
9	Warning label for high-voltage components
10	Breather for hydrogen gas
11	Reservoir for coolant circuit
12	Connection for coolant line (inlet)
13	Connection for coolant line (outlet)

4.4.2. Lithium ion battery

Similarly, the lithium-ion batteries in BMW vehicles are safe if used correctly in accordance with the manufacturer's instructions and kept free of damage. It is important to note, however, that restrictive conditions apply to correct usage. In particular it is very important to ensure that lithium-ion batteries are not overcharged and not exposed to excessively high temperatures.

Overcharging can cause metallic lithium to deposit on the positive electrode and destroy the negative electrode. If this is allowed to happen the temperatures at the battery rise and the lithium-ion battery can catch fire. In BMW vehicles the control unit of the lithium-ion battery ensures that charging and discharging cycles take place only within the specified boundary conditions. Sensors are used to monitor cell temperature and cell voltage. The control unit intervenes in the charging or discharging process as necessary. This applies both when the vehicle is in use and in circumstances in which the high-voltage battery is charged off a charger connected to the 12-V system.



Never attempt to open the casing of lithium-ion batteries. Never attempt to operate or charge a battery of this kind unless it is connected to the appropriate control unit. Failure to comply with these precautions will result in a risk of fire!

Do not expose lithium-ion batteries to excessively high temperatures. Operating temperatures higher than approx. 50 °C suffice to accelerate the ageing process and shorten battery life. If the cells are permitted to reach temperatures of 100 °C or higher a short-circuit can occur between the cells. The high currents that flow if the cells short further increase temperature and a chain reaction can occur. The entire lithium-ion battery would suffer irreversible damage and could catch fire.

It is difficult to extinguish a lithium-ion battery if it catches fire. However, there is no direct risk of the battery itself exploding. Nevertheless, the high temperatures produced by the fire could ignite objects, liquids or gases in proximity to the battery and this in turn could cause an explosion.



Only specially trained persons are permitted to tackle a fire caused by a lithium-ion battery.

The upper limit for operating temperature is very firmly defined, so lithium-ion batteries in BMW ActiveHybrid vehicles are cooled. For example, they can be connected directly to the refrigerant circuit of the air conditioning system. Consequently, it is essential to comply with the repair instructions and the instructions in the applicable safety data sheets for all work on lithium-ion batteries and in particular when opening the connections to the refrigerant circuit.

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Strict compliance with the repair instructions for work on the refrigerant circuit – and this includes lithium-ion batteries – is essential in order to preclude the possibility of a health hazard.

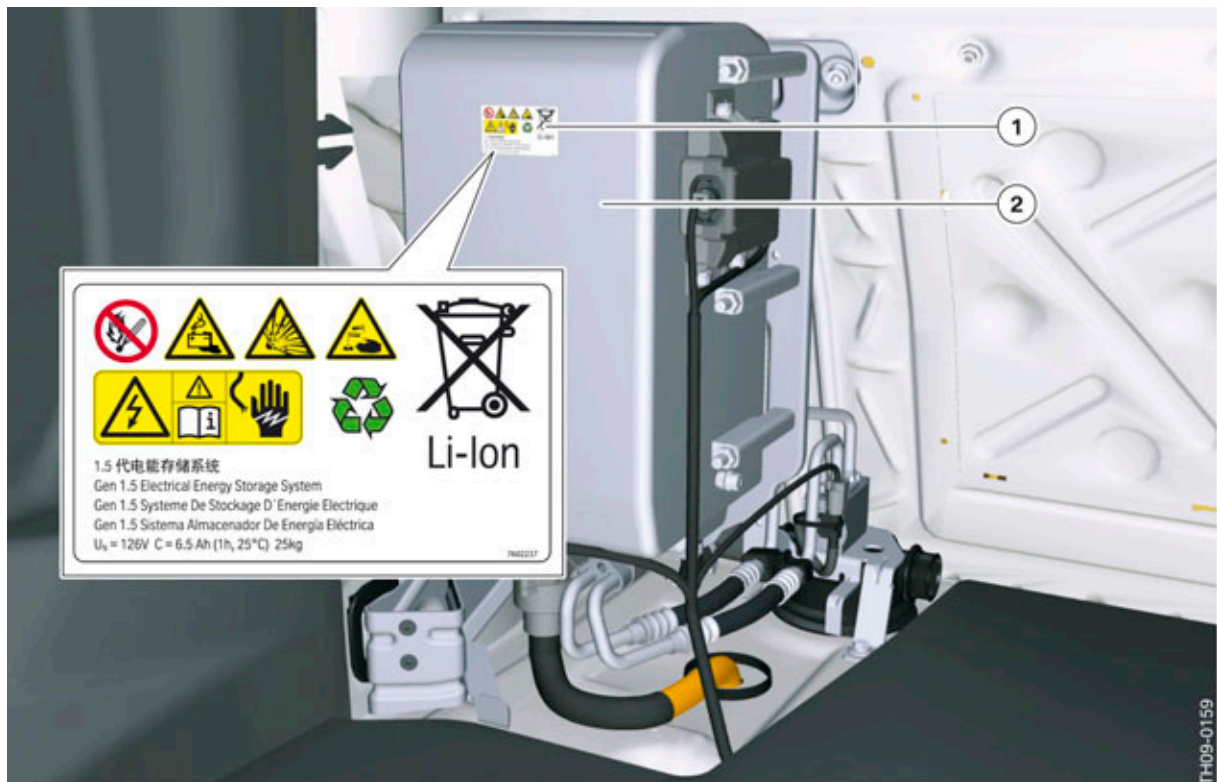


Comply with the safety data sheet for handling the refrigerants used in these systems (e.g. R134a). It is particularly important to comply with all instructions regarding the use of personal protection equipment.

The lithium-ion batteries used in BMW vehicles have hermetically sealed casings. The only external connections are the high-voltage terminals and the coolant lines. Even so, special precautions also apply to using and working with high-voltage batteries of this kind.



Always comply with the instructions in the safety data sheet and the warnings and prohibitions indicated by the labels on the lithium-ion battery.



Lithium ion battery

Index	Explanation
1	Warning label
2	Casing

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4.4.3. Electrical machines

High output is a characteristic of the electric drive in every BMW ActiveHybrid, so the electrical machines operate with very strong magnetic fields. These fields are generated by permanent magnets or electromagnets. It is important to bear in mind that the magnetic fields of permanent magnets are also permanent, in other words they are present even if the high-voltage system is switched off or the electrical machine has been removed.

These magnetic fields can interfere with electronic medical devices, particularly cardiac pacemakers. The components in question bear a prohibition label drawing attention to this specific hazard. By way of example, the illustration below shows this label affixed to an electrical machine.



Prohibition: no approach for persons with cardiac pacemakers



Persons who need a cardiac pacemaker or other electronic medical device to maintain their health are not permitted to work on components bearing the prohibition label shown here.

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4.5. First Aid

4.5.1. Basics

The engineered-safety precautions implemented for the high-voltage systems in BMW hybrid vehicles effectively prevent the risk of injury and risk to health. If an accident involving electricity occurs despite these precautions, it is important to know exactly what to do in order to help the accident victim. This is the subject matter of this part of the information bulletin.

Many people find it extremely difficult to provide first-aid assistance to accident victims or persons who have suffered injury. All too often it is fear of doing something wrong that is a major obstacle. But anyone can provide first aid – even if it "merely" entails notifying the emergency medical services that is still the right thing to do and it will certainly help the injured person or accident victim. Every individual is morally obliged to help injured persons. In some countries, in fact, this is underpinned by a legal obligation as well.

It is of course well beyond the scope of this training medium to cover all the subjects of relevance to first aid. Consequently, the following is simply a brief outline of some of the basic rules on how to administer first aid, and specifically what to do in an accident involving injury due to electric shock.

In many countries, such as the US, volunteer and professional medical services offer courses that impart in-depth knowledge of first aid and enable trainees to apply the knowledge they acquire in drills that help hone their first-aid skills.

In the vast majority of cases encounters with injured persons occur by chance and so even persons able and willing to provide first-aid assistance can be taken by surprise, so it is a good idea to consider in advance what one could or should do in a situation of this nature. In particular, it is very important not simply to rush unthinkingly to accident victim's assistance: it is much more effective to proceed in accordance with the following three step system:

- 1 Identification
- 2 Consideration
- 3 Action

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Identification

The first step - **Identification** – is extremely important in any incident in which electricity is involved, because the next actions on the part of the person or persons providing first aid depend to a very large extent on this initial assessment of what might well be a critical situation. How can you tell that this is an accident situation involving electricity? The following can be characteristic of an accident involving electricity:

- The victim is still in contact with the circuit that produced the accident. The victim is unable to move because the electric current following through his or her body has induced muscle spasms and the muscles have cramped.
- A person or persons is/are lying unconscious on the ground: a high current flowing through the victim's body will cause the heart to stop beating, the circulation of blood ceases – unconsciousness is the result.
- There are spot-like burns on the victim's body. There is always a point at which the electricity enters the body and another at which it exits.
- Persons involved in the accident might be in a state of shock. Hyperactivity or apathy are possible indicators that a person is in shock.



Step 1: identify the situation as an accident involving electricity

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Consideration

The second step is to **consider** how best to proceed. The critical aspect of an accident involving electricity is that in order to help others, it is essential for the helper to ensure that he or she is not also injured. Overanxious helpers who endanger or injury themselves are certainly not going to be in a position to help others.



Step 2: consider

Action

Once you know exactly what has to be done, it is time to **act** speedily and deliberately. If others are present, delegate specific jobs to them. This is the best way to provide assistance effectively and quickly, much more so that one person acting alone could possibly manage.

The overall objective for everything undertaken at the scene of an accident is to enable the injured person or persons to survive their injuries unscathed, if possible. Consequently, it is essential to provide assistance as rapidly as possible and this applies even if the only persons on the scene are not trained professionals. Needless to say other steps are also necessary and trained medics will have to take over at some point to assist the accident victim back to full health. All the individual steps have to be gone through in the correct sequence, because only in this way is the rescue chain complete and intact.



Step 3: act

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4.5.2. Chain of survival



Rescue chain

Index	Explanation
1	Think and Secure
2	Send out call to emergency medical services
3	Interrupt the Circuit
4	First Aid
5	Emergency Help & Follow-up Treatment

The individual links that make up this rescue chain are described in more detail below. The primary focus in these descriptions is on the aspects specific to accidents involving injuries due to electric shock.

Think and secure

The term "think and secure" refers to the actions that have to be taken right away in order to preserve the life of the accident victim. However, it is crucial for the person who arrives on the scene in order to provide assistance to assess the situation and identify this incident as an accident involving electricity, for example. So please Identify and Consider the situation before you take action.

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Send out call to emergency medical services

Professional medical assistance must be obtained in every situation in which a person suffers electric shock. Assistance of this nature is best called up by phone or mobile phone. Needless to say it is also important to call up emergency medical assistance in accident situations of any other kind, particularly if the accident victim has lost consciousness or shows visible signs of severe injuries.



Send out call to emergency medical services



Every Service employee must memorise the emergency call numbers valid at his or her place of work.

If you place an emergency call the responder in the rescue coordination center will ask you for the following information:

- Where has the accident happened?
- What has happened?
- How many persons have sustained injuries?
- What is the nature of the accident and what injuries have been sustained?

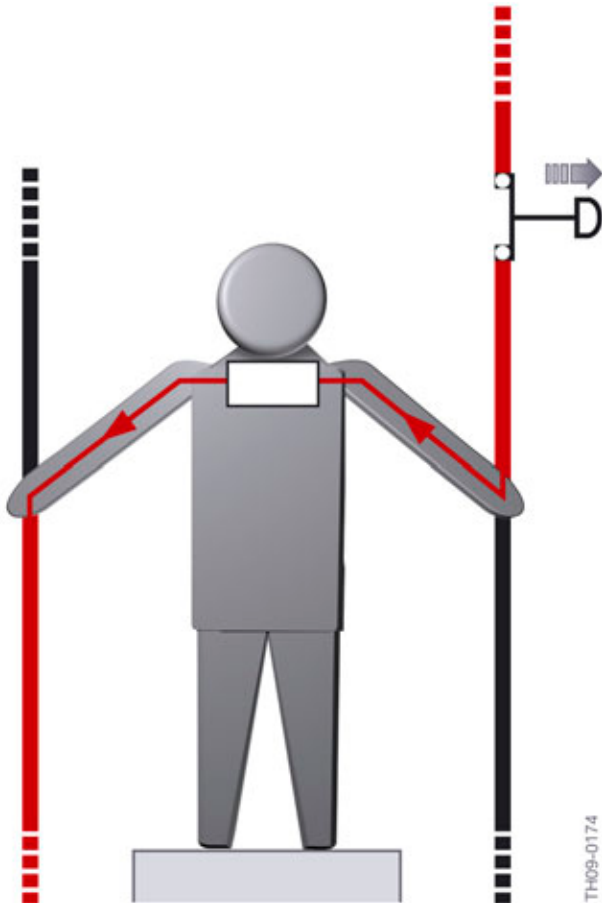
You as caller must remain on the line until the responder in the coordination center has no further questions and terminates the call.

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Interrupt the circuit

As we have already stated, an electric current flowing through the human body can cause very severe injuries. The higher the current the worse these injuries will be, but as we have also seen the same applies the longer the current is allowed to flow.



Interrupt the circuit that produced the accident

The first reaction on the part of anyone arriving on the scene of the accident is usually to try to grasp the electric-shock victim and pull them away from the live cable or component. This is the wrong thing to do and immediately puts the helper at risk!

As soon as contact is made current will flow through both bodies and injure the person who rushed unadvisedly to the assistance of the original victim. Consequently, it is essential for the helping person to assess the situation correctly right from the word go and to act in full awareness of the importance of protecting themselves and others from injury.



Protecting yourself against injury is vitally important in order for you to be able to help others. Do not attempt to lay hold of the injured person, because this attempt to interrupt the flow of electricity is futile and will be to no avail. Instead, locate the item of equipment that will enable you to switch off the power at the voltage source.

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If the electrical circuit of a hybrid vehicle has produced an accident involving electric shock, you have the following ways of switching off the voltage source:

- Pull the high-voltage safety switch (Service Disconnect)
- Disconnect the 12-V supply (e.g. by disconnecting the 12-V battery)

If you find that you cannot switch off the voltage source without putting yourself at risk, the circuit will have to be interrupted by some other means. This will entail the use of insulating tools, and in this situation electrically insulated protective gloves are best. Only with the help of equipment of this nature should you attempt to distance the injured person from the live component. In exceptional circumstances you can make use of plastic objects or dry wooden implements to manoeuvre the injured person free of the live circuit. It is only by using equipment of this nature that you can exclude or at least reduce the risk of suffering an electric shock yourself in your efforts to help the injured person.



Electrically insulated protective gloves

First aid

First aid has to be administered if the injured person is no longer conscious and/or has ceased breathing. First aid is administered as a means of maintaining the vital functions until the emergency services arrive at the scene of the accident. First aid also involves attending to injuries.



More in-depth knowledge of first aid can be obtained through the courses organized by local first-aid organizations or emergency services.

Assistance by emergency services and post-accident medical treatment

As soon as the emergency services arrive on the scene their trained personnel will instantly take over the administration of first aid and will provide other assistance as well. The condition of the accident victim is further stabilized and recovery is promoted by continuation of CPR, use of a defibrillator and/or the administration of suitable medication. This, however, is not the final link in the rescue chain.



Every accident involving electric shock requires follow-up in the form of a thorough examination by a qualified physician.

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This is because as well as producing short-term injurious effects, an electric shock also has effects that might not become apparent until hours, days or even weeks later. For example, one long-term result of an electric shock is the production in the body of excess protein that has to be flushed out by the kidneys. If the quantity of the waste products is excessive delayed kidney failure can occur even several days after the accident.

Depending on the severity of the accident, the victim will have to be treated in an outpatient center, taken into hospital for observation, or attend a physician for a series of post-accident examinations. This is the only way of avoiding subsequent complications and possible long-term injury to health.

4.6. Safe Working Practices for Working Safely on High-voltage Components

4.6.1. Preconditions

Only Service employees who satisfy the following three preconditions are permitted to work on high-voltage components.



Must be in possession of documentary proof of qualification as electrotechnician for high-voltage systems in inherently safe vehicles.

Service employees who have received only first-level training in hybrid technology (for example through the information bulletin entitled "Initial information on hybrid vehicles") are **not** sufficiently qualified to undertake work on high-voltage components.



Before commencing work on high-voltage components, you must have applied the three safety rules ("de-energise the system", "secure the system so that it cannot be reenergized without your knowledge and consent" and "check that the system is de-energized").



Must comply strictly with the repair instructions when carrying out repair work on high-voltage components.

4.6.2. Engineered-safety measures

The BMW ActiveHybrid vehicles in general and the high-voltage components in particular are designed and built to be inherently safe. This means that faults that could lead to endangerment of the vehicle user are reliably detected and identified. Such detection and identification leads to immediate shutdown of the high-voltage system, with the result that there are no longer hazardous voltages present at active components of the system. Shutdown is initiated automatically for example if a cover of a high-voltage component is removed. The high-voltage system is also of fault-tolerant design in order to ensure that the vehicle user does not have to stop on account of every minor fault. This means

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that the occurrence of a single fault does not cause direct endangerment. Self-diagnosis of the high-voltage components ascertains the presence of these faults, however, and logs them in fault memory. Nonetheless, the vehicle can proceed on its way without any risk.

The list below provides an overview of the engineered-safety measures that are used in the high-voltage system of BMW ActiveHybrid vehicles.

- Identification labelling
- Guarding to prevent accidental contact
- High voltage interlock loop
- Discharge of the high-voltage circuit
- Galvanic separation of the high-voltage electrical system from the 12 V electrical system.
- Short-circuit monitoring
- Shutdown in the event of an accident.

The sections below describe how these measures work. This is followed by a discussion of some of the specifics that apply to the safety measures for the high-voltage cables and the high-voltage battery.

Identification labelling of identification high-voltage components

Each high-voltage component has on its housing or casing an identifying label that enables Service employees and vehicle users to identify intuitively the possible hazards that can result from the high electric voltages used. The warning labels are all based on the internationally standardised and universally familiar warning signs for hazardous electrical voltage. On this basis, two different warning signs are currently used in BMW ActiveHybrid vehicles to identify high-voltage components. Both these labels contain an extra symbol prompting the Service employee to consult the Service media (e.g. the appropriate repair manual). The document in question contains the information relevant to correct procedures for safety in handling and working with the high-voltage component in question. The labels of the second type also contain a third symbol drawing attention to the possibility of an electric shock hazard.



Warning: hazardous electrical voltage



Warning for high-voltage component, type 1

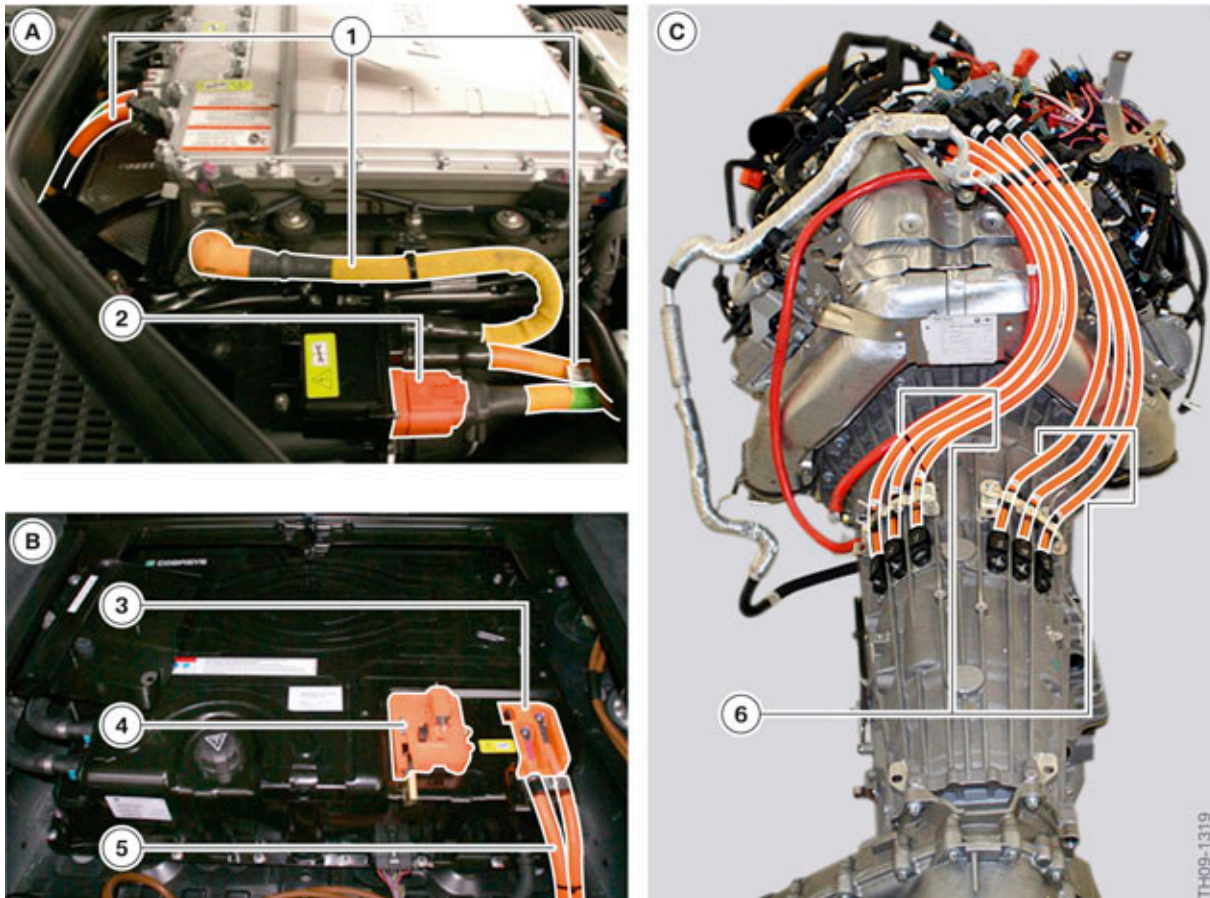


Warning for high-voltage component, type 2

The means of identifying high-voltage cables is a special instance. These cables can be several metres in length, so there would be little point in relying on one or two labels affixed at arbitrary points sufficing under all imaginable circumstances to identify a cable as carrying high voltage. Labels of this nature would be easily overlooked. Instead, all high-voltage cables have bright orange sheaths. Some connectors for high-voltage cables and the high-voltage safety switch can also be bright orange.

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Orange colouring of the outer sheath identifies the high-voltage cables

Index	Explanation
A	Orange coloring identifying cables and components in the engine compartment
B	Orange coloring identifying components at the high-voltage battery
C	Orange coloring identifying components at the active gearbox
1	High-voltage cables in the engine compartment
2	Connector on a high-voltage cable
3	Terminals on the high-voltage battery
4	High-voltage safety switch (Service Disconnect)
5	High-voltage cables at the high-voltage battery
6	High-voltage cables at the active gearbox

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The only Service employees permitted to work on the high-voltage components identified in this way are those who satisfy all the requirements for undertaking work of this nature: qualification, compliance with the safety rules, procedure strictly in accordance with the repair instructions (see also the section entitled "Preconditions").

The manufacturers of hybrid vehicles have agreed on a unified system of identification for the high-voltage components based on the warning labels shown above and the use of orange as a warning color for high-voltage cable.

Protection against direct contact

Guarding to prevent accidental contact is a term that covers a number of basic technical measures for electrical systems that are adopted to ensure that no-one can inadvertently touch a live component carrying a hazardous voltage. There are two levels, namely basic protection (against direct contact) and fault protection (against indirect contact).

Basic protection describes the level of protection under normal operating conditions, in other words in the absence of faults. Basic protection can be stated as the degree of protection, which is commonly designated by an IP code. IEC 60529 details the structure used in the system of IP codes. The first two characters (digits) after "IP" represent the level of protection against the ingress of foreign matter and moisture inside an electrical enclosure. With regard to human safety, however, it is the optional third character (letter) that is important: the minimum safety standard specified for a fully installed high-voltage system in any BMW ActiveHybrid vehicle is the IP level represented by the IP code IPxxB. The letter "B" in this code means that the housing and covers of the high-voltage system are shaped and assembled such that it is not possible to insert a finger in such a way as to come into contact with a component carrying a hazardous voltage. IPxxB is required for the engine compartment, for example, and is satisfied by the high-voltage system. The even higher classification represented by the degree of protection IPxxD is required at certain other locations in the vehicle. The letter "D" means that the enclosure is such as not to permit contact to be made with a component carrying a hazardous voltage even by insertion of a wire.

Basic protection also includes the insulation of active parts (live parts in other words) that can be inside or outside enclosures.

Protection against indirect contact

The second level, fault protection, comprises additional protective measures over and above basic protection that prevent endangerment for humans even in the event of an electrical fault occurring. They include:

- Insulation of the high-voltage cables.
- Network form of the high-voltage electrical system.

Both these protective measures are explained in this section.

The high-voltage cables invariably have an insulating plastic covering. Outside this covering there is only a shield (wire braid/film) that is used for two purposes: to reduce electromagnetic interference and primarily for the purposes of insulation monitoring (see below). The shield is protected against mechanical damage (e.g. chafing) by a bright orange plastic outer sheath. The high-voltage cables are required to meet ultra-high requirements in terms of insulation resistance. Insulation resistance ratings in the order of magnitude of several mega ohms is a must, and compliance is verified in post-

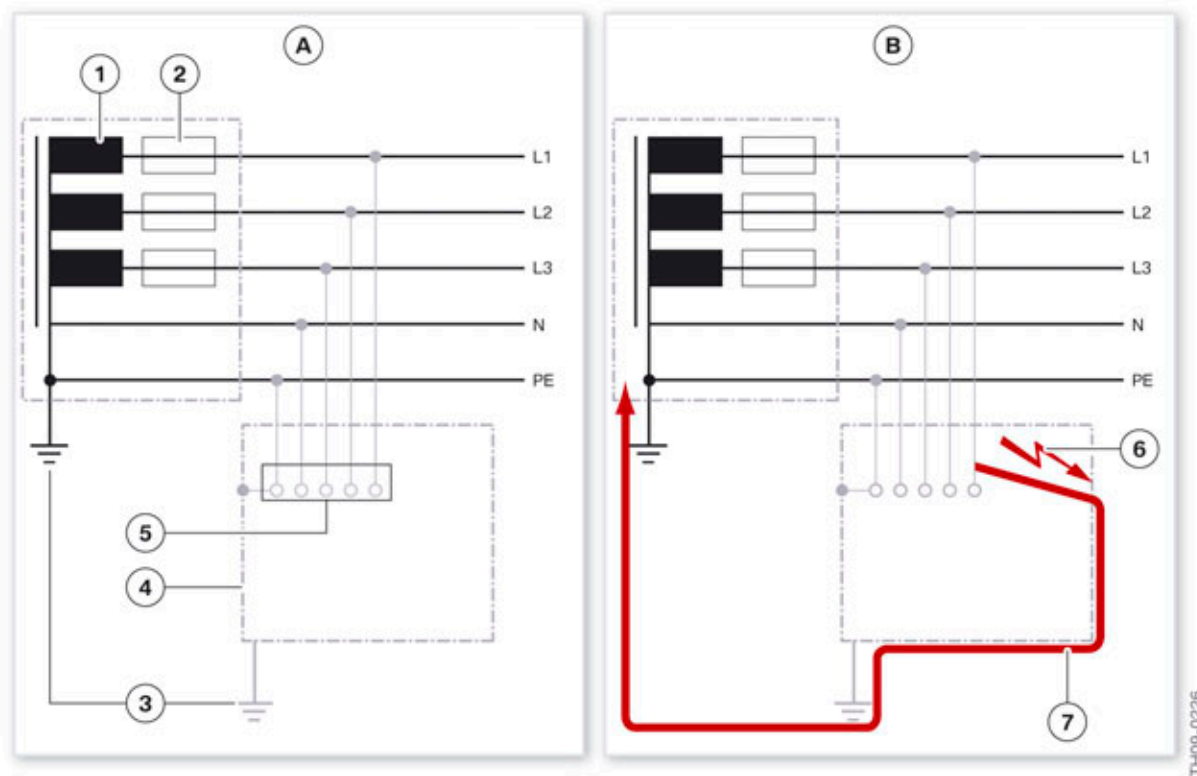
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production testing and monitored when the cables are in use. The implementation of these measures means that the high-voltage cables comply with the class II requirements for protection against direct contact.

In what way does the **network form** of the high-voltage electrical system contribute to protection in the event of an electrical fault occurring in a high-voltage component? For the purposes of illustration, the assumption is that a fault has occurred caused by a short circuit of a live cable to an electrically conductive enclosure. The first part of this explanatory description takes the form of a comparison with domestic wiring systems, and this is followed by a detailed explanation of the differences between this wiring setup and the high-voltage electrical system in a hybrid vehicle.

A domestic wiring setup is commonly a three-phase alternating current (AC) system. There are three live conductors (external conductors) and at least one conductor that is grounded (the neutral conductor and possibly a protective-earth conductor). This network form is in very widespread use and is known as a "TN" network. "TN" is derived from "Terre Neutre" (French) and signifies that the star point of the generator is grounded and a grounded neutral conductor is wired to the interface in the distributor box. Electrically conductive enclosures are also grounded.



TN network

Index	Explanation
A	Structure of a TN network
B	Fault situation in a TN network
1	Coils of the in-feed transformer or generator
2	Over-current protectors (fuses)
3	Ground

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Index	Explanation
4	Enclosure of a consumer in the TN network
5	Connections of the wires in the consumer
6	Short circuit of external conductor L1 to the enclosure of the consumer
7	Fault current from external conductor L1 flowing via enclosure and ground back to the infeed transformer
L1 ... L3	Three external conductors
N	Neutral conductor
PE	Protective earth

The enclosure or housing protects the user of a device from direct contact with live parts. If a fault occurs in which an electrically conductive connection is established between an external conductor and the enclosure, a hazardous voltage is applied to the enclosure. A current flows through the enclosure to ground and so back to the generator.

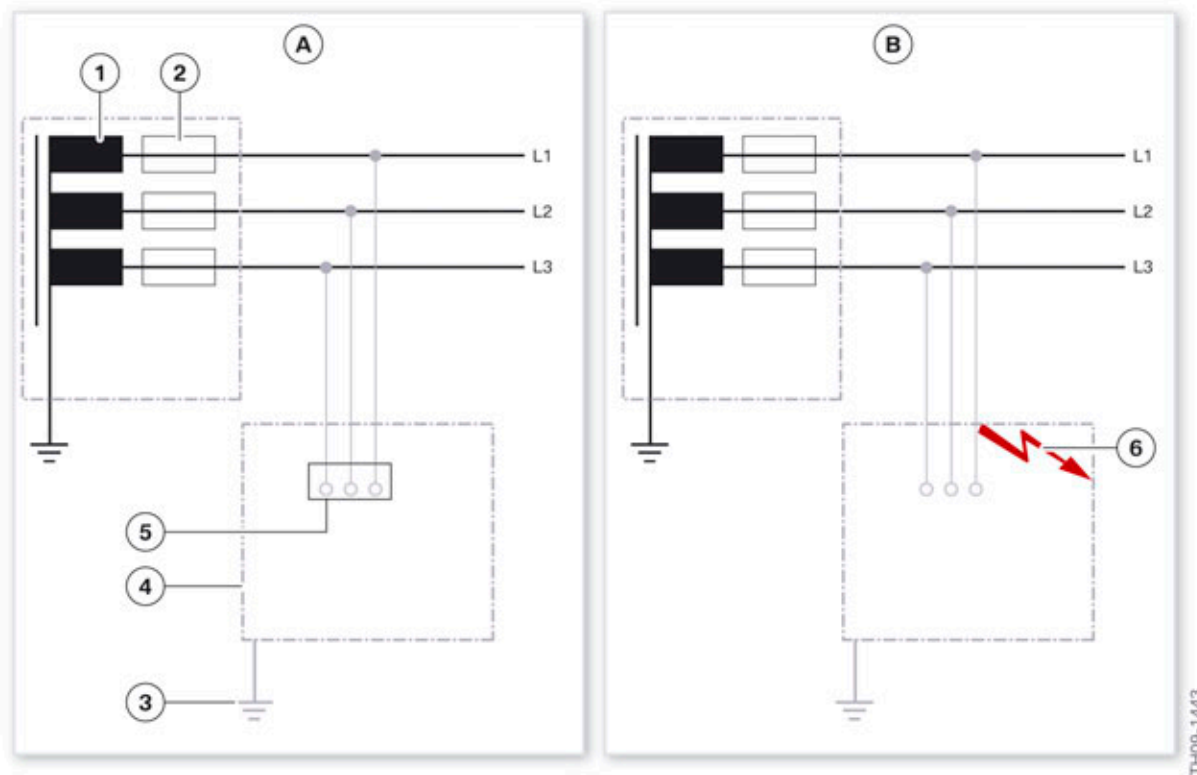
In order to protect people against the hazardous voltage at the enclosure, these TN networks employ over-current protectors in the form of fuses. The design of the fuse is such that in the above-mentioned fault situation the fuse instantly trips and the voltage is switched off. This is one way of achieving protection against indirect contact.

If a TN network with fuse-type protection were to be used in hybrid-car applications, a single fault (short circuit between external conductor and enclosure) would suffice and the high-voltage system would have to be shut down. This would severely restrict the availability of the electric propulsion unit. This is one of several reasons why the engineers who design hybrids opted for a different network form, the form commonly designated IT (abbreviation derived from the French term "Isolé Terre") in combination with effective insulation monitoring.

In an IT network there is no connection between a live conductor and earth or, in the case of automotive applications, with the vehicle ground. If we return to the fault considered above, what happens in an IT network when a live conductor shorts to an enclosure?

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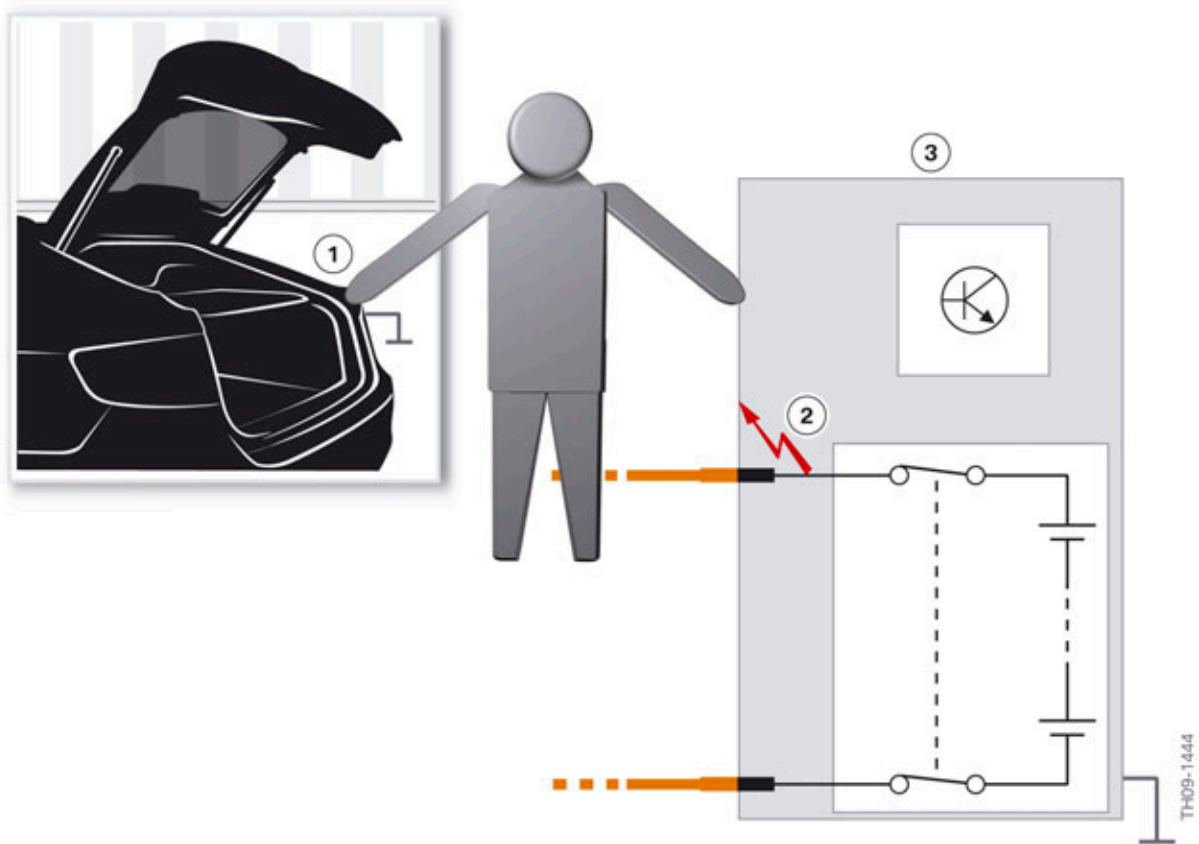
IT network

Index	Explanation
A	Structure of an IT network
B	Fault situation in an IT network
1	Coils of the in-feed transformer or generator
2	Over-current protectors (fuses)
3	Equipotential bonding compensation via earth
4	Enclosure of a consumer in the IT network
5	Connections of the wires in the consumer
6	Short circuit of external conductor L1 to the enclosure of the consumer
L1 ... L3	Three external conductors

The voltage source is not connected to earth (in automotive applications: to vehicle ground), no short-circuit current flows. Consequently, a fuse would not be tripped in this situation. This in turn means that if a fault of this nature occurs, the high-voltage system can initially remain in operation. This ensures the high availability of the high-voltage electrical system and this constitutes a significant advantage of this network form. IT networks can be used not only for three-phase systems but also for the direct-current configurations that are also used for the high-voltage systems of hybrid vehicles.

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Fault situation in the high-voltage electrical system of a hybrid vehicle

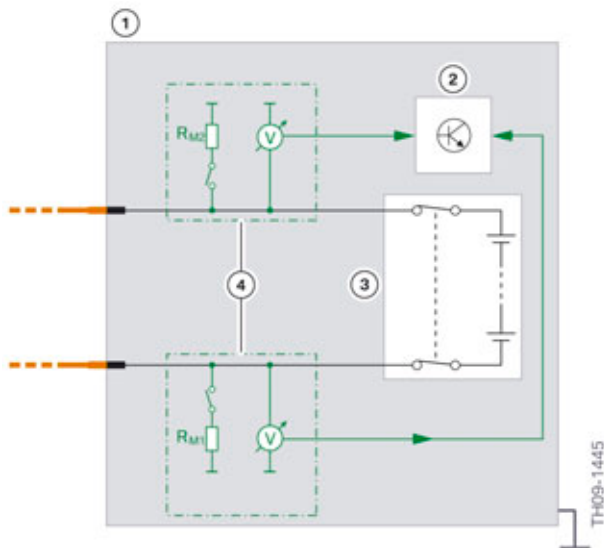
Index	Explanation
1	Vehicle body (in electrical terms: ground)
2	Short circuit of a high-voltage cable to the enclosure of the high-voltage battery unit
3	High-voltage battery unit

If a person touches the enclosure (housing of the high-voltage battery) current does not flow through his or her body because the circuit to the voltage source is not closed. In fact, the circuit is not closed even if the person simultaneously contacts vehicle ground by touching the bodywork of the vehicle with some other part of the body. The only situation in which current could flow through the person's body would be if he or she were also to simultaneously touch a second live cable of the high-voltage electrical system. This clearly illustrates a second advantage of IT networks.

But how can a fault of this kind be detected and, ultimately, how can it be rectified? This is the reason why **insulation monitoring** is implemented in the high-voltage electrical system. Its purpose is to detect hazardous insulation faults between any live high-voltage component and electrically conductive enclosures or housing or between a component and ground. A hazardous insulation fault is present when a hazardous voltage is applied between housing/ground and another live high-voltage component. Or expressed in other terms, when the insulation resistance between a high-voltage component and housing/ground drops below a defined threshold.

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Insulation monitoring in the high-voltage electrical system

Index	Explanation
1	High-voltage battery unit
2	Control unit in the high-voltage battery unit
3	High-voltage battery
4	Circuits for measuring insulation resistance

The insulation monitoring circuitry in the high-voltage electrical system measures the system's insulation resistance, for example indirectly by a series of voltage measurements. Voltage is measured at precision resistors between live components (e.g. positive and negative poles of the high-voltage battery) and vehicle ground. These measurements take place both while the high-voltage system is live and after the high-voltage system has been switched off. Insulation monitoring is generally integrated into one or two high-voltage components, for example the power electronics and/or the control unit of the high-voltage battery. But how can insulation monitoring detect an insulation fault in another high-voltage component, such as the electric A/C compressor, for example?

Insulation monitoring at one or two central points can function only if all electrically conductive housings of the high-voltage components are galvanically (electrically) connected to ground, in other words to the body of the vehicle. This galvanic connection has to be present in order for example for the monitor implemented in the power electronics to reliably detect a short circuit in a high-voltage cable of the electric A/C compressor. In the absence of this galvanic connection between housing and ground the fault would remain undetected and would therefore constitute a potential hazard. The galvanic interconnection between the housings the connections from housing to ground is referred to as "equipotential bonding". The electrical connections established for this purpose are known as "equipotential bonding conductors" or simply "bonding conductors".



The electrically conductive housings of high-voltage components must be galvanically connected to ground. As regards repairs to high-voltage components, but also when body components are replaced, this has to be taken duly into consideration during assembly: it is essential to make sure that the galvanic connection between the housing and the body is correctly restored. In this respect too,

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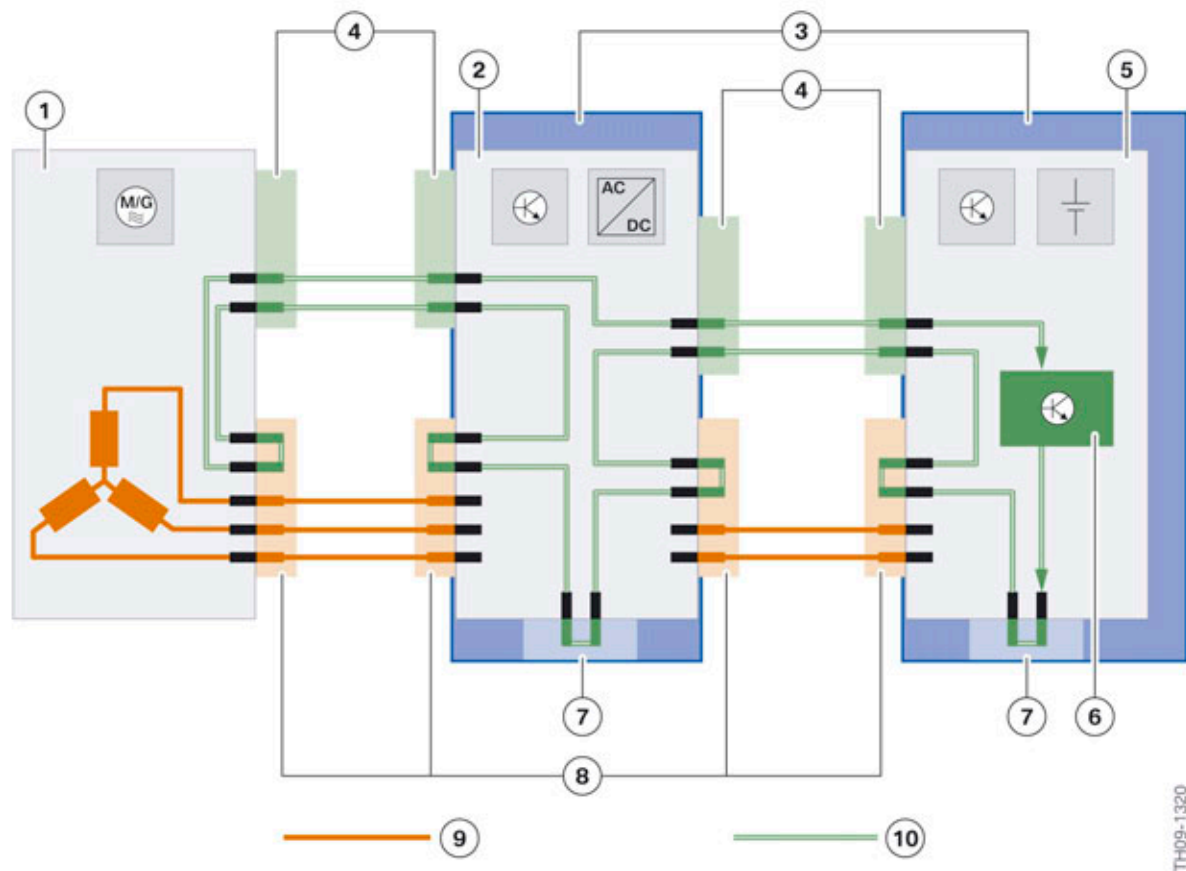
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it is very important to proceed precisely in accordance with the repair instructions. This applies most particularly to the use of the specified connecting elements (e.g. self-tapping screws) and compliance with the specifications for tightening torques.

High voltage interlock loop

The live parts of the high-voltage components are protected against direct touch contact by covers or housings. Much the same applies to the conductors of the high-voltage cables: they are protected against touch contact by their insulating sheaths or by insulating connector housings. Before working on a high-voltage component, you must apply the safety rules to shut down the high-voltage system. Once this has been accomplished according to procedure, the parts are no longer live and work can proceed in safety. There is of course a remote possibility that the correct shutdown procedure might be omitted, so an extra safety precaution is implemented as a means of imposing an automatic shutdown of the high-voltage system.

Covers over touchable live parts and plugs that have touchable contacts are integrated into the high-voltage contact monitoring circuit known as the high-voltage interlock loop. The principle of the high-voltage interlock loop is illustrated and explained below.



Principle of the high-voltage interlock loop

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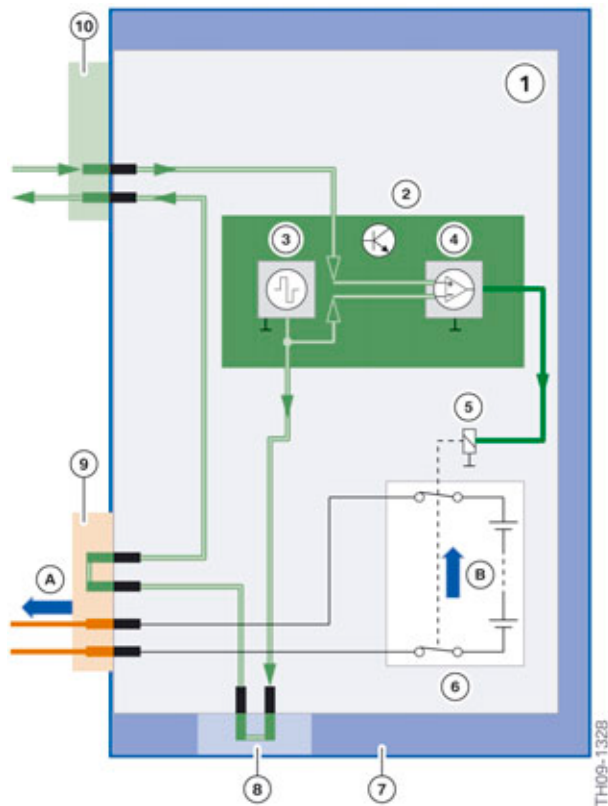
Index	Explanation
1	Electrical machine
2	Power electronics
3	High-voltage safety cover
4	Low-voltage connector for carrying the test lead of the high-voltage interlock loop
5	High-voltage battery unit
6	Electronics for high-voltage interlock loop
7	Jumper of the high-voltage interlock loop on the high-voltage safety cover
8	High-voltage connector with jumper of the high-voltage interlock loop
9	High-voltage cable
10	Test lead of the high-voltage interlock loop

The electronics of the high-voltage interlock loop discharge two primary functions. The first is to generate the interlock signal. This is an alternating voltage (or an alternating current) that is generally a square wave with low values and therefore safe. The interlock signal is tapped into a circuit that runs over the covers of the high-voltage components and/or the connectors of the high-voltage cables. There is a jumper in each cover or connector. When the cover is installed and secured, the jumper closes in the circuit of the high-voltage interlock loop. If the cover is removed and the jumper pulled, the circuit is interrupted. Much the same applies with the high-voltage connectors: is the high-voltage connector is inserted and locked the jumper closes in the circuit of the high-voltage interlock loop. If the connector is pulled, the jumper interrupts the circuit. The contacts of the high-voltage cables are recessed in the connectors. This means that when the connector is pulled the circuit of the high-voltage interlock loop is interrupted first, before the connection of the high-voltage cables is interrupted. This enhances touch-contact protection and reduces the risk of an arc flashing across the contacts when the connection is broken.

The jumpers are connected in series in the entire circuit of the high-voltage interlock loop. Consequently, removing a single cover or pulling a single connector suffices to interrupt the interlock signal. The second function discharged by electronics is to process the signal at their second connection. As soon as the incoming signal deviates significantly from the outgoing high-voltage interlock signal (signal level, short circuit to earth or to positive), the electronics trigger shutdown of the high-voltage system.

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Detailed view of the high-voltage interlock loop

Index	Explanation
1	High-voltage battery unit
2	Electronics of the high-voltage interlock loop
3	Generator for interlock signal
4	Evaluation circuit for interlock signal
5	Electromechanical switch contactor
6	High-voltage battery
7	High-voltage safety cover
8	Jumper of the high-voltage interlock loop on the high-voltage safety cover
9	High-voltage connector with jumper of the high-voltage interlock loop
10	Low-voltage connector for carrying the test lead of the high-voltage interlock loop
A	Pulling of the high-voltage connector and opening of the jumper in the circuit of the high-voltage interlock loop
B	Opening of the contacts of the switch contactor

The electronics can be integrated as part of a control unit of a high-voltage component (for example the high-voltage battery). The generator and the evaluation circuit for the interlock signal can be split across two high-voltage components (e.g. high-voltage battery and power electronics).

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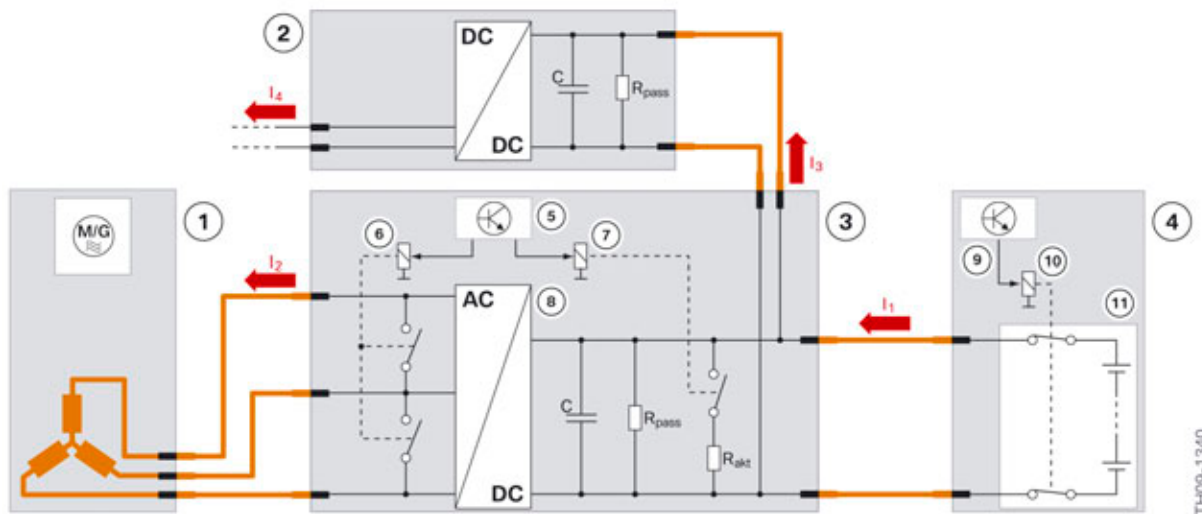
Shutdown of the high-voltage system is automatic and takes place in several steps:

- Cancellation of activation signal for the electrical machine(s)
- Short-circuit of the coils of the electrical machines
- Opening of the switch contactors in the high-voltage battery
- Discharging of the high-voltage circuit.

In this way all possible voltage sources in the high-voltage system are reliably shut down. This ensures that no later than five seconds after interruption of the high-voltage interlock loop circuit there is no longer a hazardous voltage present at any point in the entire high-voltage system.

Discharge of the high-voltage circuit

In addition to the high-voltage battery there are two other voltage sources in the high-voltage electrical system: the capacitors in the power electronics (and other high-voltage components) and the electrical machines. Even after the switch contactors of the high-voltage battery open in the power-down process, the capacitors or the electrical machines would be capable of keeping the voltage in the high-voltage electrical system at a level high enough to constitute a touch hazard. This is the reason why the high-voltage circuit is discharged each time the high-voltage system is powered down. The graphics below use a simplified circuit diagram of high-voltage components to illustrate how the discharge process takes place. The way in which the discharge circuits are actually implemented in hybrid vehicles might not necessarily be as illustrated here in this simplified representation.



Simplified circuit diagram of the high-voltage system: live state

Index	Explanation
1	Electrical machine
2	DC/DC converter
3	Power electronics
4	High-voltage battery unit
5	Control unit of the power electronics
6	Relay for short-circuiting the coils of the electrical machine

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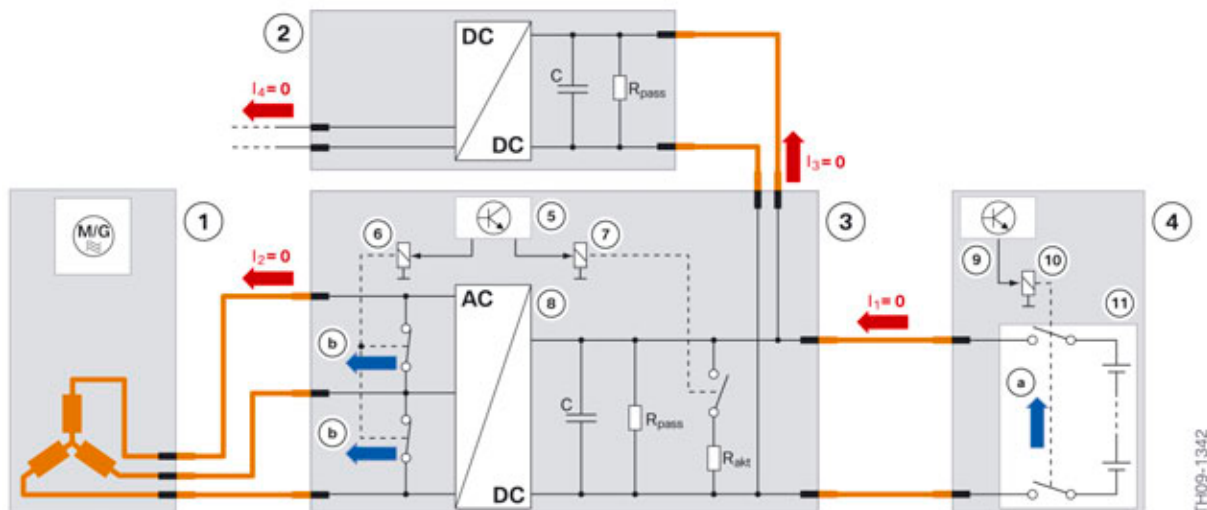
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Index	Explanation
7	Relay for active discharge of the capacitors
8	Current rectifier in the power electronics, operated in this case as an inverter
9	Control unit in the high-voltage battery unit
10	Electromechanical switch contactor in the high-voltage battery unit
11	High-voltage battery
C	Link capacitors in DC/DC converter and power electronics
$I_1 \dots I_4$	Consumer currents
R_{pass}	Passive discharge resistor
R_{akt}	Active discharge resistor

As long as the switch contactors of the high-voltage battery remain closed, the voltage of the high-voltage battery is present in the high-voltage cables. The capacitor on the DC voltage side of the power electronics carries the same voltage and is charged. The power electronics supply the high-voltage components with electrical energy, current flows through the high-voltage cables.

Before the switch contactors of the high-voltage battery are opened, the power electronics drive all the high-voltage consumers to assume a state in which they are unable to accept more current. In terms of the circuitry this state is equivalent to having no consumers connected to the high-voltage electrical system.

The electrical machines could still generate a hazardous voltage in the high-voltage electrical system, even if the switch contactors of the high-voltage battery have already opened. If the electrical machines were still rotating a voltage would be induced in their coils. This voltage would be applied to the high-voltage cables and depending on the speed of the electrical machines it could be of a level that would constitute a touch-contact hazard. To prevent this from occurring the coils of the electrical machines are short-circuited once the switch contactors of the high-voltage battery have opened.



Simplified circuit diagram of the high-voltage system: short-circuiting of the coils of the electrical machine

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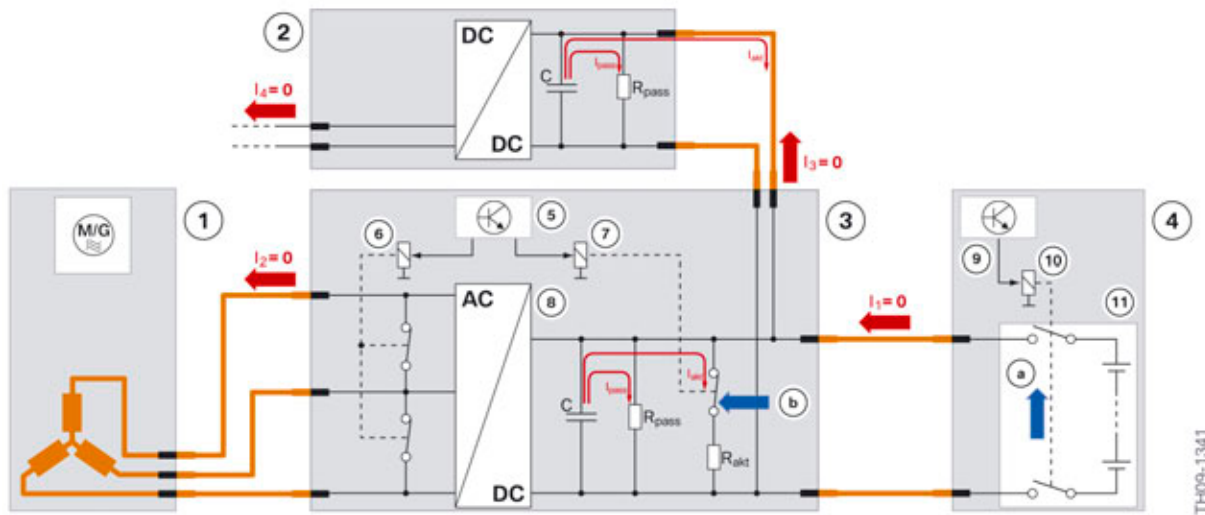
Index	Explanation
1	Electrical machine
2	DC/DC converter
3	Power electronics
4	High-voltage battery unit
5	Control unit of the power electronics
6	Relay for short-circuiting the coils of the electrical machine
7	Relay for active discharge of the capacitors
8	Current rectifier in the power electronics, operated in this case as an inverter
9	Control unit in the high-voltage battery unit
10	Electromechanical switch contactor in the high-voltage battery unit
11	High-voltage battery
a	Opening of the contacts of the switch contactor in the high-voltage battery unit
b	Closing of the contacts for short-circuiting the coils of the electrical machine
C	Link capacitors in DC/DC converter and power electronics
$I_1 \dots I_4$	Consumer currents
R_{pass}	Passive discharge resistor
R_{akt}	Active discharge resistor

In the current range of hybrid vehicles, this process is controlled by the power electronics. In all current BMW ActiveHybrid vehicles the coils of the electrical machines of the propulsion system are short-circuited. It is particularly important to ensure that the windings of these electrical machines are short-circuited because as long as the vehicle is still in motion the electrical machines in the powertrain might still be rotating. Hazardous voltages could therefore be induced in their coils. Whether or not the coils of the other electrical machines in hybrid vehicles have to be short-circuited depends on the design and how the machines are shut down (the electric A/C compressor is a case in point). If an electrical machine continues to rotate after being shut down and if this run-on can induce a hazardous voltage in the coils, the coils have to be shorted. This can differ from vehicle type to vehicle type.

The circuit diagram shows for example that capacitors used in the power electronics are connected in parallel to the high-voltage cables. When the switch contactors of the high-voltage battery are opened, the same voltage as before is still present in the high-voltage cables. The capacitors store the electrical energy and keep the voltage at this level. Without additional measures the capacitor is unable to discharge, because all the high-voltage consumers have already been shut down. A discharge circuit therefore has to be implemented for all the capacitors so that the voltage on the high-voltage cables can drop. This discharge circuit consists of a passive discharge resistor and an active discharge resistor.

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Simplified circuit diagram of the high-voltage system: discharging of the capacitors

Index	Explanation
1	Electrical machine
2	DC/DC converter
3	Power electronics
4	High-voltage battery unit
5	Control unit of the power electronics
6	Relay for short-circuiting the coils of the electrical machine
7	Relay for active discharge of the capacitors
8	Current rectifier in the power electronics, operated in this case as an inverter
9	Control unit in the high-voltage battery unit
10	Electromechanical switch contactor in the high-voltage battery unit
11	High-voltage battery
a	Opening of the contacts of the switch contactor in the high-voltage battery unit
b	Closing of the contact for active discharge of the capacitors
C	Link capacitors in DC/DC converter and power electronics
$I_1 \dots I_4$	Consumer currents
I_{pass}	Current through passive discharge resistor
I_{akt}	Current through active discharge resistor
R_{pass}	Passive discharge resistor
R_{akt}	Active discharge resistor

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Passive discharge resistor R_{pass} is permanently parallel with the capacitor. When the switch contactors in the high-voltage battery open the discharge current flows immediately from the capacitor through the passive discharge resistor. The voltage in the capacitor and thus the voltage on the high-voltage cables drops exponentially and reaches zero volts after the time $t = 5 \cdot \tau = 5 \cdot R_{\text{pass}} \cdot C$. But even when the high-voltage system is live, current flows through the passive discharge resistor. The passive discharge resistor is therefore relatively high-resistance in its design in order to keep this power loss to an acceptable level. The order of magnitude is in the tens of kilo-ohms range. The capacitors used, on the other hand, have capacitances of several hundred microfarads. This passive discharge can take up to several minutes for the voltage to drop to zero. However, the system is designed to ensure that the capacitors discharge via the passive resistor to a voltage level that is no longer hazardous within a time of no more than five minutes.

Passive discharge is only a supplementary safety measure, in case something unforeseen prevents firing of the active discharge resistor. The power electronics trigger closure of the switch on the active discharge resistor for shutdown of the high-voltage system as soon as the switch contactors of the high-voltage battery have opened. Active discharge resistor R_{akt} has a resistance of only a few tens of ohms and enables discharge at a considerably faster rate. This design ensures that the high-voltage circuit has completed discharge through the active discharge resistor within no more than five seconds.

The hybrid vehicles in the current range usually have one and only one active discharge resistor, and it is located in the power electronics. When the high-voltage system shuts down not only the capacitors of the power electronics discharge via this resistor but also the capacitors in other high-voltage components, such as the DC/DC converter and the electric A/C compressor. Central active discharge of this kind is possible because the high-voltage cables and the capacitors in the other high-voltage components are connected in parallel.

In addition to the active discharge resistor, each high-voltage component with capacitors also has passive discharge resistors. Their purpose is to ensure that the capacitors are discharged even if active discharge fails or the high-voltage cables between the high-voltage components are interrupted before active discharge can be successfully completed.

	Passive discharge	Active discharge
Type of discharge circuit	Permanent	Switched
Number of discharge resistors	One passive discharge resistor in each high-voltage component with capacitors	At least one active discharge resistor in the entire high-voltage system; more than one possible
Maximum time for discharge	5 minutes	5 seconds

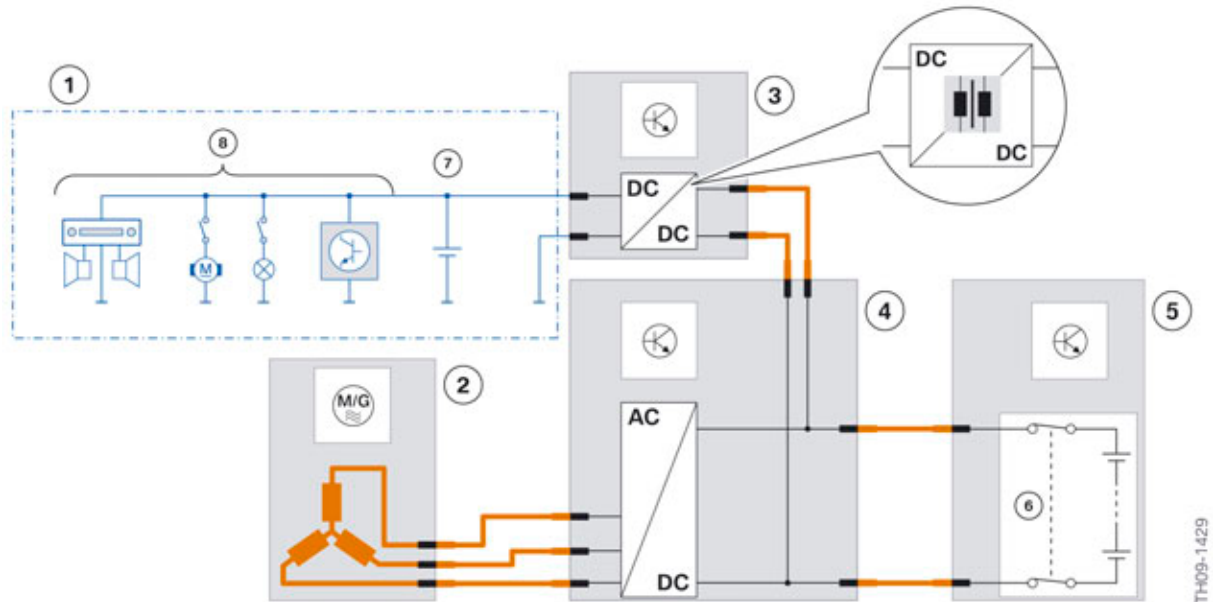
Galvanic separation of the high-voltage electrical system from the 12 V vehicle electrical system

In hybrid vehicles the high-voltage electrical system and the 12 V vehicle electrical system are interconnected by a DC/DC converter. In this way it is possible for example to charge the 12-V battery using energy from the high-voltage electrical system. This means that a 12-V generator is no longer necessary. Despite the obvious advantages of this "energy-transfer" connection between the two vehicle electrical systems, safeguards have to be in place to ensure that hazardous voltage from the high-voltage electrical system cannot be transferred to the 12-V system. If this were not the case the 12 V vehicle electrical system would be subject to application of the same rules for electrical safety as apply to the high-voltage electrical system with regard to preparation for and work on the high-voltage electronics.

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Consequently, the 12 V vehicle electrical system and the high-voltage electrical system are galvanically separated, in other words there is no conductive connection between the two systems. This is accomplished by suitably insulating all the components and wiring. The DC/DC converter requires a circuit that ensures galvanic separation while sustaining the energy-transfer capability. A circuit such as a transformer circuit is suitable in this respect. Energy cannot be transferred between two DC networks by a transformer, so the direct current voltages have to be transformed into AC voltages and vice versa.



Simplified circuit diagram: high-voltage electrical system and 12 V vehicle electrical system

Index	Explanation
1	12 V vehicle electrical system
2	Electrical machine
3	DC/DC converter
4	Power electronics
5	High-voltage battery unit
6	Contacts of the switch contactors in the high-voltage battery
7	12-V battery
8	12-V consumers

Short-circuit monitoring

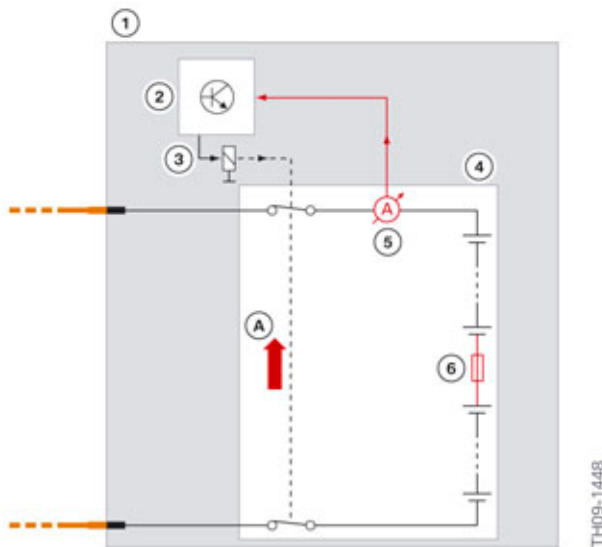
Short circuits in the high-voltage electrical system, for example between the two high-voltage battery cables would lead to very high short-circuit currents. The reasons for this are:

- High voltage
- Low internal resistance of the high-voltage battery
- Low resistance of the high-voltage cables.

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The consequences of these very high short-circuit currents would be severe. They range from arcing through irreparable damage to high-voltage cables or the high-voltage battery to fire. In order to avoid consequences of this nature, technical measures for the detection of a short circuit are engineered into the high-voltage electrical systems of hybrid vehicles. As a general rule, these measures are integrated into the high-voltage battery.



Simplified circuit diagram: short-circuit monitoring

Index	Explanation
1	High-voltage battery unit
2	Control unit in the high-voltage battery unit
3	Electromechanical switch contactor
4	High-voltage battery
5	Current sensor
6	Safety fuse
A	Opening of the switch contactor contacts in response to tripping of the short-circuit monitor

High-amperage safety fuses and electronic over-current trip breakers are used. To reduce the response time in the event of a short circuit, current is electronically monitored by current sensors in the battery cables. If the control unit of the high-voltage battery unit detects an impermissibly high current, it triggers opening of the contacts of the switch contactor in the high-voltage battery. The switch contacts are designed to open reliably even despite the high currents that occur in a short-circuit event. The downside of this design, however, is that the switch contacts are corresponding short-lived. Electronic short-circuit monitoring reduces response time vis-à-vis conventional safety fuses, particularly when the currents involved are high.

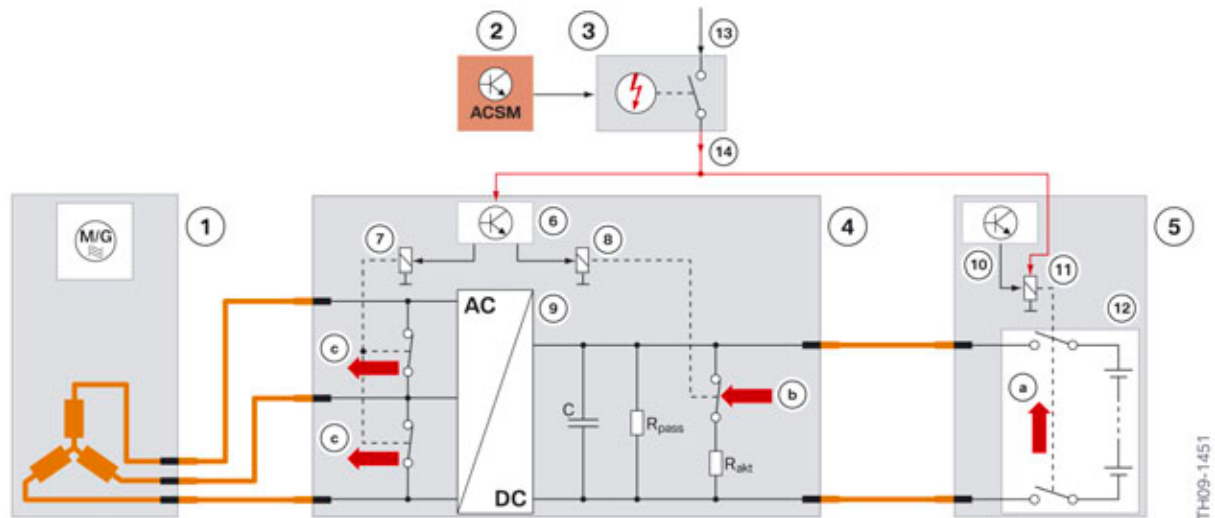
Shutdown in the event of an accident

If the vehicle is involved in an accident the high-voltage cables below the floor could be sheared off and this could cause sparking or arcing, if the bottom of the vehicle comes into contact with sharp-edged obstacles (e.g. crash barriers). In order to minimise this risk the high-voltage system

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is deactivated in the event of an accident. In all current BMW vehicles and therefore in the BMW ActiveHybrid models as well, accident detection and the co-ordination of the safety systems are handled by the Crash Safety Module (ACSM).



Shutdown of the high-voltage system in response to detection of an accident situation

Index	Explanation
1	Electrical machine
2	Crash Safety Module
3	Safety battery terminal (with an extra contact)
4	Power electronics
5	High-voltage battery unit
6	Control unit of the power electronics
7	Relay for short-circuiting the coils of the electrical machine
8	Relay for active discharge of the capacitors
9	Current rectifier in the power electronics, operated in this case as an inverter
10	Control unit in the high-voltage battery unit
11	Electromechanical switch contactor in the high-voltage battery unit
12	High-voltage battery
13	Terminal 30
14	12-V wiring interrupted by the safety battery terminal
a	Opening of the contacts of the switch contactor in the high-voltage battery unit
b	Closing of the contact for active discharge of the capacitors

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Index	Explanation
c	Closing of the contacts for short-circuiting the coils of the electrical machine
C	Link capacitors in DC/DC converter and power electronics
R_{pass}	Passive discharge resistor
R_{akt}	Active discharge resistor

If the Crash Safety Module detects a crash of corresponding severity, the 12-V battery's positive lead is pyrotechnically disconnected from the positive battery connection point. The actor used for this purpose – the safety battery terminal – has been extended to include added functionality for the BMW ActiveHybrid vehicles. Along with the positive battery cable, in these vehicles the safety battery terminal interrupts another 12-V conductor. This conductor is used in two different ways to shut down the high-voltage system in hybrids:

- 1 Opening the switch contactor contacts in the high-voltage battery.
- 2 Active discharge of the high-voltage circuit.

The switch contactor contacts are opened without additional action on the part of the control unit in the high-voltage battery unit. The 12-V cable interrupted by the safety battery terminal is used in the high-voltage battery for the supply voltage of the electromagnet of the electromechanical switch contactor in the high-voltage battery unit. If the supply voltage drops out, the contacts of the switch contactor automatically open.

The power electronics use the 12-V cable interrupted by the safety battery terminal as a signal input for active discharge of the high-voltage circuit. Interruption of this cable triggers the control unit of the power electronics to:

- short-circuit the coils of the electrical machines
- initiate active discharge of the capacitors.

Triggering of the safety battery terminal is a signal in response to which the high-voltage system is shut down in the event of a crash. The Crash Safety Module's bus telegrams are also processed: as soon as the Crash Safety Module signals a crash of corresponding severity by means of a bus telegram, the high-voltage system is shut down.

These engineered-safety measures suffice to ensure that the high-voltage system is shut down reliably and in a very short space of time in the event of a crash. This virtually excludes the possibility of high voltages presenting hazards in the course of or subsequent to a crash.



Nevertheless, it is important to bear in mind that special care always has to be exercised when dealing with high-voltage components if the component housings have been damaged in a crash. In case of doubt, always contact BMW Group's Technical Support (PUMA).

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4.6.3. Points to bear in mind for working on high-voltage system

Necessary work

This section draws attention to the following specifics of handling the high-voltage system in hybrid vehicles:

- Visual inspection of high-voltage components
- Diagnostics
- Establishing electrical connections between body components.

Before commencing work on a hybrid car, always subject the high-voltage components to close visual inspection. It is always advisable to perform this check and in the case of vehicles that have been involved in accidents, it is absolutely essential. The visual inspection enables you to identify damaged high-voltage components at an early juncture, even before a self-diagnosis function has detected a fault in the high-voltage system. If, for example, the insulation of a high-voltage cable has been damaged but there is no conductive connection between the live wire and a body component, the insulation monitoring will not yet have detected a fault condition. A conscientious visual inspection paying close attention to every aspect of the system will however reveal the presence of the defect. Similarly, particular attention should be paid to close visual examination of the housing of the high-voltage battery when a car that has been involved in a crash is checked over. This is a prophylactic measure that can help avoid health and safety issues such as the possibility of health endangerment due to spilled electrolyte, for example.

On completion of repair work on the high-voltage system, use the diagnosis system to perform a check of the high-voltage components. Depending on the vehicle involved, this can entail simply reading out the fault memory or starting special Service functions. This check requests the results of numerous self-diagnosis routines in the high-voltage system. This is one way of ensuring that the high-voltage system has been correctly repaired and is again in full working order and perfectly safe.



If there is any doubt or if the findings relating to the high-voltage system are equivocal, contact BMW Group's Technical Support (PUMA). This applies to both visual inspection and to the check with the diagnosis system.

The body of the vehicle is used as a "large equipotential bonding conductor" between the housings of the individual high-voltage components. This electrically conductive connection is crucial as a precondition for correct operation of the insulation monitoring function of the high-voltage system.



If coated or painted parts of the bodywork have been replaced, it is absolutely essential to proceed strictly in accordance with the instructions in the repair manual, and this is particularly important in the case of hybrid vehicles.

For example, a high tightening torque given in the repair manual might be necessary because of the need to ensure that an electrically conductive connection is established between a new body panel and the supporting structure even though the new panel is coated and therefore the connection would not be suitably conductive if the high torque were not applied to the threaded fastener or fasteners. This is of crucial importance with regard to correct operation of the engineered safety measures

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implemented for the high-voltage system. Along with the tightening torques, another aspect that requires strict compliance with the repair instructions is the use of the specified connecting elements such as self-tapping screws, for example.

Permissible work and prohibited work

Given the complexity of the high-voltage system and the possible hazards that could be posed by the high electrical voltage, it is absolutely essential for the safety measures in particular to be in full working order at all times. In order to ensure this complete functionability, certain restrictions apply to maintenance and repair work. The basic prerequisite applicable to all work on the high-voltage system is that each Service employees who undertakes work of this nature must have been trained as an electro-technician for high-voltage systems in inherently safe vehicles. In addition, it is absolutely essential to apply the three rules for electrical safety in all instances and to proceed invariably strictly in accordance with the applicable test schedule and repair manual.

It is permissible to disconnect and re-establish the electrical connections between high-voltage cables and high-voltage components.

It is prohibited to attempt to repair a damaged high-voltage cable. For example, do not attempt to repair a localized defect in the insulation. Instead, you must replace the entire high-voltage cable.

It is prohibited to open the housing of a high-voltage component that evinces a defect (the housing of the power electronics, for example) in order to attempt repair of the component. Instead, you must proceed in accordance with the applicable test schedule and replace the entire high-voltage component.



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